



Pathlock Native Cyber Security and GRC Suite Dynamic Access Controls Deployment Guide

Version: 2025 Q4



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1 Overview

SAP ECC/S4HANA maintains security with its role-based access controls, assisting organizations in preventing unauthorized access and mitigating risk violations.

Pathlock DAC functions as an additional security layer for SAP users within the ECC/S4HANA application. This extra layer, acting as an add-on, provides users with a granular approach to secure their ECC/S4HANA environments.

1.1 Purpose

This guide provides information regarding Pathlock DAC deployment.

1.2 Mandatory Upgrade Requirements (Existing DAC Customers)

Customers upgrading from a DAC release earlier than DAC 2025 Q3 must run the Policy Migration Utility before using the DAC Policy Studio

Mandatory Prerequisite:

Before configuring policies in DAC Policy Studio (/APPSDM/ABAC), you must execute the policy migration. Skipping this step may lead to missing or inconsistent policy behavior.

Action Required:

1. Execute migration using transaction - /APPSDM/POL_MIGRATE
2. After successful migration, proceed with configuration in DAC Policy Studio(/APPSDM/ABAC).

Note: This is a **one-time step** for upgrades only. This step is not required for first time installations.

1.3 Required Roles and Authorization Objects

The following PFCG Roles are essential for accessing the Pathlock DAC application:

PFCG Roles	Purpose	T-Code Assigned
/APPSDM/DAC_ADMIN	This super-user role grants extensive access for performing DAC configurations, centralization processes, master data maintenance, and viewing logs.	▪ /APPSDM/ABAC - ABAC configuration ▪ /APPSDM/FF - Field Finder ▪ /APPSDM/SR - System Readiness Check ▪ /APPSDM/CENTRAL - ABAC Centralization Process ▪ /APPSDM/USER_MASTER - User Master Maintenance

PFCG Roles	Purpose	T-Code Assigned
		▪ /APPSDM/DC - Data Classification ▪ /APPSDM/EXEC_LOGS - Policy Execution Logs ▪ /APPSDM/MAINT_ACTION – Maintain Actions ▪ /APPSDM/C_SETTINGS – Centralization Settings ▪ /APPSDM/LIST - List of Static Values ▪ /APPSDM/DC_SIMU – Data Classification Simulation ▪ /APPSDM/POLVH – Policy Version History ▪ /APPSDM/POL_SIMU – Policy Simulator ▪ /APPSDM/DISC – Data Discovery ▪ /APPSDM/TECH_CONFIG - Additional Technical Configuration ▪ /APPSDM/ALVCONFIG - ALV Grid Technical Configuration
/APPSDM/POL_CHANGE	Business users require this role to execute administrative tasks such as DAC configurations, centralization processes, and the Field Finder utility.	▪ /APPSDM/ABAC - ABAC configuration ▪ /APPSDM/FF - Field Finder ▪ /APPSDM/CENTRAL - ABAC Centralization Process
/APPSDM/DC_CHANGE	Business users need this role to manage Data Classification configurations.	▪ /APPSDM/DC - Data Classification
/APPSDM/UM_CHANGE	Business users require this role to maintain the user master.	▪ /APPSDM/USER_MASTER - User Master Maintenance
/APPSDM/ADV_CHANGE	Business users require this role to maintain T-codes with actions and view the advanced configuration tab.	▪ /APPSDM/MAINT_ACTION - Maintain Actions
/APPSDM/IT_CHANGE	Business users require this role to access System Readiness and Centralization Settings.	▪ /APPSDM/SR - System Readiness Check ▪ /APPSDM/C_SETTINGS - ABAC Centralization Settings
/APPSDM/LOG_CHANGE	Business users require this role to access the Policy Execution Logs Display.	▪ /APPSDM/EXEC_LOGS - Policy Execution Logs

Authorization objects required to access Pathlock DAC UI:

Authorization Object	Activity	PFCG Roles
Y&DAC_POL	02 (Change) 03 (Display)	/APPSDM/POL_CHANGE - Pathlock ABAC: Change Role for Policy Admin
Y&DAC_UM	02 (Change) 03 (Display)	/APPSDM/UM_CHANGE - Pathlock ABAC: Change Role for User Master Maintenance
Y&DAC_ADV	02 (Change)	/APPSDM/ADV_CHANGE - Pathlock ABAC: Change Role for Advanced Config
Y&DAC_LOG	03 (Display)	/APPSDM/LOG_CHANGE - Pathlock ABAC: Access Logs
Y&DAC_IT	02 (Change) 03 (Display)	/APPSDM/IT_CHANGE - Pathlock ABAC: Change Role for IT
Y&DAC_DC	02 (Change) 03 (Display)	/APPSDM/DC_CHANGE - Pathlock ABAC: Change Role for Data Classification
Y&DAC_LT	02 (Change) 03 (Display)	/APPSDM/LT_CHANGE - Pathlock ABAC: Change Role for List

2 Configuring Pathlock DAC

To enable field masking / restriction in the system for unauthorized users, configure the Pathlock DAC using the tool - **Pathlock: ABAC**. This tool can be accessed via transaction - **/N/APPSPDM/ABAC** or through the user menu path - **Pathlock DAC > Pathlock ABAC**.

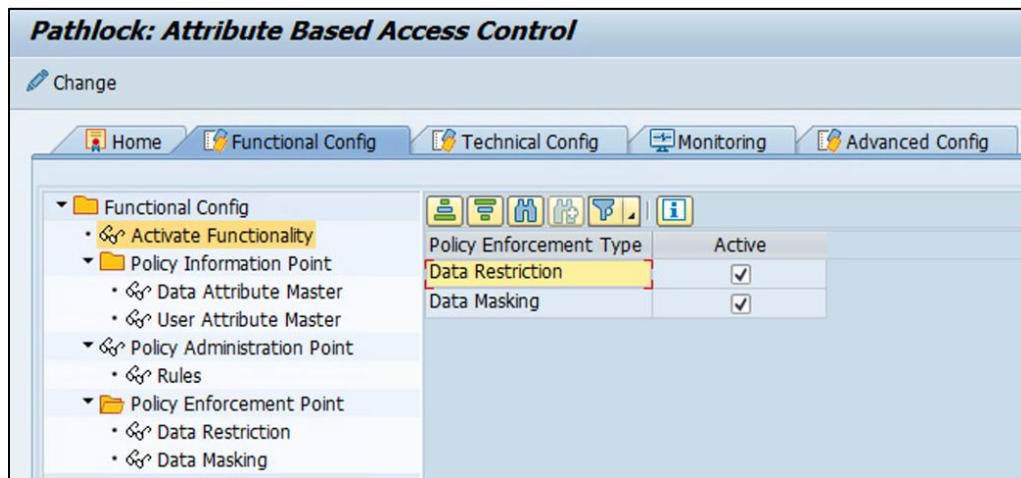
The configurations are divided into two sections:

- [Functional Configuration](#)
- [Technical Configuration](#)

2.1 Functional Configuration

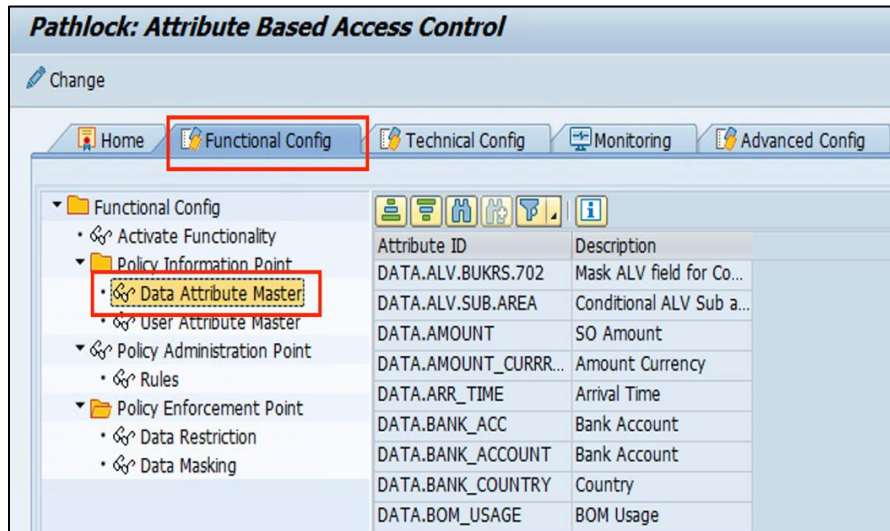
Business users perform the functional configuration using the configuration steps below:

- **Activate 'Masking/Data Restriction' Functionality:** This functionality can be activated by enabling the 'Active' checkbox and vice versa. This configuration activity requires customizing transport requests to capture and move the configuration into the respective target systems.



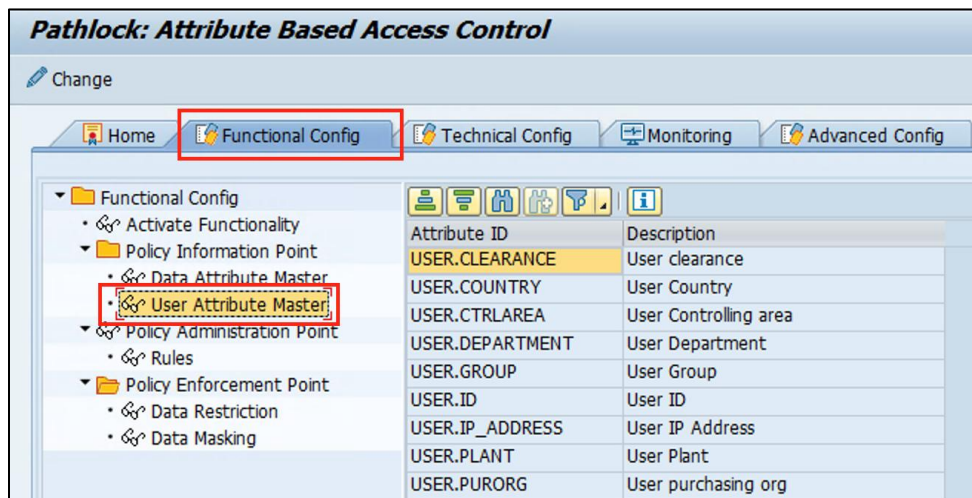
- **Data Attribute Master:** Define data attributes for data objects such as Business partner, Sales document, Employee Salary, Country, etc.

Note: Use 'DATA.' as a prefix to define all Data Attributes.

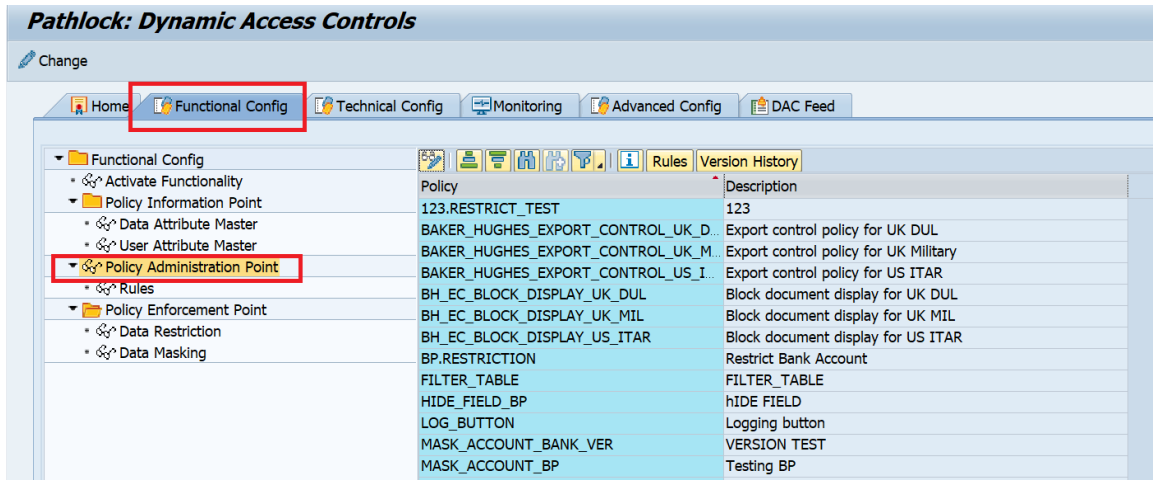


- **User Attribute Master:** Define user attributes for user-specific data, including User ID, PFCG role, User country, User IP, etc.

Note: Use 'USER.' as a prefix to define all user attributes.

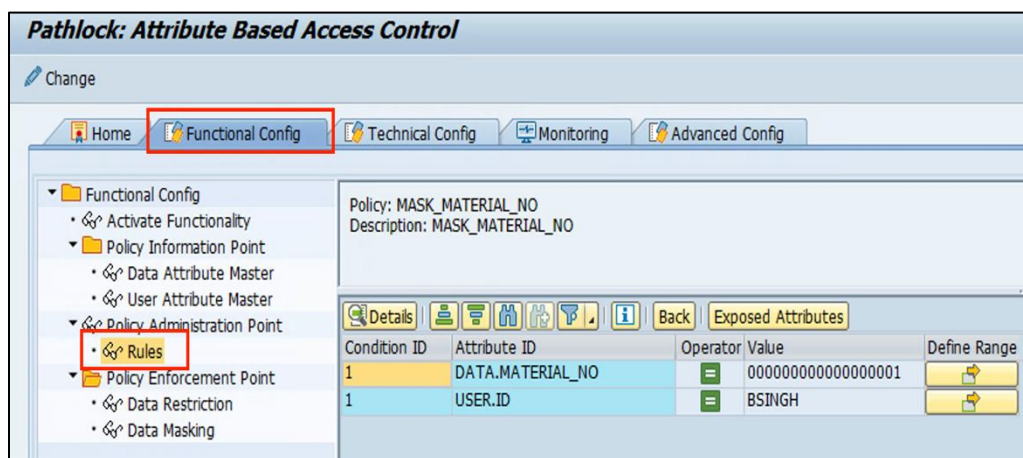


- **Policy Administration Point:** It involves defining a set of rules using one or more conditions that rely on user and Data Attributes within the system. These rule conditions are dynamically evaluated based on the available user and Data Attributes during runtime. Additionally, the DAC policy engine offers a customer-specific implementation that allows the derivation of Data & User Attributes as required. In the Policy Administration process, User and Data Attributes are assigned to the policy, forming the foundation for defining policy rule conditions.



Policy	Description
123.RESTRICT_TEST	123
BAKER_HUGHES_EXPORT_CONTROL_UK_D...	Export control policy for UK DUL
BAKER_HUGHES_EXPORT_CONTROL_UK_M...	Export control policy for UK Military
BAKER_HUGHES_EXPORT_CONTROL_US_I...	Export control policy for US ITAR
BH_EC_BLOCK_DISPLAY_UK_DUL	Block document display for UK DUL
BH_EC_BLOCK_DISPLAY_UK_MIL	Block document display for UK MIL
BH_EC_BLOCK_DISPLAY_US_ITAR	Block document display for US ITAR
BP.RESTRICTION	Restrict Bank Account
FILTER_TABLE	FILTER_TABLE
HIDE_FIELD_BP	hIDE FIELD
LOG_BUTTON	Logging button
MASK_ACCOUNT_BANK_VER	VERSION TEST
MASK_ACCOUNT_BP	Testing BP

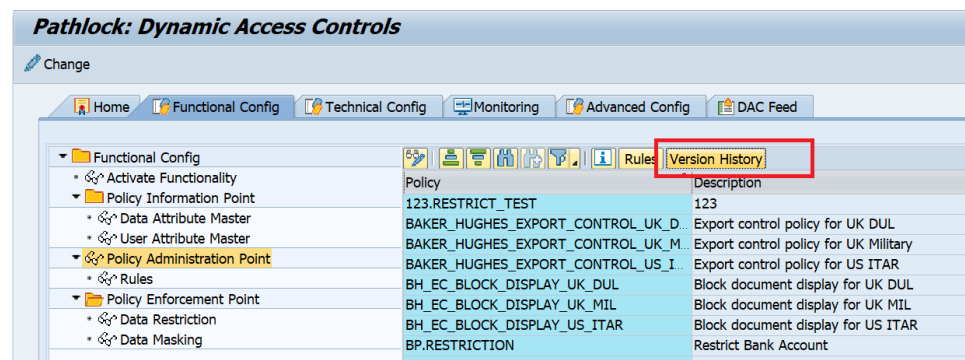
- **Rules:** Set of rules evaluated based on available User and Data Attributes at runtime.



Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.MATERIAL_NO	=	000000000000000001	
1	USER.ID	=	BSINGH	

○ Version History

The users can now see all the versions that are created in policy configurations. So if any policy or its related functions like data classification or list has been created, modified or deleted, it can then be viewed in the version history using the version history button shown below:



Policy	Description
123.RESTRICT_TEST	123
BAKER_HUGHES_EXPORT_CONTROL_UK_D...	Export control policy for UK DUL
BAKER_HUGHES_EXPORT_CONTROL_UK_M...	Export control policy for UK Military
BAKER_HUGHES_EXPORT_CONTROL_US_I...	Export control policy for US ITAR
BH_EC_BLOCK_DISPLAY_UK_DUL	Block document display for UK DUL
BH_EC_BLOCK_DISPLAY_UK_MIL	Block document display for UK MIL
BH_EC_BLOCK_DISPLAY_US_ITAR	Block document display for US ITAR
BP.RESTRICTION	Restrict Bank Account
FILTER_TABLE	FILTER_TABLE

Alternatively, the version can also be viewed using transaction /APPSDM/POLVH.

ECD(1)/100 Pathlock DAC: Policy Version History

Policy: BP.RESTRICTION

Version	Description	Created By	Created On	Changed By	Changed On
00018	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.020.257
00017	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.020.024
00016	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.015.707
00015	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.015.707
00014	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.013.246
00013	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.011.435
00012	Restrict Bank Account	NKHATRI	20.250.528.005.657	MVASHIST	20.250.616.011.044
00011	Restrict Bank Account	NKHATRI	20.240.319.103.143	NKHATRI	20.250.527.004.508
00010	Restrict Bank Account	NKHATRI	20.240.319.103.143	NKHATRI	20.250.525.101.537
00009	Restrict Bank Account	NKHATRI	20.240.319.103.143	NKHATRI	20.250.525.101.537

- **Exposed Attribute:** The "Exposed Attributes" option facilitates linking screen attributes, enabling the derivation of the actual Data Attribute (used in policy rule conditions) through the implementation of the Data Attribute BADI# /APPSDM/DATA_ATTRIBUTES.

For Example - Masking 'Material Description' based on 'Material Type' in t-code# MM03. As 'Material Type' is not available on same screen as 'Material Description', the additional attribute 'Material Number' (available on-screen) can be used to derive 'Material Type' by implementing Data Attribute BADI# /APPSDM/DATA_ATTRIBUTES. This 'Material Number' needs to be exposed to the Policy via the button - "Exposed Attributes".

Pathlock: Attribute Based Access Control

Change

Home **Functional Config** Technical Config Monitoring Advanced Config

Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules**
- Policy Enforcement Point
 - Data Restriction
 - Data Masking

Policy: MASK_MATERIAL_NO
Description: MASK_MATERIAL_NO

Details Back **Exposed Attributes**

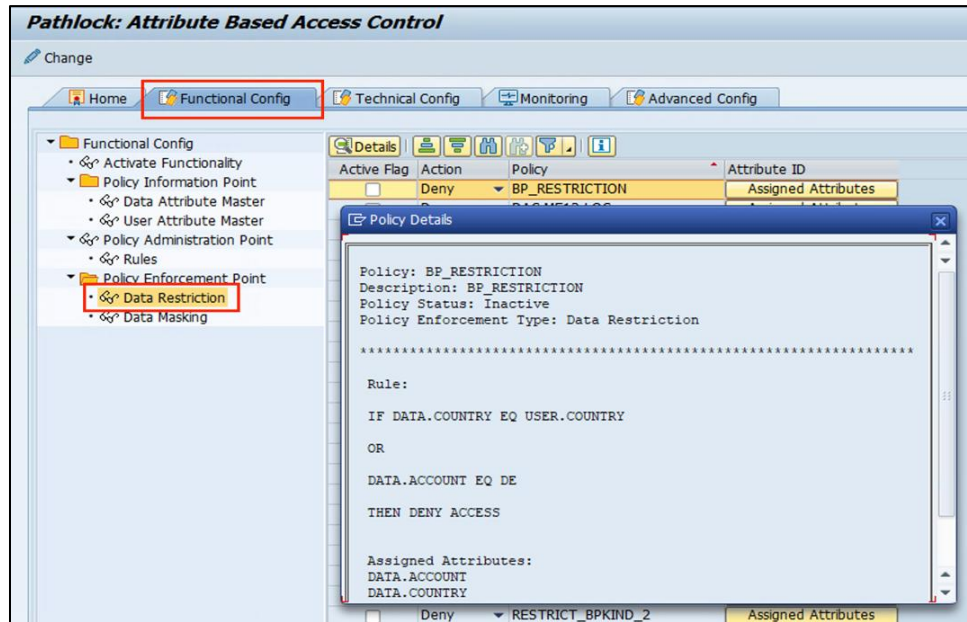
Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.MATERIAL_NO	=	00000000000000000001	
1	USER.ID	=	BSINGH	

Exposed Attributes

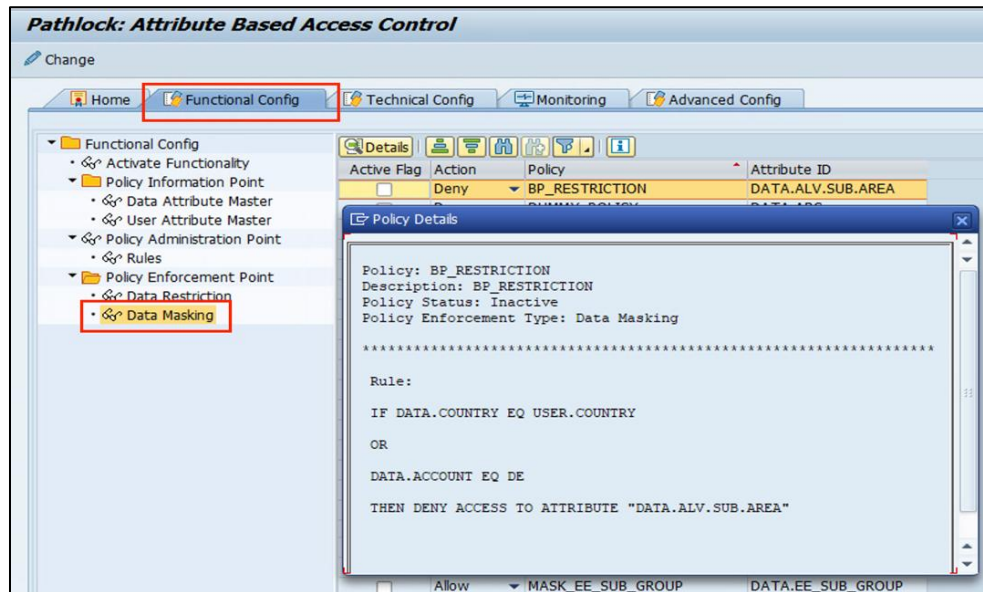
Policy: MASK_MATERIAL_NO
Description: MASK_MATERIAL_NO

Attribute ID	Description
DATA.COND_VALUE	Sales item condition value

- **Policy Enforcement Point:** Assignment of Policy to the Attribute on which actions will be performed.
 - **Data Restriction:** Assignment of policies to the Attribute on which action (Allow/Deny) will be performed for Conditional Transaction/Data access.



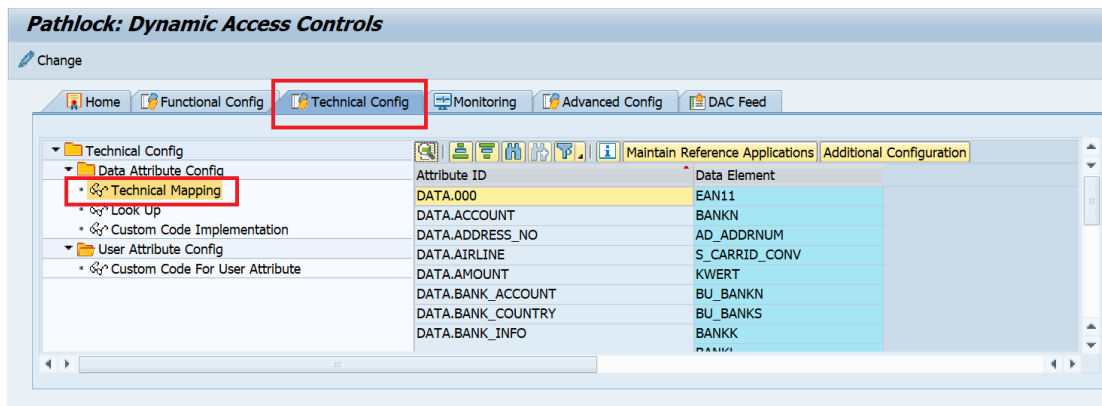
- **Data Masking:** Assignment of policies to the Attribute on which action (Allow/Deny) will be performed for Masking.



2.2 Technical Configuration

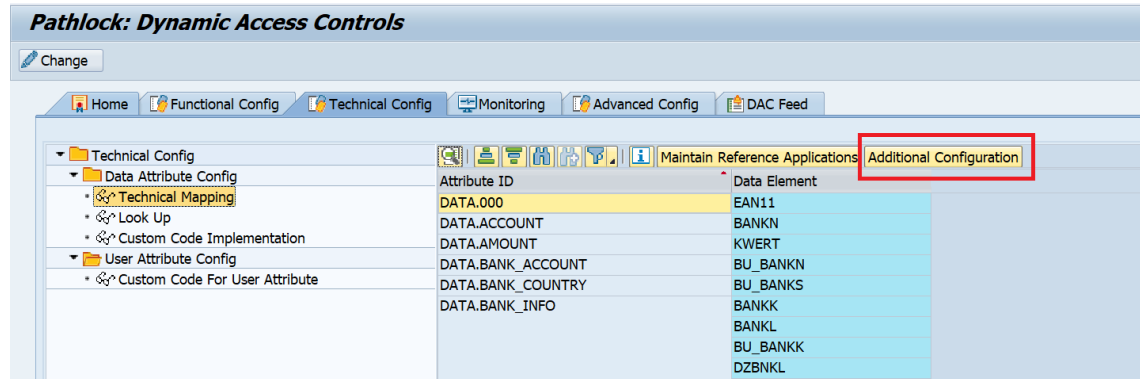
Technical users perform the technical configuration using the following steps:

- **Data Attribute Configuration:**
 - **Technical Mapping:** Assign Data Elements to the Data Attributes using this configuration node.

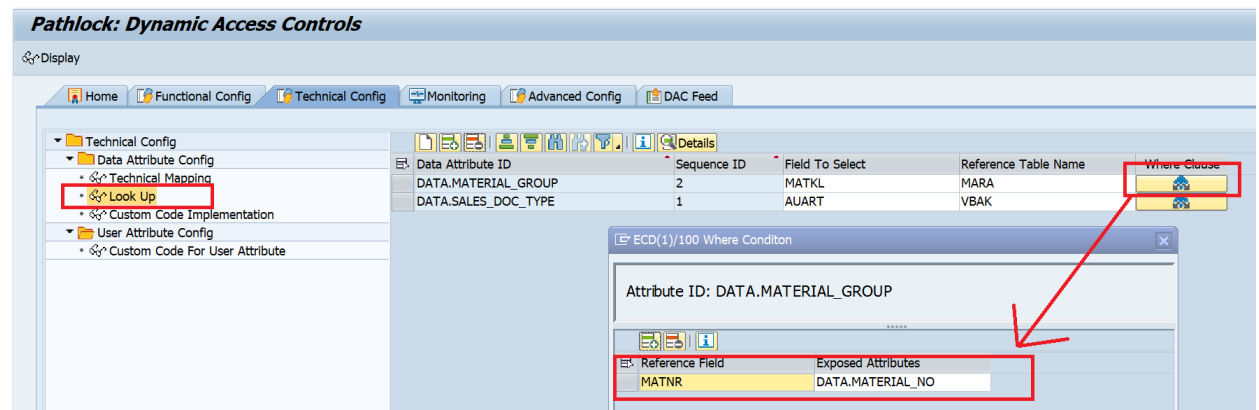


- **Additional Configuration:** Is a functionality that enables system users to specify technical addresses for a field within specific output options. The output options include Module Pool Programs (GUI transactions), ALVs/Tables, Search Helps, and UI5/Fiori scenarios.

Notes: 'Additional Configuration' becomes necessary when there is no data element available in the screen design.



- **Look Up:** This feature enables the user to specify the context to be derived from database tables based on conditions specified under where condition as shown below. In the below given screenshot, material group is derived from the table MARA based on Material number. Here, the context for data restriction / masking is Material Group which is derived from table MARA using the field Material Number. In scenarios where only a simple SELECT query is needed, users can use Look Up functionality.



- **Custom Code for Data Attribute:** This activity shows the Data Attribute BADI implementations implemented via custom code in the system. Implement the Data Attribute BADI# /APPSDM/DATA_ATTRIBUTES to derive the Context Attributes value in scenarios where complex logic is needed to derive context. In case of simple SELECT queries, the look up functionality can be used instead. If user-maintained configuration for both LOOK UP and BADIs, then the logic in BADI takes precedence.

Pathlock: Dynamic Access Controls

Change

Home

Functional Config

Technical Config

Monitoring

Advanced Config

DAC Feed

Technical Config

Data Attribute Config

Technical Mapping

Look Up

Custom Code Implementation

User Attribute Config

Custom Code For User Attribute

Attribute ID

Implementation Name

DATA.BP_TYPEX

ZBADI_BP_TYPE

DATA.COMPUTER_NAME

ZBADI_GET_COMPUTER

DATA.MATERIAL_TYPE

ZBADI_GET_MATERIAL_TYPE

DATA.MB.MATGRP

ZBADI_GET_MB_MATGRP

DATA.PLANT

ZBADI_GET_PLANT

DATA.UUID

ZBADI_GET_UUID

DATA.MAINTAINED_USER

ZBADI_USER_DETAILS

DATA.NON_REGISTERED_DEVICE

ZBADI_USER_DETAILS1

DATA.BUTTON

ZBI_DATA_BUTTON

DATA.TCODE

ZBI_DATA_TCODE

DATA.MATERIAL GROUP 1

ZENH BADI MATERIAL GROUP

▪ User Attribute Configuration:

- **Custom Code for User Attribute:** This activity shows the User Attribute BADI implementations implemented via custom code in the system. Implement the User Attribute BADI# /APPSDM/USER_ATTRIBUTES to derive the User Attributes value.

Pathlock: Dynamic Access Controls

Change

Home

Functional Config

Technical Config

Monitoring

Advanced Config

DAC Feed

Technical Config

Data Attribute Config

Technical Mapping

Look Up

Custom Code Implementation

User Attribute Config

Custom Code For User Attribute

Attribute ID

Implementation Name

USER.AUTHORISED_DUL

/APPSDM/BADI_BH_UK_DUL

USER.MATERIAL

ZBADI_USER_ATTRIBUTES

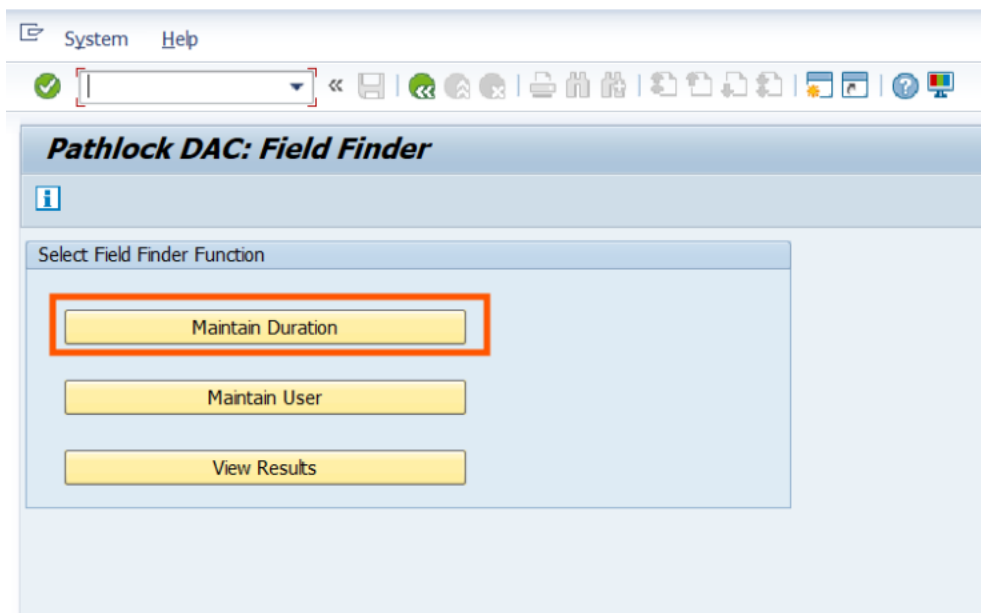
3 Other Functionalities in DAC

3.1 Field Finder

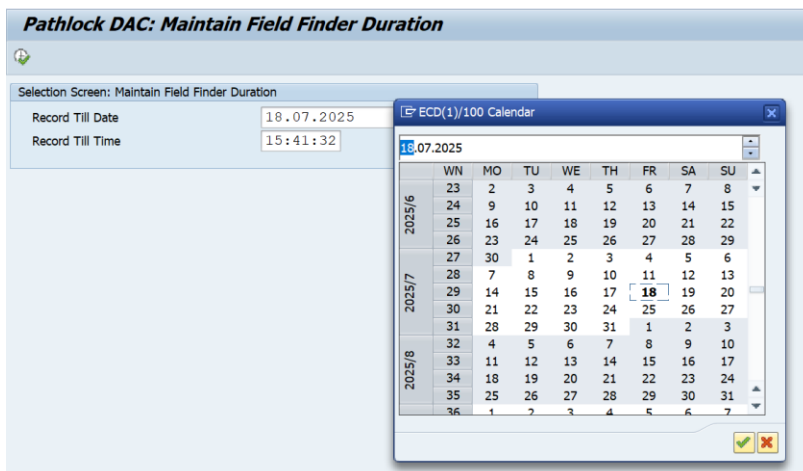
Transaction Code to run Field Finder is /APPSPDM/FF. To achieve Data Masking, users will need to find out the technical address of those fields that need to be masked on the screen to use it in a policy. Field finder is a tool that helps in the above process by recording technical details about fields. It is basically a recording tool.

To start recording the technical address of each field in a given transaction, execute the following steps:

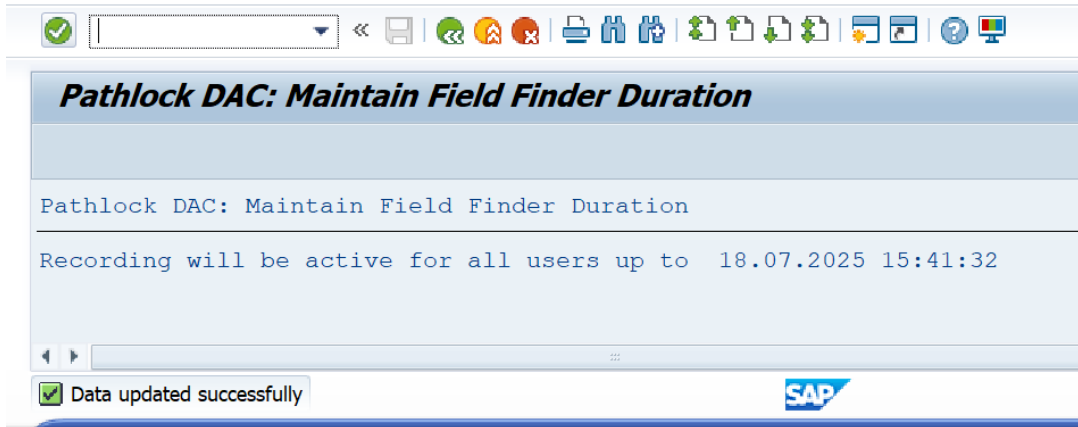
1. Run the field finder application (/N/APPSPDM/FF)
2. Set the duration as shown below. Click on Maintain Duration Button.



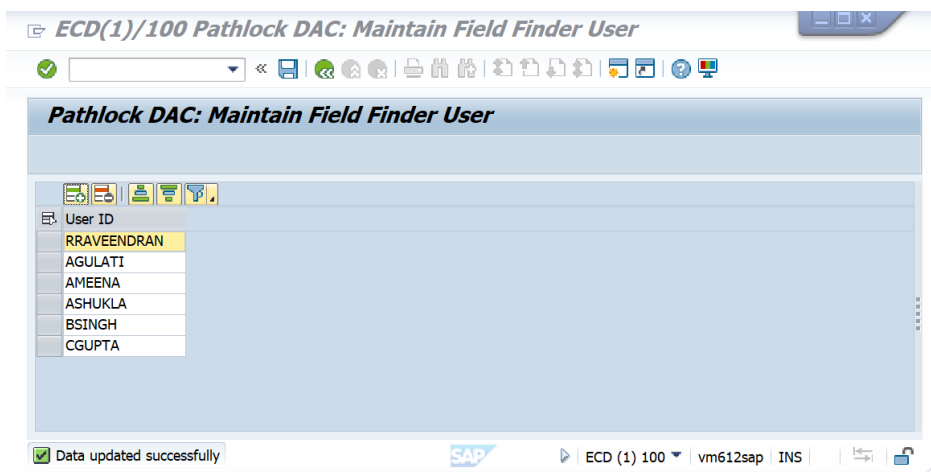
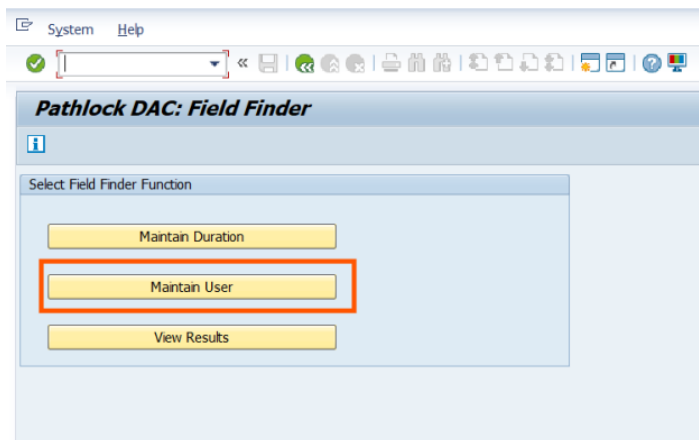
Click on F4 help provided in Record till Date and Time Field. Select the Date and Time from the pop-up.



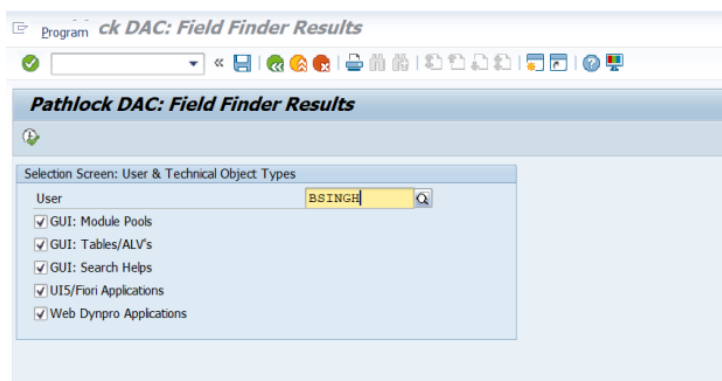
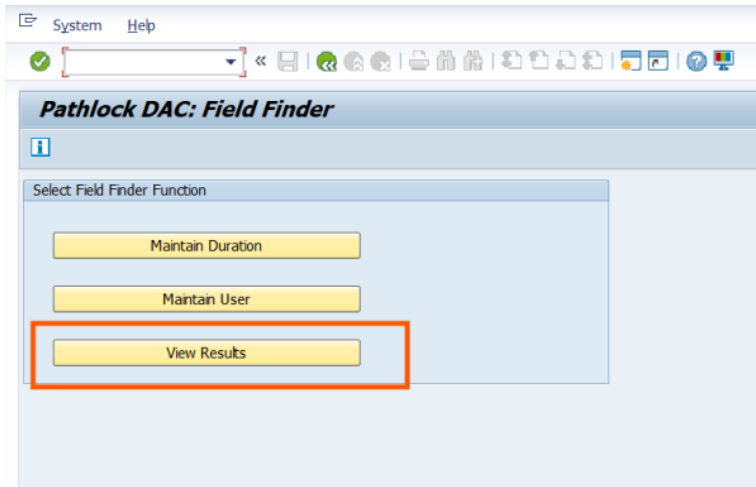
After selecting the date and time from pop-up click on execute button.



- After setting the duration, add the users for whom the recording will start. See the screenshots below.



4. After maintaining users, execute the transaction for which you want to do the recording.
5. You can view the recording results in 'View Results' button in field finder.



Below Screenshot shows the recording of Transaction BP.



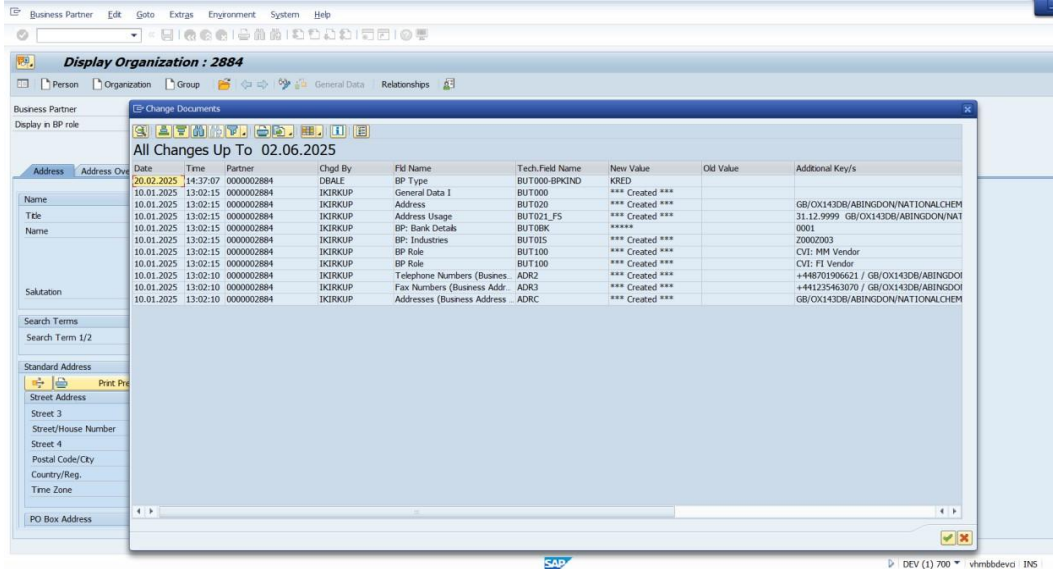
The screenshot shows the 'Pathlock DAC: Field Finder Results' window with a table of recording results. The table has columns: 'Time Stamp', 'Field Name', 'Data Element', 'Description', 'Value', 'Mapped Via', and 'Data Attribute'. The table contains 15 rows of data.

Time Stamp	Field Name	Data Element	Description	Value	Mapped Via	Data Attribute
29.05.2025 13:03:49	UDATE	CDATUM	Creation Date of Change Document	20250220		
29.05.2025 13:03:49	UTIME	CDZERT	Time of Change	143707		
29.05.2025 13:03:49	OBJECTID	COBJECTV	Object Value	0000002884		
29.05.2025 13:03:49	USERNAME	CDUSERNAM	User Name of Person Making Change in Cha.	DBALE		
29.05.2025 13:03:49	FILNAME	CHAR50	Comment	BP Type		
29.05.2025 13:03:49	TFLDNAME	CHAR50	Comment	BUT000-BPKIND	Additional Configuration	DATA.FIELD_NAME
29.05.2025 13:03:49	F_NEW	CDFLDVALN	New Content of Changed Field	KRED	Additional Configuration	DATA.FIELD_VALUE
29.05.2025 13:03:49	F_OLD	CDFLDVALD	Old Content of Changed Field		Additional Configuration	DATA.FIELD_VALUE
29.05.2025 13:03:49	KEYVAL	BU_KEYVALLOW	Central BP: Formatted table key string with L.			
29.05.2025 13:03:22	PARTNER	BU_PARTNER	Business Partner Number	0000002884		
29.05.2025 13:03:22	DESCRIPTION	BU_DESCRIP	Short name of business partner	AEA TECHNOLOGY S.		
29.05.2025 13:03:22	MCNAME1	BU_MCNAME1_GK	Search Help Field 1 (Name 1/Last Name, Ca.	AEA TECHNOLOGY S.		
29.05.2025 13:03:22	MCNAME2	BU_MCNAME2_GK	Search Help Field 2 (Name 2/First Name, Ca.			

Use Case	Capture technical details of the fields in “Change History” screen of the transaction “BP”
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Enable recording as described above.

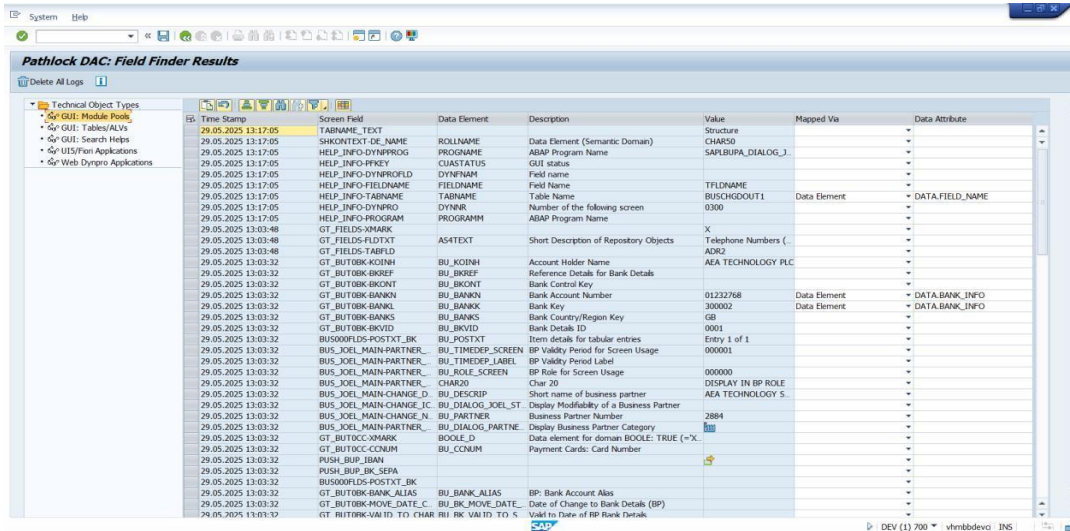
1. Execute the transaction BP and navigate to all those screens for which recording needs to be done. See the screenshot below.



The screenshot shows the SAP Business Partner Change Documents screen for organization 2884. The table displays changes up to 02.06.2025. The columns include Date, Time, Partner, Chgd By, Fld Name, Tech. Field Name, New Value, Old Value, and Additional Key/s.

Date	Time	Partner	Chgd By	Fld Name	Tech. Field Name	New Value	Old Value	Additional Key/s
20.02.2025	14:37:07	0000002884	DBALE	BP Type	BUT000-BPKIND	KRED		
10.01.2025	13:02:15	0000002884	IKIRKUP	General Data 1	BUT100	*** Created ***		
10.01.2025	13:02:15	0000002884	IKIRKUP	Address	BUT120	*** Created ***		
10.01.2025	13:02:15	0000002884	IKIRKUP	Address Usage	BUT021_F5	*** Created ***		GB/OK1430B/ABINGDON/NATIONALCHEM
10.01.2025	13:02:15	0000002884	IKIRKUP	BP: Bank Details	BUT08K	*****		31.12.9999 GB/OK1430B/ABINGDON/NAT
10.01.2025	13:02:15	0000002884	IKIRKUP	BP: Industries	BUT105	*** Created ***		0001
10.01.2025	13:02:15	0000002884	IKIRKUP	BP Role	BUT100	*** Created ***		200002003
10.01.2025	13:02:15	0000002884	IKIRKUP	BP Role	BUT100	*** Created ***		CVI: MM Vendor
10.01.2025	13:02:15	0000002884	IKIRKUP	BP Role	BUT100	*** Created ***		CVI: FI Vendor
10.01.2025	13:02:10	0000002884	IKIRKUP	Telephone Numbers (Business)	ADR3	*** Created ***		+448701906621 / GB/OK1430B/ABINGDOI
10.01.2025	13:02:10	0000002884	IKIRKUP	Fax Numbers (Business Address)	ADR3	*** Created ***		+44125463070 / GB/OK1430B/ABINGDOI
10.01.2025	13:02:10	0000002884	IKIRKUP	Addresses (Business Address)	ADRC	*** Created ***		GB/OK1430B/ABINGDON/NATIONALCHEM

2. Open the field finder using transaction ‘/n/APPSPDM/FF’ and click on ‘View Results’. Enter the username to see the recordings. The results are organized under different nodes based on technical objects



The screenshot shows the Pathlock DAC: Field Finder Results screen. The table displays field recordings organized by technical object types. The columns include Time Stamp, Screen Field, Data Element, Description, Value, Mapped Via, and Data Attribute.

Time Stamp	Screen Field	Data Element	Description	Value	Mapped Via	Data Attribute
29.05.2025 13:17:05	TABNAME_TEXT	ROLLNAME	Data Element (Semantic Domain)	Structure		
29.05.2025 13:17:05	SARORTTEXT_OIE_NAME	CHAR50	CHAR50	SAPLBUPA_DIALOG_1		
29.05.2025 13:17:05	HELP_INFO-DYNPROG	PROGRAMME	ABAP Program Name			
29.05.2025 13:17:05	HELP_INFO-IPKEY	CUASTATUS	GUI status			
29.05.2025 13:17:05	HELP_INFO-DYNPROFLD	DYNPRNAM	Field name			
29.05.2025 13:17:05	HELP_INFO-FIELDNAME	FIELDNAME	Field Name	TFLDNAME		
29.05.2025 13:17:05	HELP_INFO-TABNAME	TABNAME	Table Name	BUSCHGOUT1	Data Element	DATA.FIELD_NAME
29.05.2025 13:17:05	HELP_INFO-DYNPRD	DYNPRR	Number of the following screen	0300		
29.05.2025 13:17:05	HELP_INFO-PROGRAM	PROGRAMM	ABAP Program Name			
29.05.2025 13:03:48	GT_FIELDS-XMARK			X		
29.05.2025 13:03:48	GT_FIELDS-FLOTXT	AS4TEXT	Short Description of Repository Objects	Telephone Numbers (. . .)		
29.05.2025 13:03:48	GT_FIELDS-TAFID			ADR3		
29.05.2025 13:03:32	GT_BUT0BK-KOINH	BU_KOINH	Account Holder Name	AEA TECHNOLOGY PLC		
29.05.2025 13:03:32	GT_BUT0BK-BKREF	BU_BKREF	Reference Details for Bank Details			
29.05.2025 13:03:32	GT_BUT0BK-BKONT	BU_BKONT	Bank Control Key			
29.05.2025 13:03:32	GT_BUT0BK-BANKN	BU_BANKN	Bank Account Number	01232768	Data Element	DATA.BANK_INFO
29.05.2025 13:03:32	GT_BUT0BK-BANKL	BU_BANKK	Bank Key	300002	Data Element	DATA.BANK_INFO
29.05.2025 13:03:32	GT_BUT0BK-BANKS	BU_BANKS	Bank Country/Region Key	GB		
29.05.2025 13:03:32	GT_BUT0BK-BKVID	BU_BKVID	Bank Details ID	0001		
29.05.2025 13:03:32	BUS000FLDS-POSTXT_IBK	BU_POSTXT	Item details for tabular entries	Entry 1 of 1		
29.05.2025 13:03:32	BUS_JOEL-MAIN-PARTNER...	BU_TIMEDEP_SCREEN	BP Validity Period for Screen Usage	000001		
29.05.2025 13:03:32	BUS_JOEL-MAIN-PARTNER...	BU_ROLE_SCREEN	BP Role for Screen Usage	000000		
29.05.2025 13:03:32	BUS_JOEL-MAIN-CHANGE...	BU_DESCRIP	Short name of business partner	DISP		
29.05.2025 13:03:32	BUS_JOEL-MAIN-CHANGE...	BU_DIALOG_JOEL_ST	Display Modifiability of a Business Partner	2884		
29.05.2025 13:03:32	BUS_JOEL-MAIN-CHANGE...	BU_PARTNER	Business Partner Number			
29.05.2025 13:03:32	BUS_JOEL-MAIN-PARTNER...	BU_DIALOG_PARTNE...	Display Business Partner Category			
29.05.2025 13:03:32	GT_BUT0CC-XMARK	BOOLE_D	Data element for domain BOOLE: TRUE (=X)			
29.05.2025 13:03:32	PUSH_BUP_BK_SEPA	BU_CCONUM	Payment Cards: Card Number			
29.05.2025 13:03:32	PUSH_BUP_BK_SEPA					
29.05.2025 13:03:32	GT_BUS00FLDS-POSTXT_IBK	BU_BANK_ALIAS	BP: Bank Account Alias			
29.05.2025 13:03:32	GT_BUT0BK-BANK_ALIAS	BU_BK_MOVE_DATE...	Date of Change to Bank Details (BP)			
29.05.2025 13:03:32	GT_BUT0BK-MOVE_DATE...		Valid to Date of BP Bank Details			

Pathlock DAC: Field Finder Results

Delete All Logs

Time Stamp	Field Name	Data Element	Description	Value	Mapped Via	Data Attribute
29.05.2025 13:03:49	UDATE	CDATUM	Creation Date of Change Document	20250220		
29.05.2025 13:03:49	UTIME	CDUZEIT	Time of Change	143707		
29.05.2025 13:03:49	OBJECTID	COBJECTV	Object Value	000002884		
29.05.2025 13:03:49	USERNAME	CUSENAME	User Name of Person Making Change in Cha	DBALE		
29.05.2025 13:03:49	FLDNAME	CHAR50	Comment	BP Type		
29.05.2025 13:03:49	TFLDNAME	CHAR50	Comment	BUT000-BPKIND	Additional Configuration	DATA.FIELD_NAME
29.05.2025 13:03:49	F_NEW	CDFLDVALN	New Content of Changed Field	KRED	Additional Configuration	DATA.FIELD_VALUE
29.05.2025 13:03:49	F_OLD	CDFLDVALO	Old Content of Changed Field		Additional Configuration	DATA.FIELD_VALUE
29.05.2025 13:03:49	KEYVAL	BU_KEYVALLOW	Central BP: Formatted table key string with L			
29.05.2025 13:03:22	PARTNER	BU_PARTNER	Business Partner Number	000002884		
29.05.2025 13:03:22	DESCRIPTION	BU_DESCRIP	Short name of business partner	AEA TECHNOLOGY S.		
29.05.2025 13:03:22	MCNAME1	BU_MCNAME1_GK	Search Help Field 1 (Name 1/Last Name, Ca	AEA TECHNOLOGY S.		
29.05.2025 13:03:22	MCNAME2	BU_MCNAME2_GK	Search Help Field 2 (Name 2/First Name, Ca			

- Find the record you want to map and add it here. If it has a data element, use the Data Element from drop down under 'Mapped via' field to map it with Data Attribute. If there's no data element, use 'Additional Configuration' to map with Data Attribute. See the screenshots below.

Pathlock DAC: Field Finder Results

Delete All Logs

Time Stamp	Screen Field	Data Element	Description	Value	Mapped Via	Data Attribute
29.05.2025 13:17:05	TABNAME_TEXT	ROLNAME	Data Element (Semantic Domain)	CHAR50		
29.05.2025 13:17:05	SHKORTTEXT-DE_NAME	PROGNAME	ABAP Program Name	SAPLBUPA_DIALOG_3		
29.05.2025 13:17:05	HELP_INFO-PFKEY	CUASTATUS	GUI status			
29.05.2025 13:17:05	HELP_INFO-DYNPROFLD	DYNFNAM	Field name			
29.05.2025 13:17:05	HELP_INFO-FIELDNAME	FIELDNAME	Field Name	TFLDNAME	Data Element	DATA.FIELD_NAME
29.05.2025 13:17:05	HELP_INFO-TABNAME	TABNAME	Table Name	BUSCHGOUT1		
29.05.2025 13:17:05	HELP_INFO-DYNPRO	DYNNR	Number of the following screen	0300		
29.05.2025 13:17:05	HELP_INFO-PROGRAM	PROGRAMM	ABAP Program Name			
29.05.2025 13:03:48	GT_FIELDS-XMARK			X		
29.05.2025 13:03:48	GT_FIELDS-FLDTXT	AS4TEXT	Short Description of Repository Objects	Telephone Numbers (		
29.05.2025 13:03:48	GT_FIELDS-TABFLD			ADR2		
29.05.2025 13:03:32	GT_BUT0BK-KOINH	BU_KOINH	Account Holder Name	AEA TECHNOLOGY PLC		
29.05.2025 13:03:32	GT_BUT0BK-BKREF	BU_BKREF	Reference Details for Bank Details			
29.05.2025 13:03:32	GT_BUT0BK-BKONT	BU_BKONT	Bank Control Key			
29.05.2025 13:03:32	GT_BUT0BK-BANKN	BU_BANKN	Bank Account Number	01232768	Data Element	DATA.BANK_INFO
29.05.2025 13:03:32	GT_BUT0BK-BANKL	BU_BANKL	Bank Key	300002	Data Element	DATA.BANK_INFO
29.05.2025 13:03:32	GT_BUT0BK-BANKS	BU_BANKS	Bank Country/Region Key	GB		
29.05.2025 13:03:32	GT_BUT0BK-BKVID	BU_BKVID	Bank Details ID	0001		
29.05.2025 13:03:32	BUS000FLDS-POSTXT_BK	BU_POSTXT	Item details for tabular entries	Entry 1 of 1		
29.05.2025 13:03:32	BUS_JOEL_MAIN-PARTNER...	BU_TIMEDEP_SCREEN	BP Validity Period for Screen Usage	000001		
29.05.2025 13:03:32	BUS_JOEL_MAIN-PARTNER...	BU_TIMEDEP_LABEL	BP Validity Period Label			
29.05.2025 13:03:32	BUS_JOEL_MAIN-PARTNER...	BU_ROLE_SCREEN	BP Role for Screen Usage	000000		
29.05.2025 13:03:32	BUS_JOEL_MAIN-PARTNER...	CHAR20	Char 20	DISPLAY IN BP ROLE		
29.05.2025 13:03:32	BUS_JOEL_MAIN-CHANGE_D	BU_DESCRIP	Short name of business partner	AEA TECHNOLOGY S.		
29.05.2025 13:03:32	BUS_JOEL_MAIN-CHANGE_IC	BU_DIALOG_JOEL_ST	Display Modifiability of a Business Partner			
29.05.2025 13:03:32	BUS_JOEL_MAIN-CHANGE_N	BU_PARTNER	Business Partner Number	2884		
29.05.2025 13:03:32	BUS_JOEL_MAIN-PARTNER...	BU_DIALOG_PARTNE	Display Business Partner Category			
29.05.2025 13:03:32	GT_BUT0CC-XMARK	BOOLE_D	Data element for domain BOOLE: TRUE (=X)			
29.05.2025 13:03:32	GT_BUT0CC-COJUM	BU_CONUM	Payment Cards: Card Number			
29.05.2025 13:03:32	PUSH_BUP_IBAN					
29.05.2025 13:03:32	PUSH_BUP_BK_SEPA					
29.05.2025 13:03:32	BUS000FLDS-POSTXT_BK					
29.05.2025 13:03:32	GT_BUT0BK-BANK_ALIAS	BU_BANK_ALIAS	BP: Bank Account Alias			
29.05.2025 13:03:32	GT_BUT0BK-MOVE_DATE_C	BU_BK_MOVE_DATE	Date of Change to Bank Details (BP)			
29.05.2025 13:03:32	GT_BUT0BK-VAID ID	BU_CHR_RU BK VAID ID	Valid to Date of BP Bank Details			

The configuration that is maintained in Field Finder will reflect automatically in Technical Mapping. (/N/APPSPDM/ABAC > Technical Config > Technical Mapping > Additional configuration). See the screenshot below.

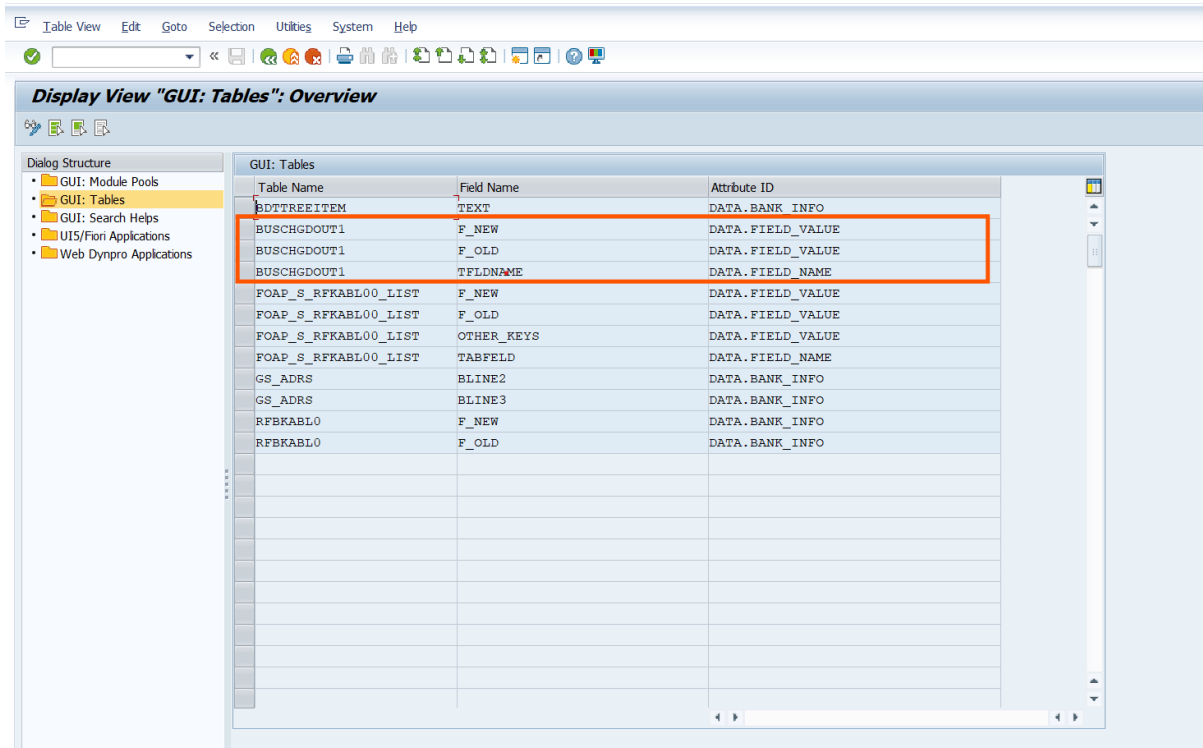


Table Name	Field Name	Attribute ID
BTTTREEITEM	TEXT	DATA.BANK_INFO
BUSCHGDOUT1	F_NEW	DATA.FIELD_VALUE
BUSCHGDOUT1	F_OLD	DATA.FIELD_VALUE
BUSCHGDOUT1	TFLDNAME	DATA.FIELD_NAME
FOAP_S_RFKABLO0_LIST	F_NEW	DATA.FIELD_VALUE
FOAP_S_RFKABLO0_LIST	F_OLD	DATA.FIELD_VALUE
FOAP_S_RFKABLO0_LIST	OTHER_KEYS	DATA.FIELD_VALUE
FOAP_S_RFKABLO0_LIST	TABFELD	DATA.FIELD_NAME
GS_ADRS	BLINE2	DATA.BANK_INFO
GS_ADRS	BLINE3	DATA.BANK_INFO
RFBKABLO	F_NEW	DATA.BANK_INFO
RFBKABLO	F_OLD	DATA.BANK_INFO

3.2 Data Masking / Restriction API

In scenarios where Masking / Data Restriction is not supported by default by DAC, we call an API at an appropriate code, to achieve masking. Before implementing the API, the given prerequisites must be completed:

With the help of an ABAP Developer, Identify the place where the API should be inserted to enable masking.

Before calling the API class, there should be a code to check the RZ11 parameter value with 'ALL'.

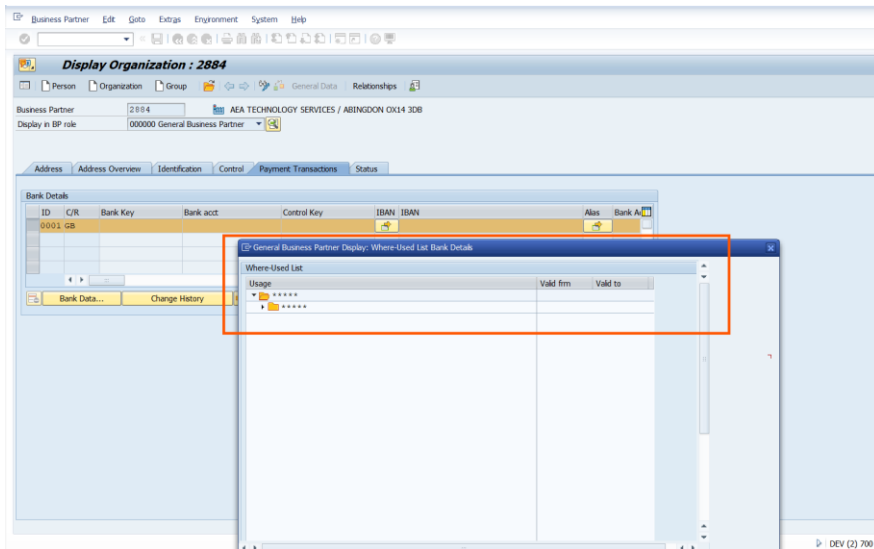
Inserting an API at an appropriate place involves calling the below class-method via an implicit enhancement or modification.

CLASS	/APPSDM/CL_INTERNAL_ABAC_API
METHOD	HANDLE_ABAC_CONTROL

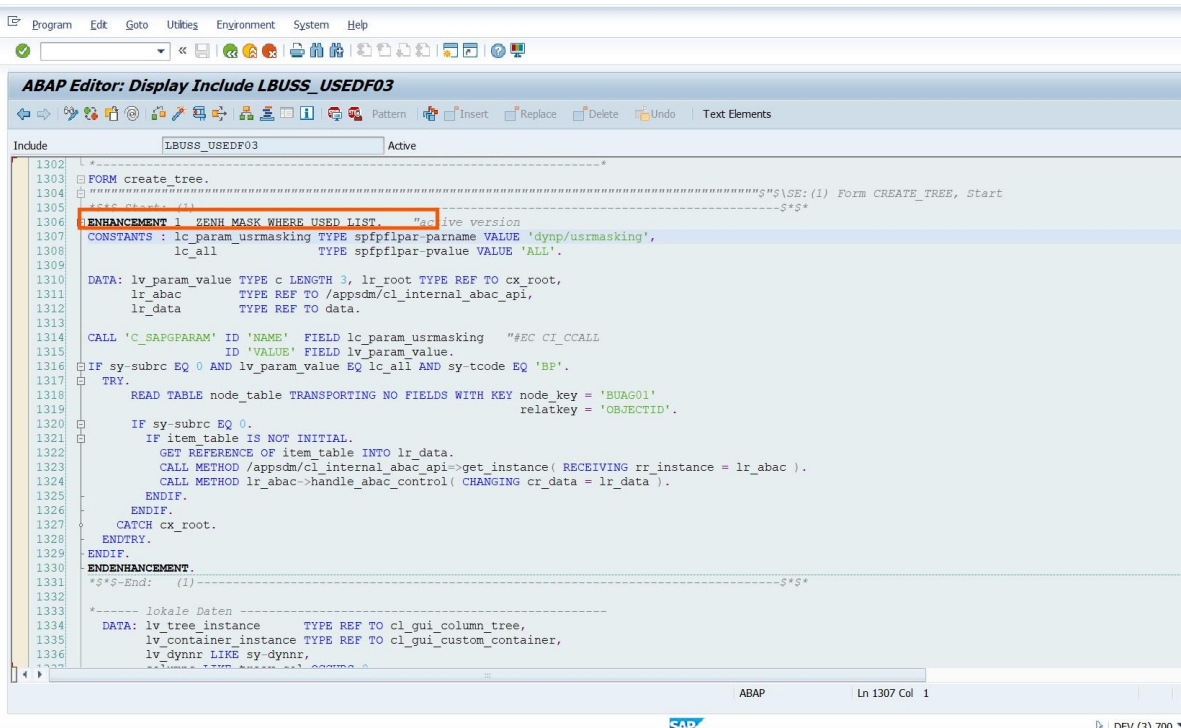
In the method 'HANDLE_ABAC_CONTROL', pass the reference of the internal table or work area or local variable that contains the data to be masked.

Use Case	In the BP T-Code to enable masking in Where-Used list under payment transaction.
----------	--

In the above scenario, the results are displayed as a simple tree where masking is not supported by DAC by default. For such cases, we call an API to achieve masking.



The API is implemented via an implicit enhancement as shown in the below screenshot.

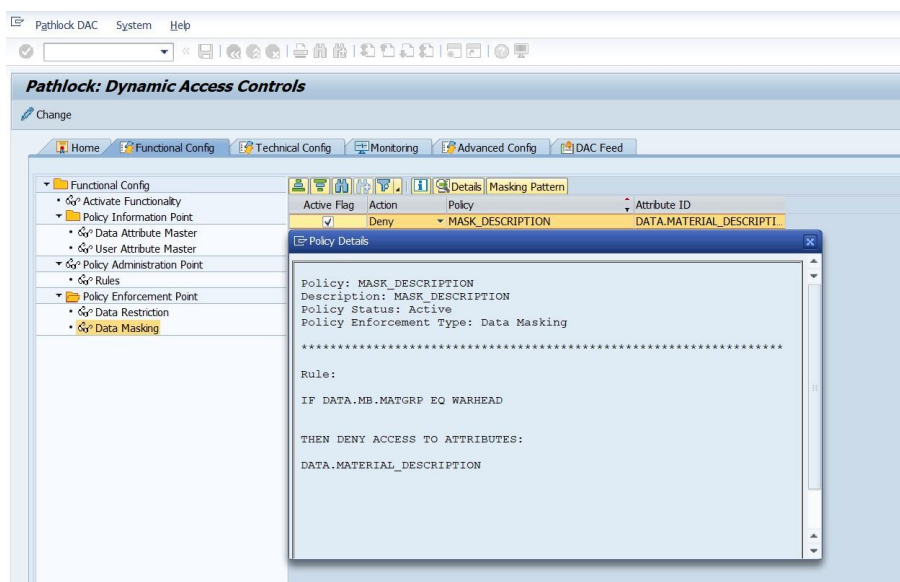


3.3 Data Attributes BADI Implementation

In some scenarios masking / restriction needs to be done based on the context of another field. In some scenarios, this required context may not be present on the screen and will then need to be derived programmatically. BADI is used in such cases to derive the data context.

To understand this better, please refer to the below example:

Use Case 1	Mask the Material Description in table MAKT for those materials whose Material Group is 'WARHEAD'.
------------	--

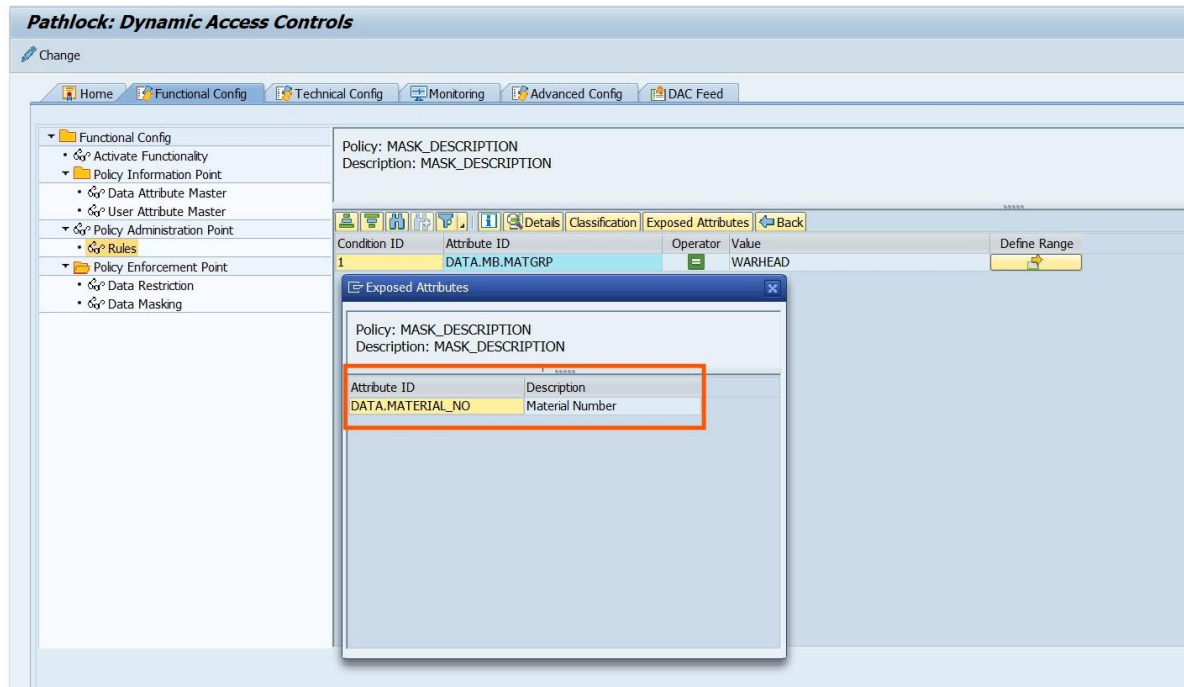


In the above use case, the context for masking material description is Material group. However, the field material group is not present in table MAKT and will then have to be derived programmatically based on material number.

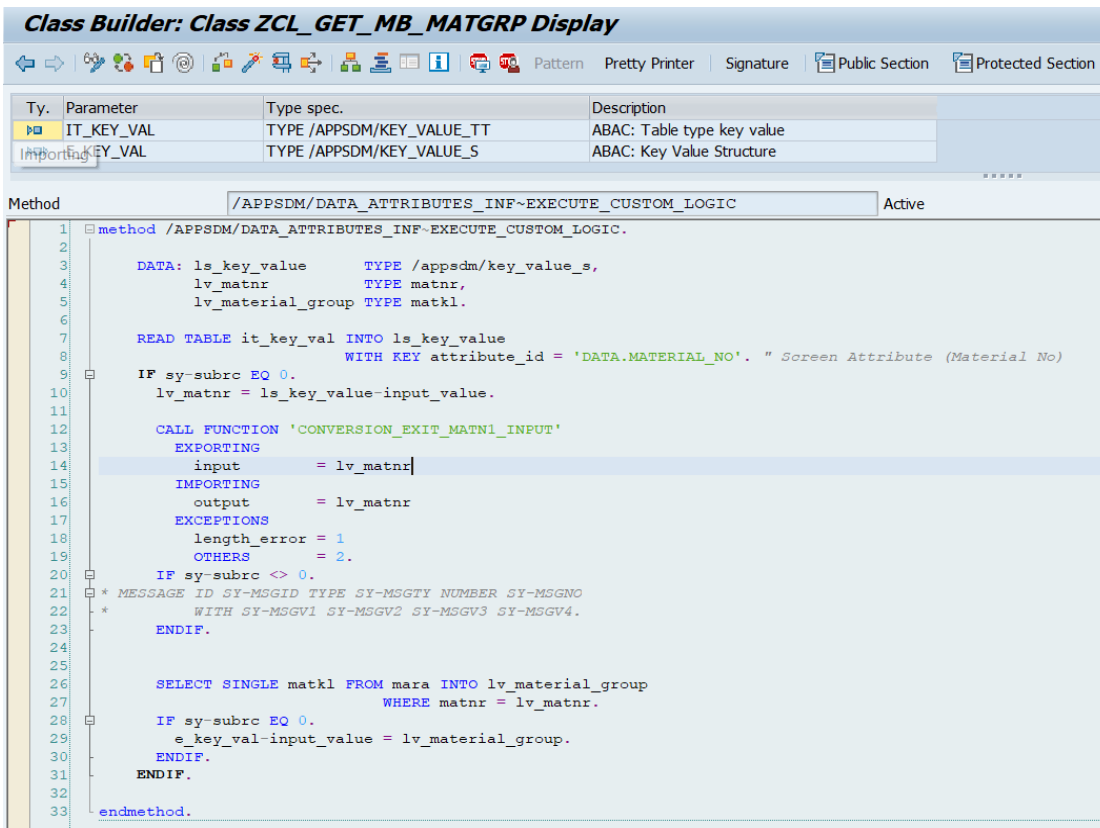
To achieve this, we need to create a custom implementation for the Badi definition

/APPSDM/DATA_ATTRIBUTES. The attribute to be derived should be passed as the Badi filter. In the above example, the attribute for material group will be assigned to the badi filter.

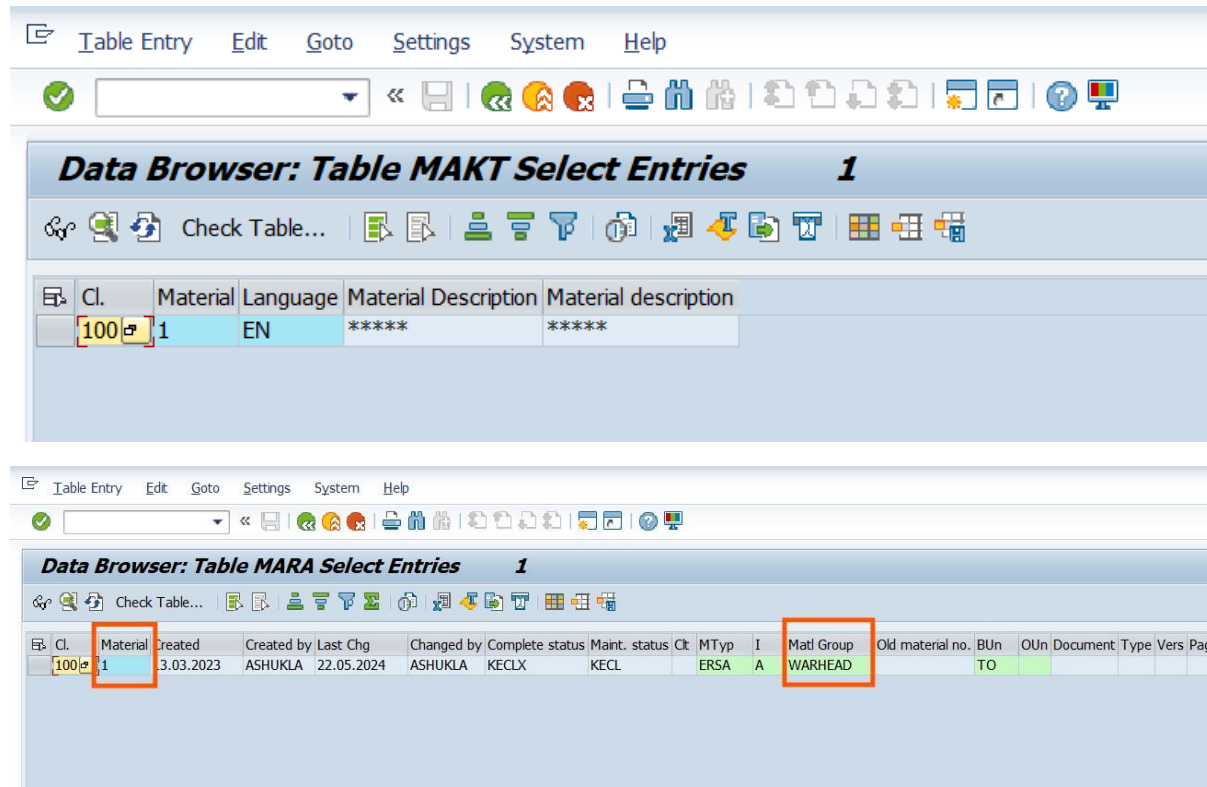
In this custom Badi implementation, the context value for data masking / restriction can be derived via code. The attributes required to derive this context should be maintained under "Exposed Attributes". When the user clicks on exposed attribute button, a screen pops up which is where it should be maintained. The exposed attributes are attributes which are already present on the screen and used to derive the context via code. In the above example, the attribute for Material number is the exposed attribute.



See the screenshot below for reference.



Evidence showing data is masked.



The first screenshot shows the SAP Data Browser interface for Table MAKT. The title bar reads "Data Browser: Table MAKT Select Entries 1". The table has five columns: Cl., Material, Language, Material Description, and Material description. The first row shows the value "100" in the Cl. column, which is highlighted with a red box. The Material column contains the value "1".

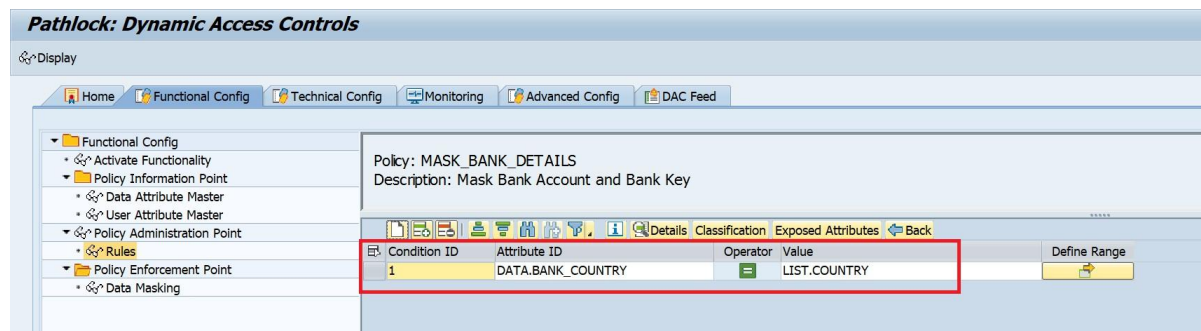
The second screenshot shows the SAP Data Browser interface for Table MARA. The title bar reads "Data Browser: Table MARA Select Entries 1". The table has many columns, including Cl., Material, Created, Created by, Last Chg, Changed by, Complete status, Maint. status, Ct, MTyp, I, Matl Group, Old material no., BUn, OUn, Document, Type, Vers, and Page. The first row shows the value "100" in the Cl. column, which is highlighted with a red box. The Material column contains the value "1". The Matl Group column contains the value "WARHEAD", which is also highlighted with a red box.

3.4 Lists

The transaction for List is /APPSDM/LIST and is used to maintain a set of static values against a List Attribute which can then be used in ABAC policies (in the transaction "/APPSDM/ABAC") and data classification (in the transaction /APPSDM/DC").

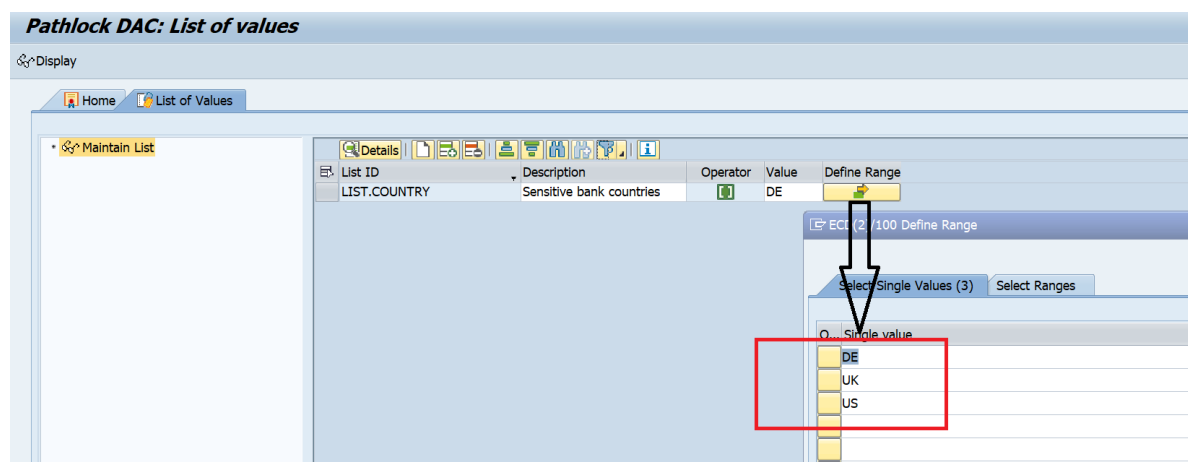
An example of using Lists in DAC policy is as below:

Given a policy MASK_BANK_DETAILS which masks sensitive bank details if the bank country is sensitive.



Here, the bank country of the account that the customer is trying to view is validated against a list of countries to evaluate the given policy.

The lists of countries are maintained in transaction /APPSDM/LIST under the list attribute LIST.COUNTRY as shown below:



Using a list instead of maintaining static values directly in the policy have the following advantages:

Reuse of lists in multiple policies and hence maintenance becomes easier.

Sensitive values are not accessed by users viewing policies. They can be accessed only using the transaction /APPSDM/LIST the access of which can be given to users who can see sensitive values thereby protecting sensitive data.

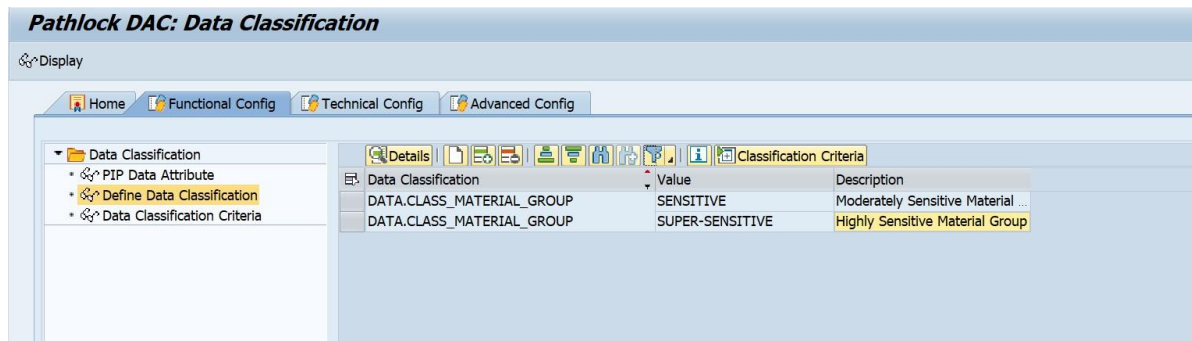
How to create a list:

As mentioned above, the transaction to create / edit / display a list is /APPSDM/LIST. A List attribute should be created here, and a set of static values should be maintained against it. Click on the “Define Range” button to maintain multiple values for a given list. A List attribute name should always start with ‘LIST.’

3.5 Data Classification

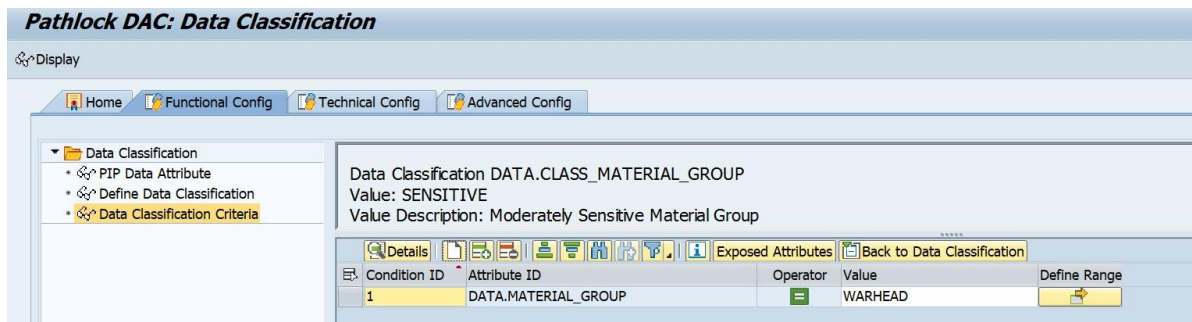
Data Classification is used to classify data based on a set of rules (Data Classification Criteria) which are defined using one or more conditions of data attributes. This classification value of data can then be used in an ABAC policy. The transaction code used to access Data classification is /N/APPSDM/DC

An example of Data classification is as shown below:

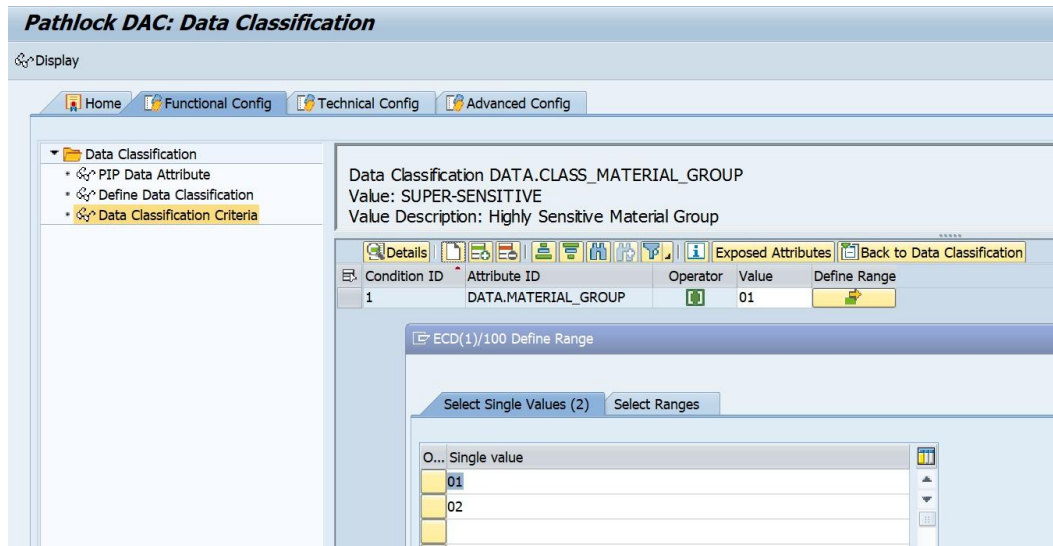


Here, two data classifications are defined for Sensitive and Non-Sensitive materials. And these material classification values are determined based on material groups, used in data classification criteria.

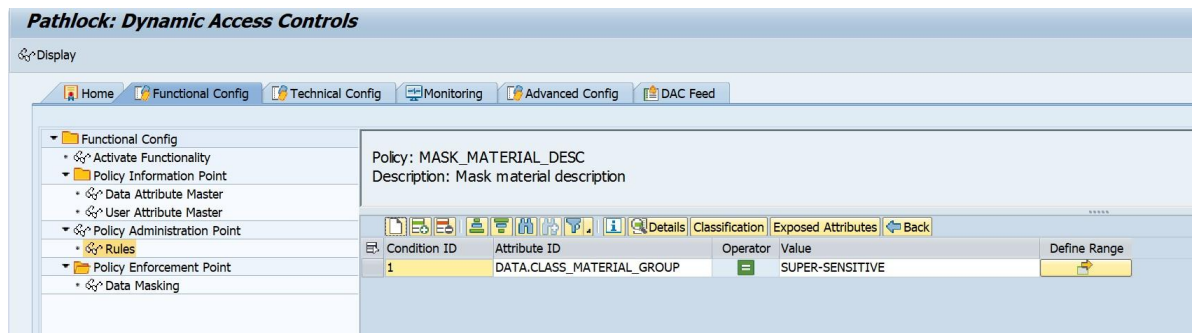
In the below screenshot, the materials belonging to groups WARHEAD are classified as "SENSITIVE"



In the below screenshot, the materials belonging to groups 01 and 02 are classified as “SUPER- SENSITIVE”



These classifications are then used in the policies as shown below:



In the above shown policy ‘MASK_MATERIAL_DESC’, the class of material group decides if the

Material description should be masked or not. The advantage of using data classification is:

Reduce complexity of policies – The policy looks neat as the classification of data happens in the separate transaction for data classification /APPDM/DC. So, creation of multiple policy rule conditions inside the policy can be avoided.

How to create a data classification:

As mentioned above, the transaction to create / edit and display a list is /APPSDM/DC. A Data Classification attribute should be created here (Consider the attribute ‘DATA.CLASS_MATERIAL_GROUP’ given in the above example) and its value should be assigned (Consider the value SUPER-SENSITIVE in the above example).

The name of a classified attribute must start with ‘DATA.CLASS_’

Click on the “Classification Criteria” button to define rules for classification. In the above

example, the rule is

-1-

IF DATA.MATERIAL_GROUP EQ 01, 02

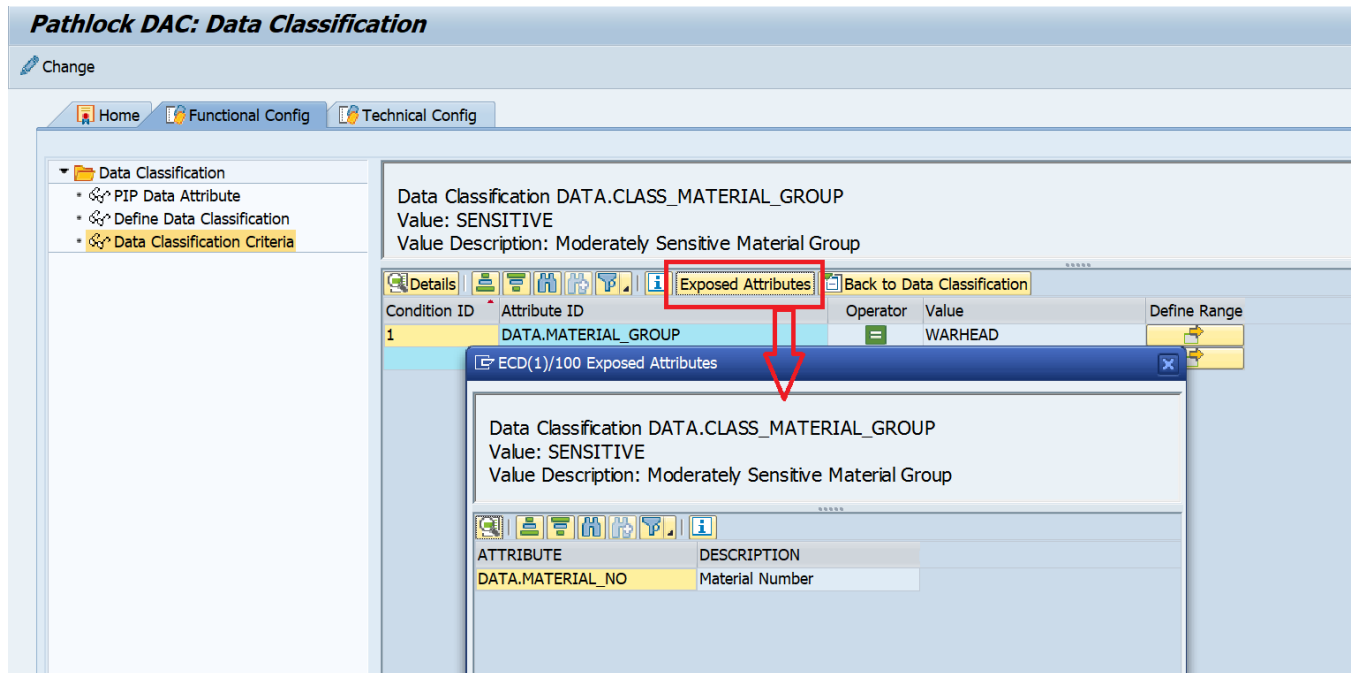
THEN DATA.CLASS_MATERIAL_GROUP EQ SUPER-SENSITIVE

- 2 -

IF DATA.MATERIAL_GROUP EQ WARHEAD

THEN DATA.CLASS_MATERIAL_GROUP EQ SENSITIVE

If the fields used in Data Classification is not present on the screen of a transaction, then they need to be derived via a BADI or LOOK UP. Hence that field needs to be added in the exposed attribute of data classification screen:



Pathlock DAC: Data Classification

Change

Home Functional Config Technical Config

Data Classification

- PIP Data Attribute
- Define Data Classification
- Data Classification Criteria

Data Classification DATA.CLASS_MATERIAL_GROUP
Value: SENSITIVE
Value Description: Moderately Sensitive Material Group

Details Exposed Attributes Back to Data Classification

Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.MATERIAL_GROUP	=	WARHEAD	

ECD(1)/100 Exposed Attributes

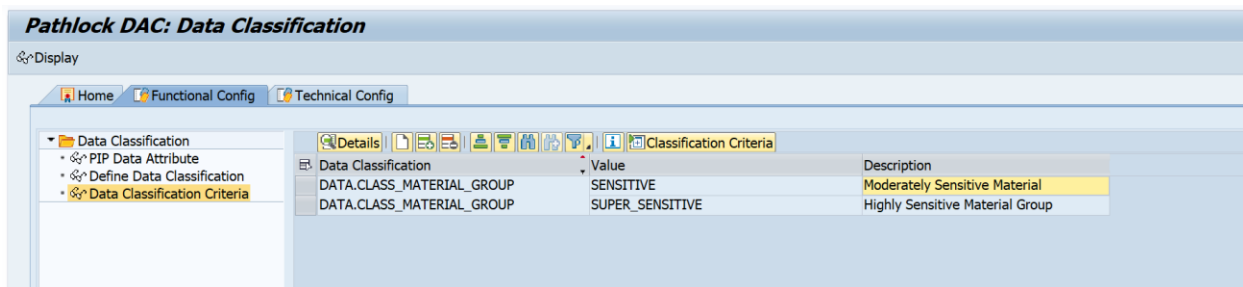
Data Classification DATA.CLASS_MATERIAL_GROUP
Value: SENSITIVE
Value Description: Moderately Sensitive Material Group

ATTRIBUTE	DESCRIPTION
DATA.MATERIAL_NO	Material Number

3.6 Data Classification Simulation

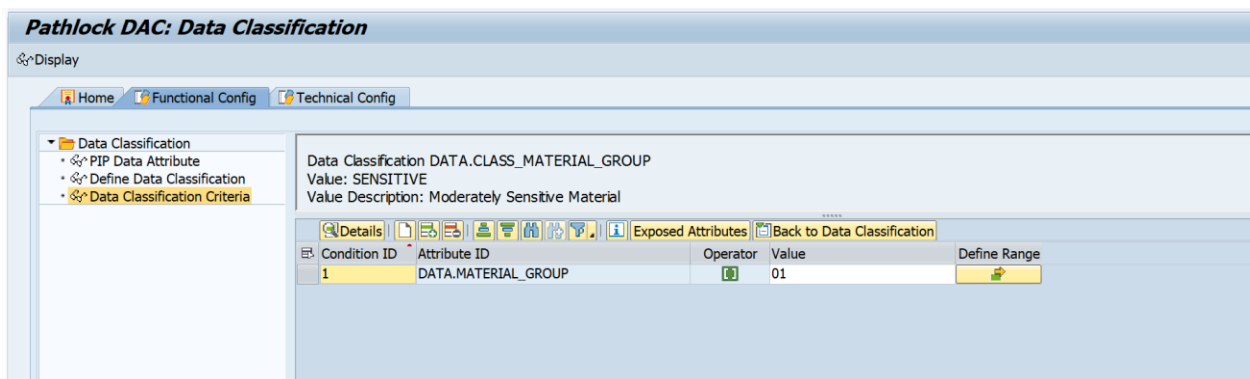
Data Classification Simulation works on table data. The Simulator will evaluate the conditions maintained in the Data Classification Criteria in transaction /APPSDM/DC. Data Classification simulation can be executed via transaction code /APPSDM/DC_SIMU.

An example of Data Classification Simulation is as shown below:

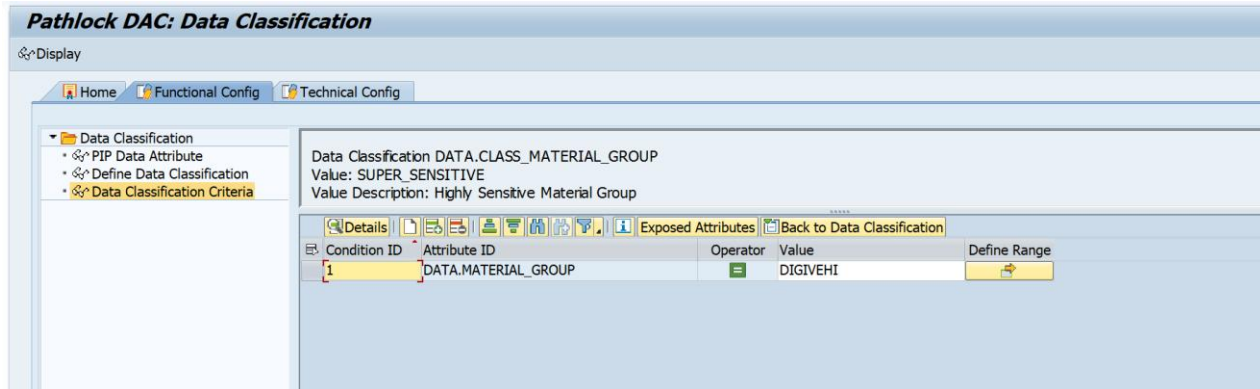


Here, materials are classified into 2 classes – Sensitive and Non-Sensitive. A set of rules shown below are defined to classify the material data.

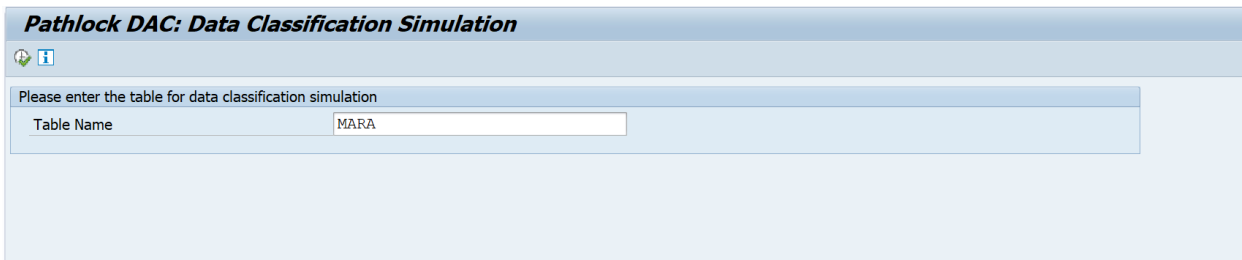
In the below screenshot, the materials with material groups 01 and 02 are classified as “SENSITIVE”



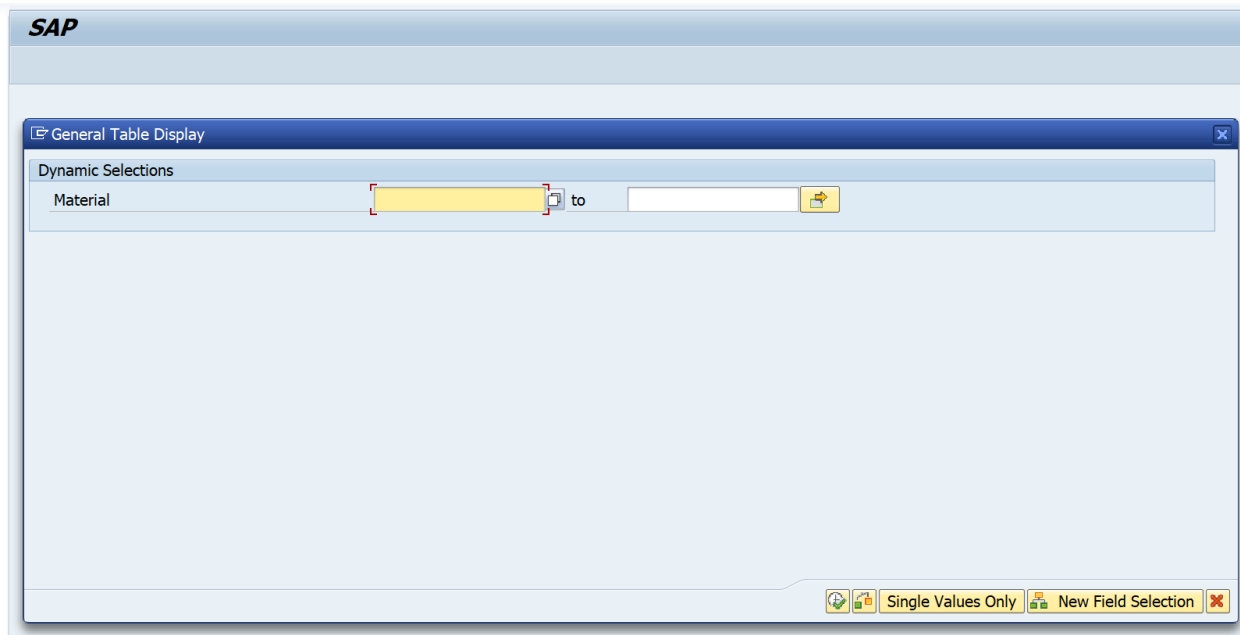
In the below screenshot, the material group DIGIVEHI is classified as “SUPER-SENSITIVE”



Now, to execute the data classification simulation on material data stored in mara table, execute data classification simulation transaction – /N/APPSPDM/DC_SIMU by entering table name i.e. MARA for this case.



On pressing the execute button, a selection screen opens up where the user can use selection criteria to filter results and then view the classified materials as per data classification configuration in /N/APPSPDM/DC.



Result: The output of the table display will carry out a simulation of data classification for each row of the table and will show the result of simulation with all data classifications and their values against each record. Only the primary key fields and the fields involved in data classification criteria will be displayed along with the simulation results.

Pathlock DAC: Table Display

Table Name: MARA
Number of Entries: 7

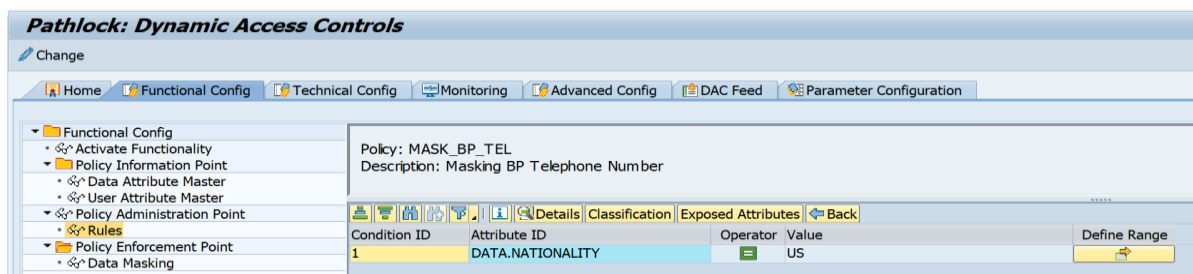
Material	MTyp	Matl Group	Data Classification_1	Value_1
21	FERT	01	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
22	FERT	01	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
23	FERT	01	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
24	FERT	01	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
25	FERT	01	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
30	HIBE	02	DATA.CLASS_MATERIAL_GROUP	SENSITIVE
DVAUTO01	VEHI	DIGIVEHI	DATA.CLASS_MATERIAL_GROUP	SUPER_SENSITIVE

3.7 Policy Simulator

The Policy Simulator enables policy administrators to simulate policy conditions for a specific SAP transaction and the currently logged-in user. This allows validation and testing of complex policies in a non-productive (test or quality) environment before deployment to production. The Policy Simulator can be accessed using transaction code - /APPSDM/POL_SIMU

Important Behavior: If multiple policies are active in the system, only the policy explicitly selected and provided to the Policy Simulator is evaluated during simulation. All other active policies are ignored, even if they are enabled and applicable for the same transaction or user. This ensures isolated and accurate testing of individual policies without interference from other active configurations.

An example of Policy Simulator is shown below with two different policies:



Pathlock: Dynamic Access Controls

Change

Home Functional Config Technical Config Monitoring Advanced Config DAC Feed Parameter Configuration

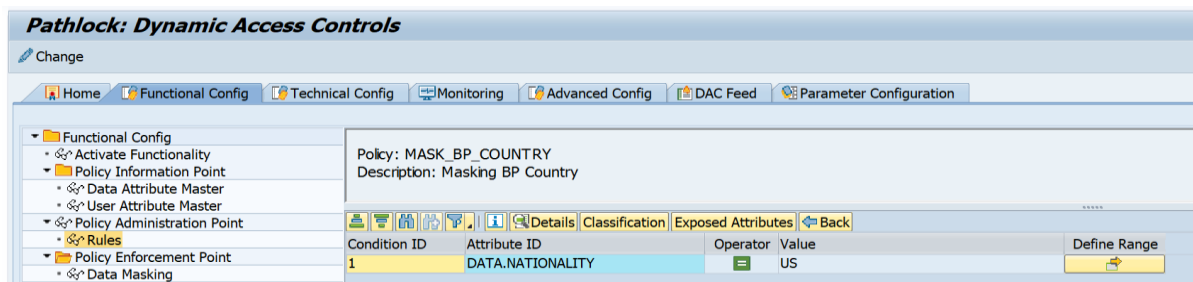
Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Masking

Policy: MASK_BP_TEL
Description: Masking BP Telephone Number

Details Classification Exposed Attributes Back

Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.NATIONALITY	=	US	



Pathlock: Dynamic Access Controls

Change

Home Functional Config Technical Config Monitoring Advanced Config DAC Feed Parameter Configuration

Functional Config

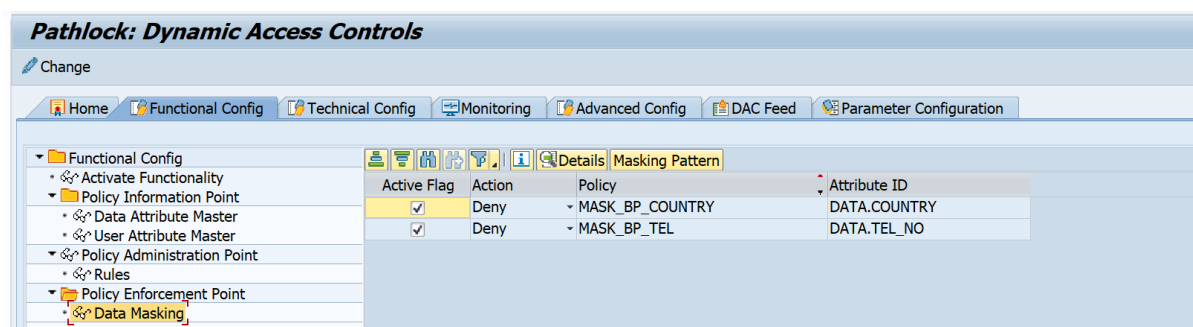
- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Masking

Policy: MASK_BP_COUNTRY
Description: Masking BP Country

Details Classification Exposed Attributes Back

Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.NATIONALITY	=	US	

Here we have two policies with the same rules. Policy Enforcement is active for both the policies under Data Masking node as shown below:



Pathlock: Dynamic Access Controls

Change

Home Functional Config Technical Config Monitoring Advanced Config DAC Feed Parameter Configuration

Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Masking

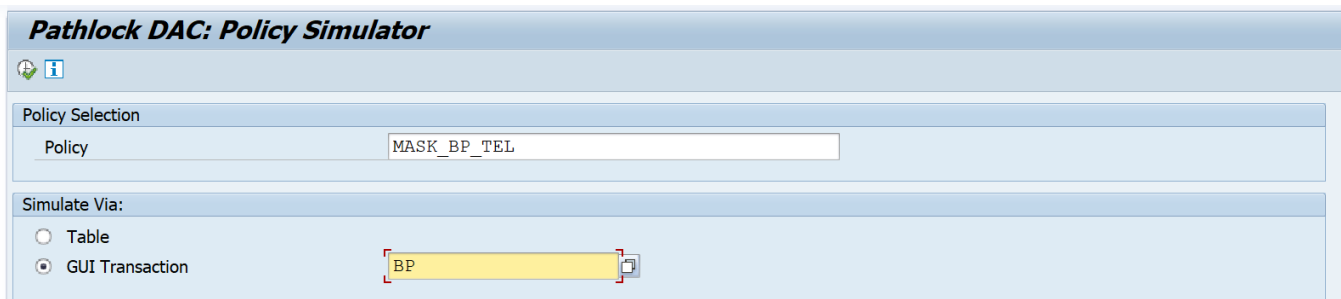
Details Masking Pattern

Active Flag	Action	Policy	Attribute ID
☒	Deny	MASK_BP_COUNTRY	DATA.COUNTRY
☒	Deny	MASK_BP_TEL	DATA.TEL_NO

Example Behavior:

If the Policy Simulator is executed for the policy MASK_BP_TEL, the masking effect will be applied only to the Business Partner telephone number. Even if another policy (for example, a policy masking the country data attribute) is active in the system, it will not be considered during the simulation unless it is explicitly selected in the Policy Simulator.

Now if policy simulator is executed for policy MASK_BP_TEL for GUI Transaction – BP, then the effect of masking can only be seen on telephone number, even policy on country data attribute is active.



Pathlock DAC: Policy Simulator

Policy Selection

Policy: MASK_BP_TEL

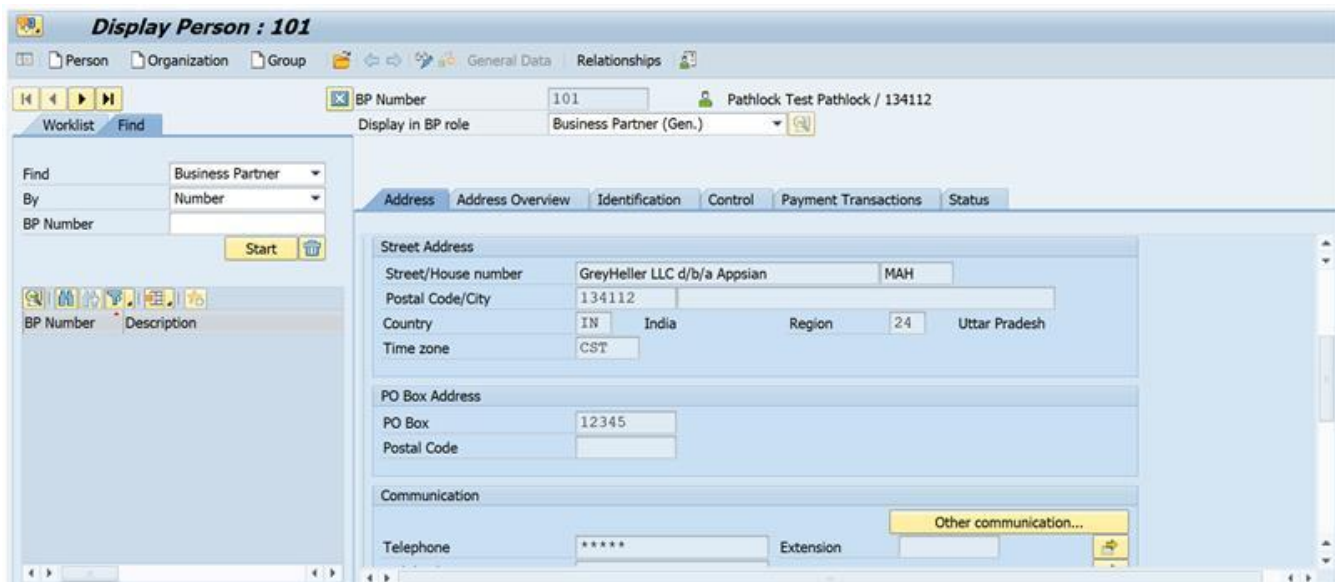
Simulate Via:

☐ Table

☒ GUI Transaction

BP

Result: Masking is applied exclusively to the telephone number for business partner. The country field remains unmasked, as only the policy MASK_BP_TEL is evaluated during the Policy Simulator execution.



Display Person : 101

Person Organization Group General Data Relationships

BP Number: 101 Pathlock Test Pathlock / 134112

Display in BP role: Business Partner (Gen.)

Worklist Find

Find: Business Partner

By: Number

BP Number: []

Start

BP Number Description

Address Address Overview Identification Control Payment Transactions Status

Street Address

Street/House number: GreyHeller LLC d/b/a Apsian MAH

Postal Code/City: 134112

Country: IN India Region: 24 Uttar Pradesh

Time zone: CST

PO Box Address

PO Box: 12345

Postal Code: []

Communication

Telephone: ***** Extension: []

Other communication...

3.8 Data Discovery

Data Discovery is used to identify all SAP artifacts where a specific data definition is used. Based on the selected search criteria, it retrieves the corresponding GUI applications, database tables, and OData services.

Data Discovery supports searches using:

- Data Element
- Data Attribute
- Field Description

Only one search criterion can be selected at a time using the corresponding radio button.

Data Discovery can be executed using transaction code: /N/APPSPDM/DISC

Search Scope and Behavior:

Based on the selected radio button, Data Discovery searches for:

- GUI Applications
- Database Tables
- OData Services

where the specified data element, data attribute, or field description is used.

Example: Data Discovery Using Data Element

1. On the selection screen, select the Data Element radio button.
2. Enter the required data element value for which data discovery needs to be performed.
3. Execute the report to retrieve all related GUI applications, tables, and OData services.

Similarly, users can perform Data Discovery by selecting:

- Data Attribute, or
- Description (description of the data element or a field)

using their respective radio buttons and input fields.

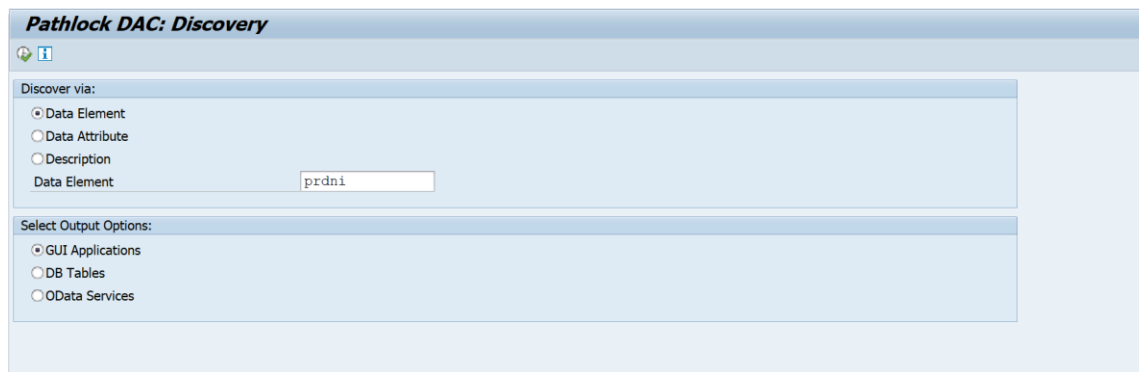
Prerequisite for OData Service Discovery

To enable discovery of OData services, the following prerequisite must be completed:

- Program /APPSS4H/DD_ODATA_SRV must be scheduled and executed in the background.
- This is a one-time activity required to collect and store OData service metadata for the relevant data elements.

Note: Without completing this prerequisite, OData services will not appear in the Data Discovery results.

Here, for instance PRDNI data element is used to search GUI Applications, discovery report will show the possible applications wherever data element is used in the system.



Pathlock DAC: Discovery

Discover via:

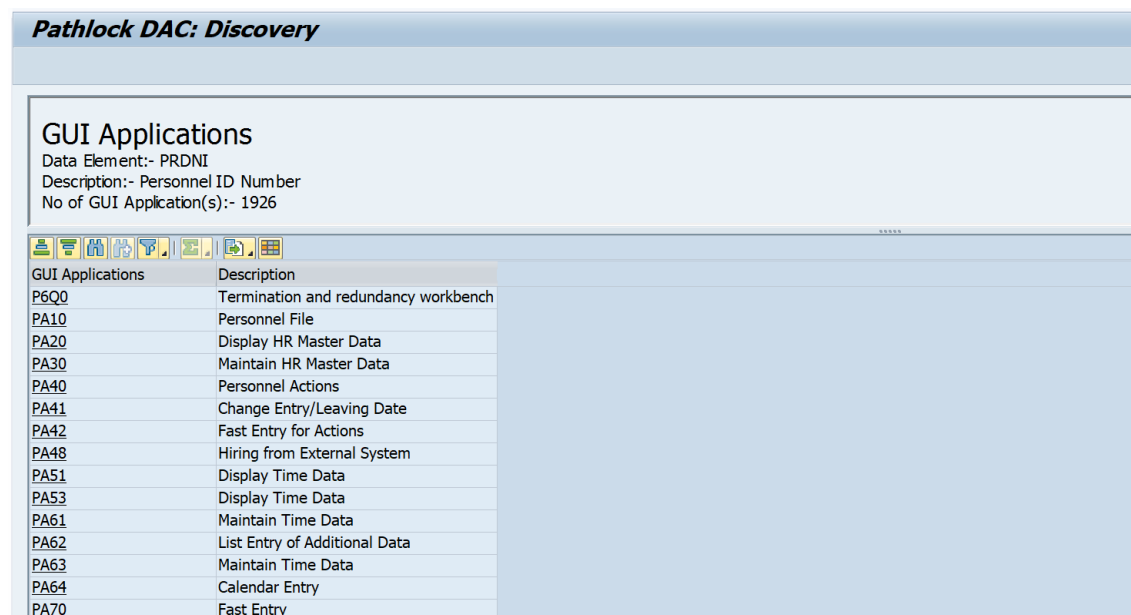
- ☒ Data Element
- ☐ Data Attribute
- ☐ Description

Data Element:

Select Output Options:

- ☒ GUI Applications
- ☐ DB Tables
- ☐ OData Services

Result: We get all the Transaction codes where this data element is used. **Similarly**, we can **execute** discovery for DB Tables and OData Services.



Pathlock DAC: Discovery

GUI Applications
Data Element:- PRDNI
Description:- Personnel ID Number
No of GUI Application(s):- 1926

GUI Applications	Description
P6Q0	Termination and redundancy workbench
PA10	Personnel File
PA20	Display HR Master Data
PA30	Maintain HR Master Data
PA40	Personnel Actions
PA41	Change Entry/Leaving Date
PA42	Fast Entry for Actions
PA48	Hiring from External System
PA51	Display Time Data
PA53	Display Time Data
PA61	Maintain Time Data
PA62	List Entry of Additional Data
PA63	Maintain Time Data
PA64	Calendar Entry
PA70	Fast Entry

4. Limitations

Pathlock DAC has the following current limitations:

1. Processing on masked data available in ALV output cannot be used for subsequent actions, which can lead to runtime errors. For example, creating a subsequent document using masked fields available in ALV output.
2. Masking in download options is only supported in ALV Outputs. For example, Spreadsheet, export to a local file, etc.

5. Sample Implementation

This section explains the use cases that demonstrate data masking and data restriction in a business setting. These implementations protect sensitive information based on user attributes and data attributes.

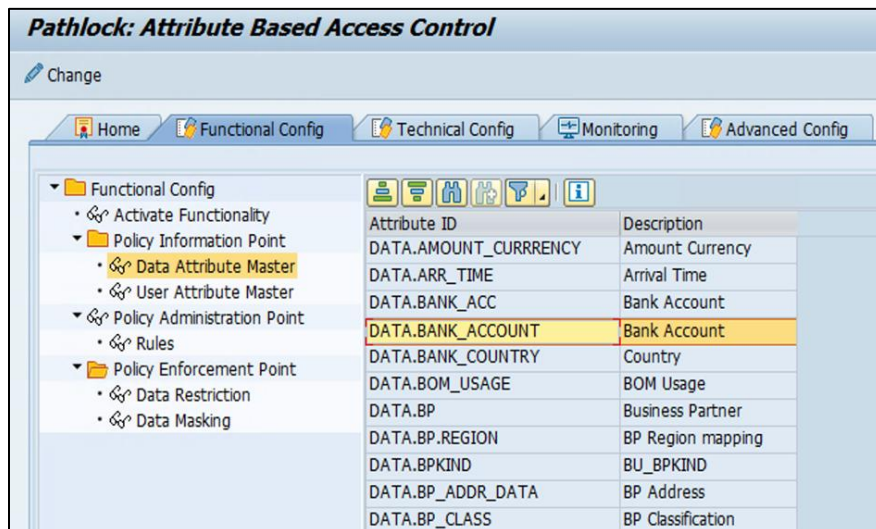
- [Use Case 1: Masking 'Bank Account' for Business Partners based on User PFCG Role](#)
- [Use Case 2: Masking 'Material Description' based on 'Material Group'](#)
- [Use Case 3: Masking 'Material Description' based on 'Material Group' for Specific Transactions](#)
- [Use Case 4: Restricting Sensitive Material Group Changes with Data Restriction \(Input blocking, Filtering, and Output blocking\)](#)
- [Use Case 5: Restrict Bank Account access based on User Country and Bank Country via Data Restriction in native UI5/Fiori Applications](#)
- [Use Case 6: Restrict Material Access based on Data Plant and User Plant, maintained in User Master via Data Restriction](#)
- [Use Case 7: Restrict Material BOM Access based on Material Classification determined by Material Number and Plant via Data Restriction](#)
- [Use Case 8: Masking 'Starting Balance' based on 'GL Account'](#)
- [Use Case 9: Simulate Data Classification on Database Tables](#)

5.1 Use Case 1

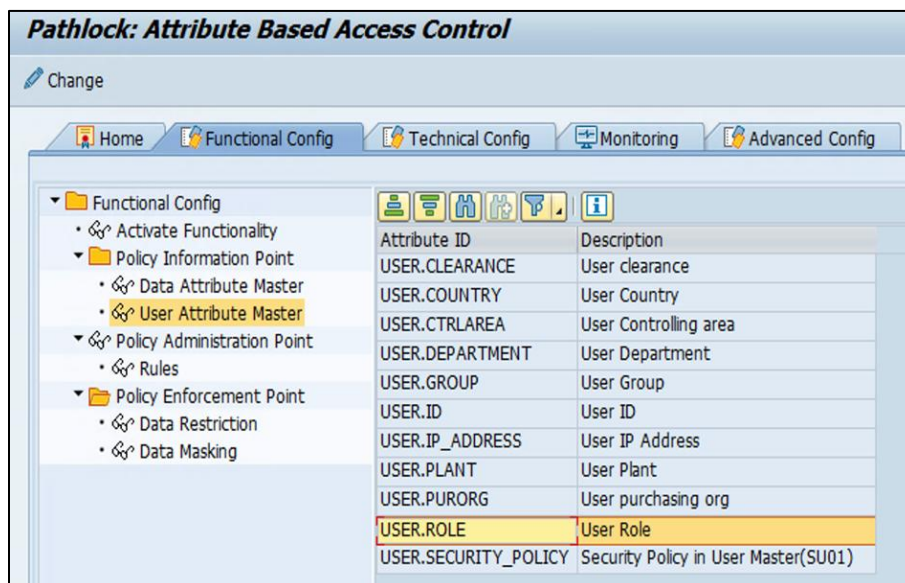
Masking 'Bank Account' for Business Partners based on User PFCG Role

To implement the masking of the business partner's 'Bank Account,' the following configuration is required:

1. Define a Data Attribute with the Attribute ID 'DATA.BANK_ACCOUNT' and its description as 'Bank Account.'



- Define a User Attribute named 'USER.ROLE' with the description 'User Role.'



- Create a Policy named 'RESTRICT_PII_DATA' along with its description.

Pathlock: Attribute Based Access Control

Change

Home Functional Config Technical Config Monitoring Advanced Config

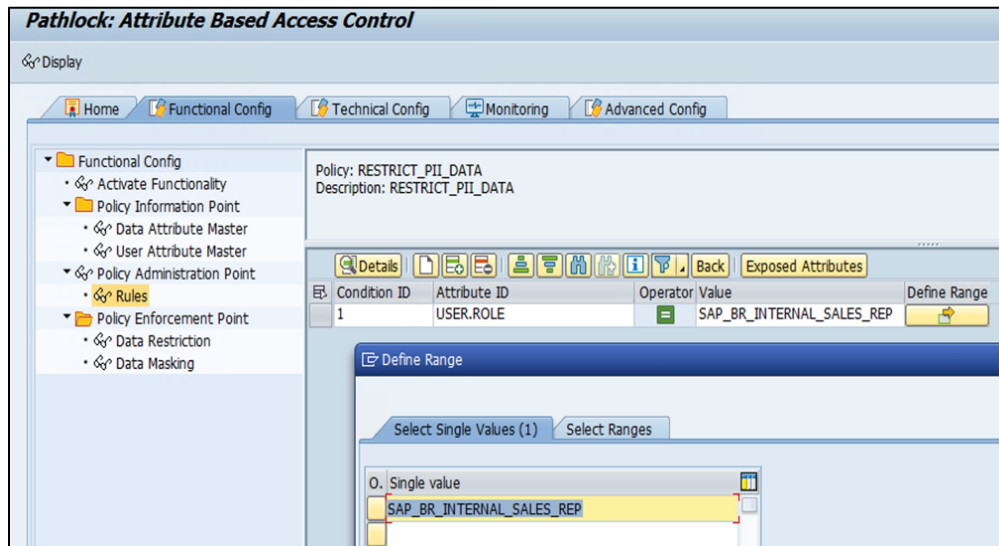
Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
 - Policy Administration Point**
 - Rules
- Policy Enforcement Point
 - Data Restriction
 - Data Masking

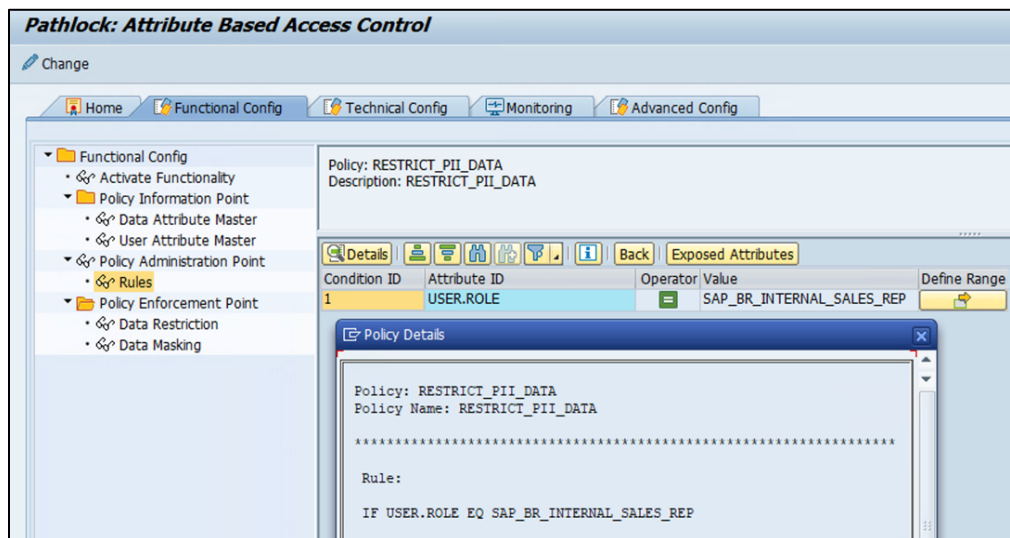
Rules Maintain Custom Message Status

Policy	Description
RESTRICT_CTRLAREA	RESTRICT_CTRLAREA
RESTRICT_CUSTOMER_DATA	RESTRICT_CUSTOMER_DATA
RESTRICT_DATA	RESTRICT_DATA
RESTRICT_FLIGHT_DETAILS	RESTRICT_FLIGHT_DETAILS
RESTRICT_GL_LOCAL_AMOUNT	RESTRICT_GL_LOCAL_AMOUNT
RESTRICT_ITAR	RESTRICT ITAR MATERIAL
RESTRICT_MAT_DESC	RESTRICT_MAT_DESC
RESTRICT_MATERIAL	RESTRICT_MATERIAL
RESTRICT_MATERIAL_ACCESS	RESTRICT_MATERIAL_ACCESS
RESTRICT_MATERIAL_BOM	RESTRICT_MATERIAL_BOM
RESTRICT_PA20.EMPLOYEE	RESTRICT_PA20.EMPLOYEE
RESTRICT_PII_DATA	RESTRICT_PII_DATA
RESTRICT_PII_SSN	RESTRICT_PII_SSN
RESTRICT_PLANT	RESTRICT_PLANT
RESTRICT_PURORG	RESTRICT_PURORG
RESTRICT_SALES	RESTRICT_SALES

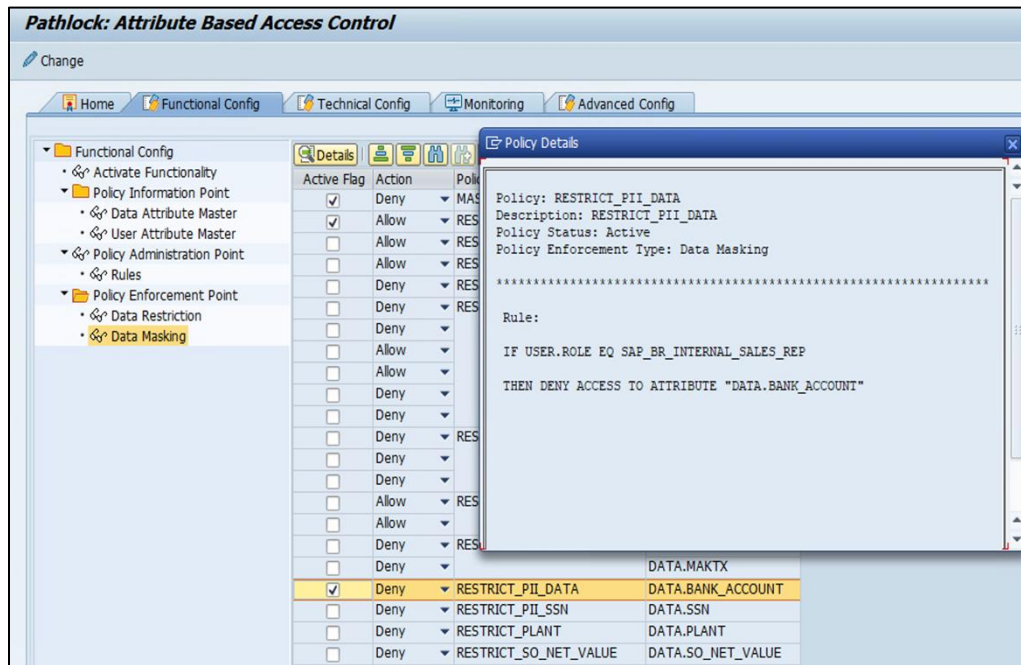
- Define rule condition using User Attribute 'USER.ROLE' to the Policy 'RESTRICT_PII_DATA'.



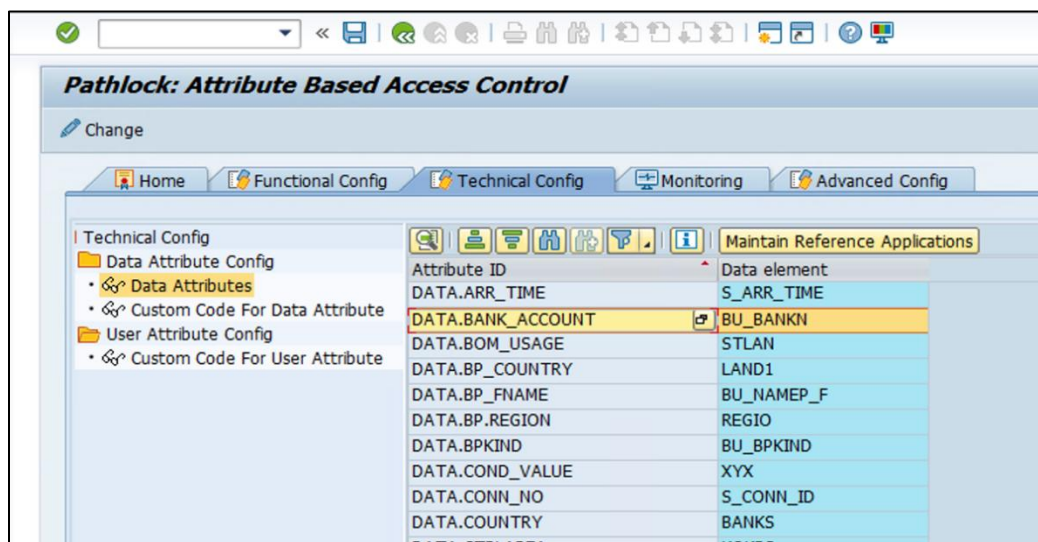
- Click on the Details button to view the Policy details.



6. Configure a Policy Enforcement Point for 'Data Masking' and link it to the Policy 'RESTRICT_PII_DATA' and the Attribute 'DATA.BANK_ACCOUNT.'



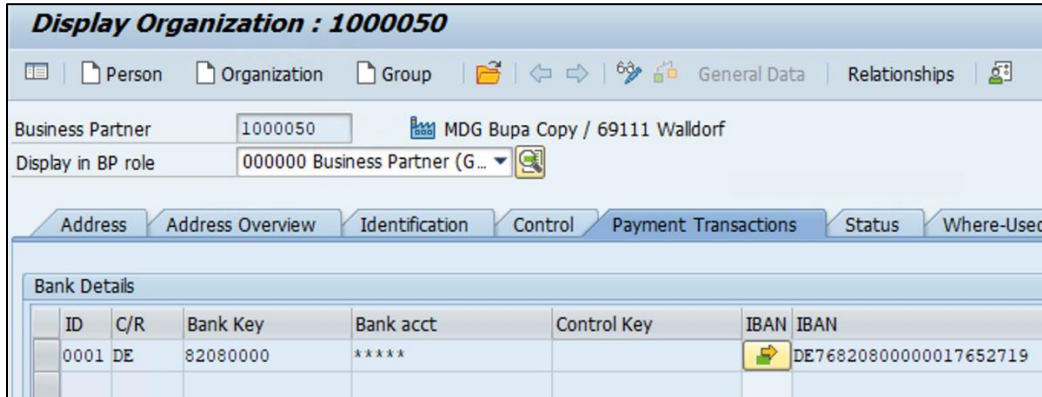
7. Under the **Technical Configuration** tab, assign the Attribute 'DATA.BANK_ACCOUNT' to the Data Element 'BU_BANKN'.



Expected Results:

Upon implementing the Policy, the following screenshots demonstrate the expected outcomes of the enforcement. The masking will be applied across all output options, such as GUI Transactions (BP), SE16 table outputs, downloaded data in ALVs, search help, and Fiori/UI5 apps, wherever the Data Element 'BU_BANKN' is used:

- **Masking in Transaction – 'BP':** The 'Bank Account' field will appear masked within the BP transaction.



Display Organization : 1000050

Business Partner: 1000050 MDG Bupa Copy / 69111 Walldorf

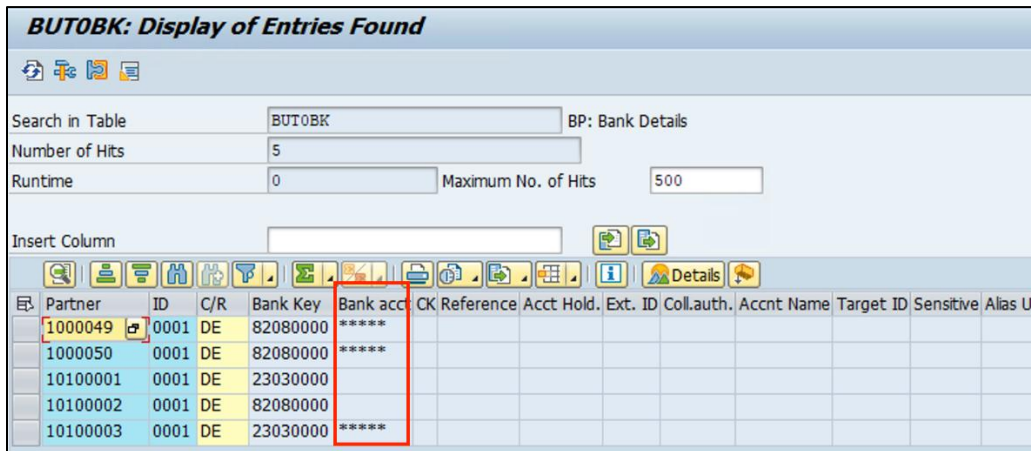
Display in BP role: 000000 Business Partner (G...)

Address Address Overview Identification Control **Payment Transactions** Status Where-Used

Bank Details

ID	C/R	Bank Key	Bank acct	Control Key	IBAN	IBAN
0001	DE	82080000	*****			DE76820800000017652719

- **Masking in SE16 ALV Grid, ALV List, and SE16 Standard List:** The 'Bank Account' field will be masked in the SE16 data browser.



BUTOBK: Display of Entries Found

Search in Table: BUTOBK BP: Bank Details

Number of Hits: 5

Runtime: 0 Maximum No. of Hits: 500

Insert Column

Partner	ID	C/R	Bank Key	Bank acct	CK Reference	Acct Hold.	Ext. ID	Coll.auth.	Acct Name	Target ID	Sensitive	Alias	Us
1000049	0001	DE	82080000	*****									
1000050	0001	DE	82080000	*****									
10100001	0001	DE	23030000	*****									
10100002	0001	DE	82080000	*****									
10100003	0001	DE	23030000	*****									

Additionally, any data downloaded from these sources will also display the 'Bank Account' field as masked.

	Business Partner	Bank Detail s ID	Bank Country/Regio n	Bank Key	Bank acct	Bank Control Key	Refere
2	1000049	0001	DE	82080000	*****		
3	1000050	0001	DE	82080000	*****		
4	10100001	0001	DE	23030000			
5	10100002	0001	DE	82080000			
6	10100003	0001	DE	23030000	*****		

- **Masking in Search Help:** In search help results, the 'Bank Account' field will be masked.

Business Partner Number (2) 152 Entries found						
W: Partners by Different Attributes B: Partners by bank details I: Partners by Identification Number U: Partner Accordi.						
Bank Key	Bank acct	Bank Account Alias	IBAN	C/R Name 1/Last Name	Name 2/First Nam	
000124622	*****			CA	FOREIGN SUPPLIER US 50 (CA)	
01100039	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	ABC CUSTOMER 01	
011000390	*****			US	BANK 1 - CITI BANK 2021 FS00	
011000390	*****			US	BANK 1 - CITI BANK 2021 FS00	
011000390	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	BANK 1 - SAMPLE BANK	
011000390	*****			US	DOMESTIC CUSTOMER INVOICE LIST	
011000390	*****			US	DOMESTIC CUSTOMER INVOICE LIST	
011000390	*****			US	DOMESTIC CUSTOMER US 3	
011000390	*****			US	DOMESTIC US CPD	

- **Masking in Fiori/UI5 App:** Within Fiori/UI5 applications, the 'Bank Account' field in the results will also be masked.

<

SAP

Business Partner

>

MDG Bupa Copy

1000050

>

Edit

<

Communication

>

Bank Accounts

Payment Cards

Identification

>

Contacts

Relationships

Classification

Bank Accounts

Bank Accounts (1)

Search

Q

Copy

ID	Bank Key	Account Number	Payment System	
☐ 0001	82080000	*****		

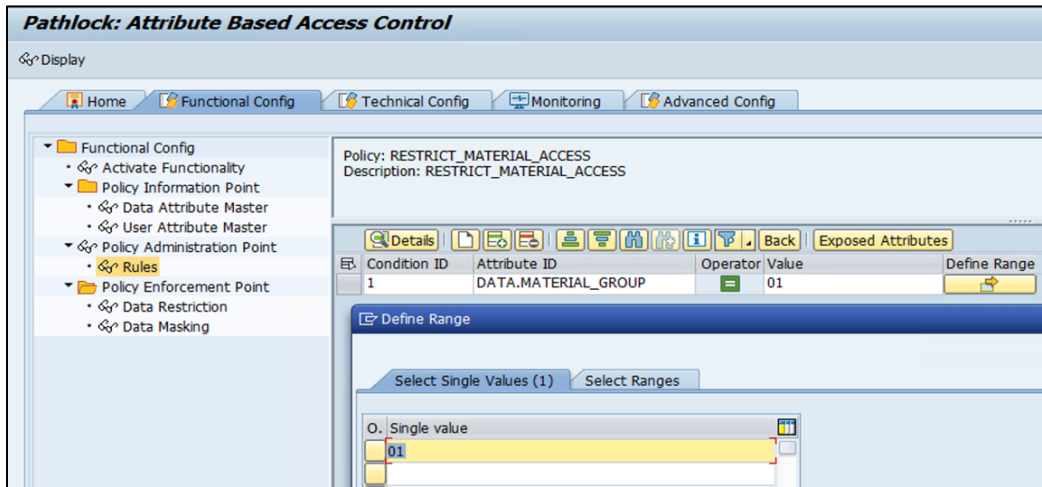
Alias Type:

5.2 Use Case 2

Masking 'Material Description' based on 'Material Group.'

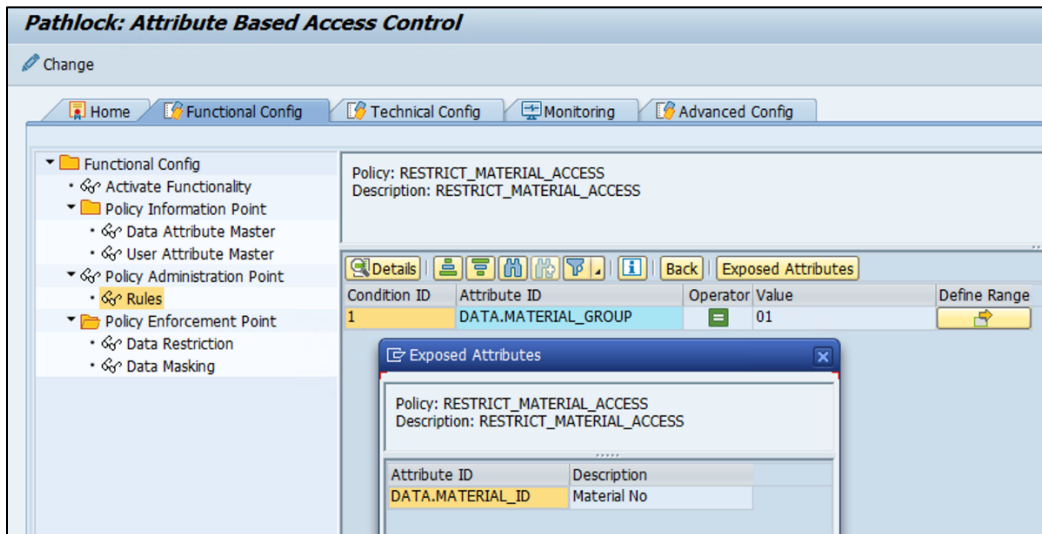
To mask the 'Material Description,' follow the similar steps outlined in [Use Case 1](#) to define Data Attributes, User Attributes, and Technical Configuration for Data Attributes. Additionally, implement the following steps:

1. Define a rule condition in the new Policy 'RESTRICT_MATERIAL_ACCESS' defined to restrict material access.



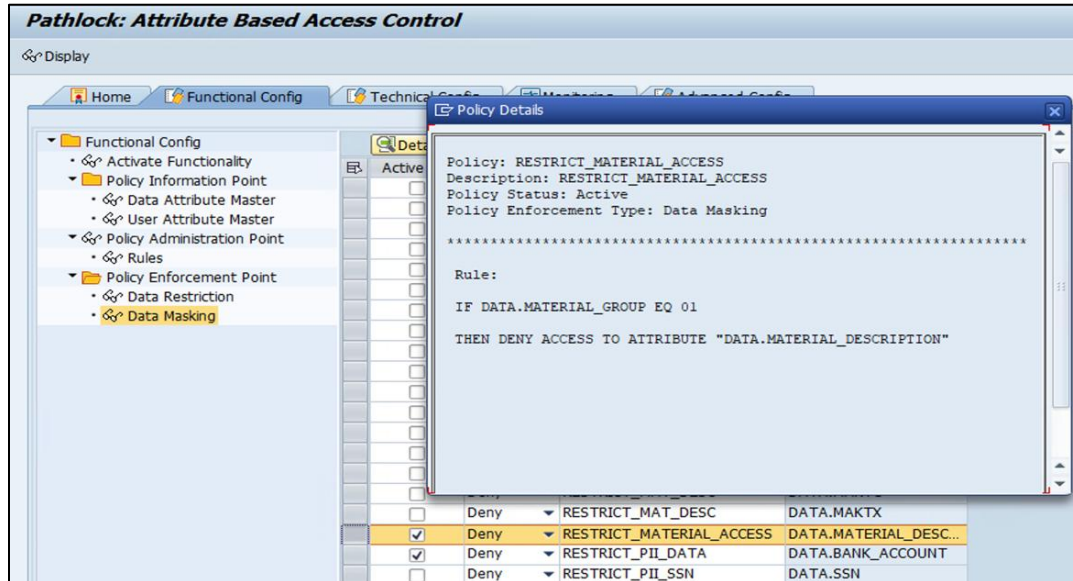
The screenshot shows the 'Pathlock: Attribute Based Access Control' interface. The left sidebar contains a tree view with 'Functional Config' expanded, showing 'Policy Information Point' > 'Data Attribute Master' > 'Rules'. The main area displays the 'Policy: RESTRICT_MATERIAL_ACCESS' with description 'RESTRICT_MATERIAL_ACCESS'. Below this is a table with columns: Condition ID, Attribute ID, Operator, Value, and Define Range. A single row is visible with Condition ID '1', Attribute ID 'DATA.MATERIAL_GROUP', Operator '=', and Value '01'. The 'Define Range' dialog is open, showing 'Select Single Values (1)' and 'Select Ranges' tabs. Under 'Select Single Values (1)', there is a list of values with '01' selected.

To enable material group using material ID, DATA.MATERIAL_ID is added in the exposed attribute and Data Attribute BADI is implemented for DATA.MATERIAL_GROUP.

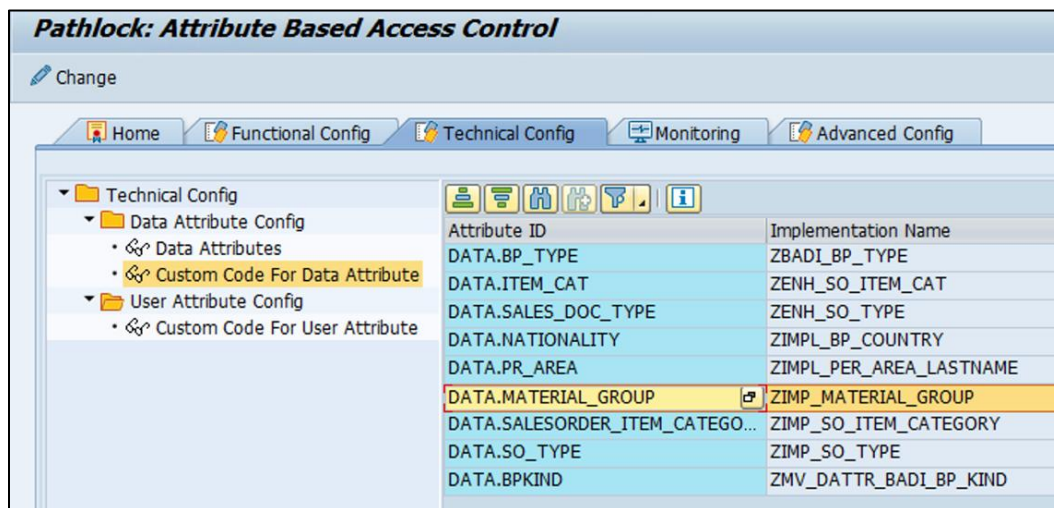


The screenshot shows the 'Pathlock: Attribute Based Access Control' interface. The left sidebar is the same as the previous screenshot. The main area displays the 'Policy: RESTRICT_MATERIAL_ACCESS' with description 'RESTRICT_MATERIAL_ACCESS'. Below this is a table with columns: Condition ID, Attribute ID, Operator, Value, and Define Range. A single row is visible with Condition ID '1', Attribute ID 'DATA.MATERIAL_GROUP', Operator '=', and Value '01'. The 'Exposed Attributes' dialog is open, showing the 'Policy: RESTRICT_MATERIAL_ACCESS' with description 'RESTRICT_MATERIAL_ACCESS'. Below this is a table with columns: Attribute ID and Description. A single row is visible with Attribute ID 'DATA.MATERIAL_ID' and Description 'Material No'.

2. Configure Data Masking.



3. Create a BADI implementation 'ZIMP_MATERIAL_GROUP' to add code for BADI '/APPSDM/DATA_ATTRIBUTES.' This implementation will appear in **Technical Config > Custom Code for Data Attribute**.



Code Snippet

```

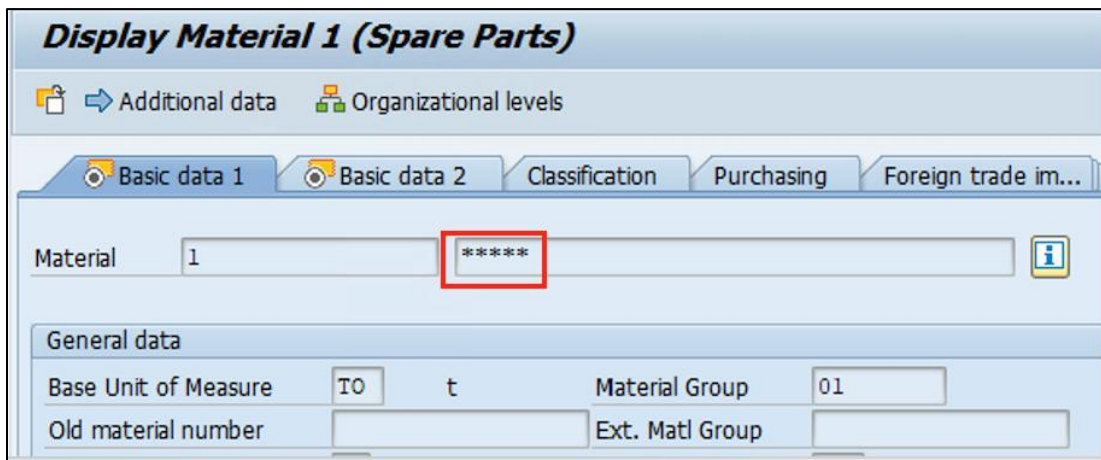
Class Builder: Class ZCL_IMPL_MATERIAL_GROUP Display
Method /APPSDM/DATA_ATTRIBUTES_INF-EXECUTE_CUSTOM_LOGIC Active
1 METHOD /appsdm/data_attributes_inf-execute_custom_logic.
2
3 DATA: ls_key_value TYPE /appsdm/key_value_s,
4       lv_matnr TYPE matnr,
5       lv_material_group TYPE matkl.
6
7 READ TABLE it_key_val INTO ls_key_value
8       WITH KEY attribute_id = 'DATA.MATERIAL_ID'. " Screen Attribute (Material No)
9
10 IF sy-subrc EQ 0.
11     lv_matnr = ls_key_value-input_value.
12
13 CALL FUNCTION 'CONVERSION_EXIT_MATN1_INPUT'
14     EXPORTING
15         input      = lv_matnr
16     IMPORTING
17         output     = lv_matnr
18     EXCEPTIONS
19         length_error = 1
20         OTHERS      = 2.
21
22 SELECT SINGLE matkl FROM mara INTO lv_material_group
23       WHERE matnr = lv_matnr.
24
25 IF sy-subrc EQ 0.
26     e_key_val-input_value = lv_material_group.
27 ENDIF.
28 ENDMETHOD.

```

Expected Results:

Upon applying the Policy, the following screenshots exhibit the expected results of the enforcement. The masking will be applied across all output options (GUI Transaction – MM03, SE16 table outputs, downloaded data in ALVs, search help, and Fiori/UI5 apps) wherever the Data Element 'MAKTX' is used:

- **Masking 'Material Description':** The field 'Material Description' will be masked in MM02/MM03 transactions for materials belonging to the material group '01'.



Display Material 1 (Spare Parts)

Additional data Organizational levels

Basic data 1 Basic data 2 Classification Purchasing Foreign trade im...

Material 1 *****

General data

Base Unit of Measure TO t Material Group 01

Old material number Ext. Matl Group

- **Masking in SE16 ALV Grid, ALV List, and SE16 Standard List:** The field 'Material Description' will be masked in the data browser (SE16N).

Display of Entries Found

Table to be searched: Material Descriptions

Number of hits:

Runtime: Maximum no. of hits

	Material	Language	Description	Description
1	EN	*****	*****	
3	EN	L&T Excavator	L&T EXCAVATOR	
5	EN	L&T Excavator	L&T EXCAVATOR	
24	EN	Test Material	TEST MATERIAL	

Also, any downloaded data files will contain the masked data.

Material	Language Key	Material Description	Material description
1	EN	*****	*****
3	EN	L&T Excavator	L&T EXCAVATOR
5	EN	L&T Excavator	L&T EXCAVATOR
24	EN	Test Material	TEST MATERIAL

- **Masking in Search Help:** In search help results, the 'Material Description' field will be masked.

Material Number (1) 4 Entries found

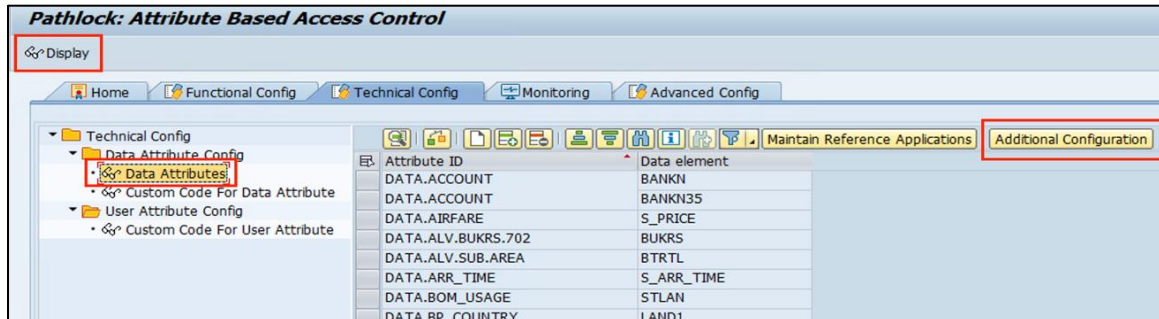
Material Number/Material Description

Material description	Language	Material
*****	EN	1
L&T EXCAVATOR	EN	3
L&T EXCAVATOR	EN	5
TEST MATERIAL	EN	24

5.3 Use Case 3

Masking 'Material Description' based on 'Material Group' for Specific Transactions

To enable masking of the 'Material Description' based on the Material group for specific transactions, tables, and search helps, utilize the 'Additional configuration' feature in addition to the config from [Use Case 2](#).



Maintain Program name and screen field (using F1 help) related to Material transactions. Here Data Element is not maintained, masking will be applicable only for the maintained Program and screen field.

Change View "GUI: Module Pools": Overview

New Entries

Dialog Structure

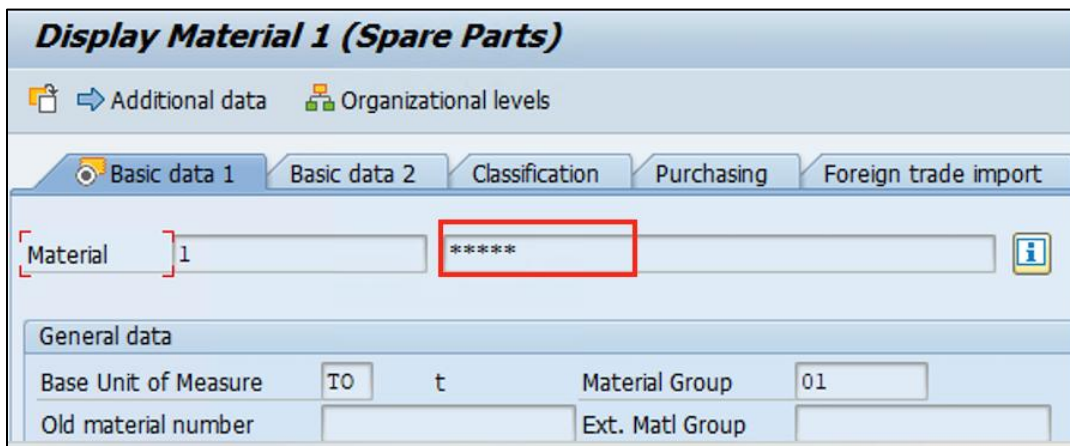
- GUI: Module Pools
- GUI: Tables
- GUI: Search Helps
- UIS/Fiori Applications

Program Name (Screen Data)	Screen Field	Attribute ID
SAPLMGD1	MAKT-MAKTX	DATA.MATERIAL_DESCRIPTION
SAPMV45A	KUAGV-KUNNR	DATA.TC34_MODULEPOOL
SAPMV45A	RV45A-KWMENG	DATA.TC34_MODULEPOOL
SAPMV45A	RV45A-MABNR	DATA.MATERIAL_NO
SAPMV45A	VBAP-VBELN	DATA.SALESORDER_NO
SAPMV45A	VBAP-ARKTX	DATA.TC34_MODULEPOOL
SAPMV45A	VBAP-PSTYV	DATA.SALESORDER_ITEM_CATEGORY
SAPMV45A	VBKD-BSTDK	DATA.PO_DATE

Expected Results:

Upon implementing the Policy, the following screenshots demonstrate the expected results of the enforcement. The masking will be applied only in the transaction code for which details have been maintained in the additional configuration, and in other places, the field will not be masked.

- **Masking 'Material Description':** The field 'Material Description' will be masked in the MM02/MM03 transaction for materials within the material group '01'.



Display Material 1 (Spare Parts)

Additional data Organizational levels

Basic data 1 Basic data 2 Classification Purchasing Foreign trade import

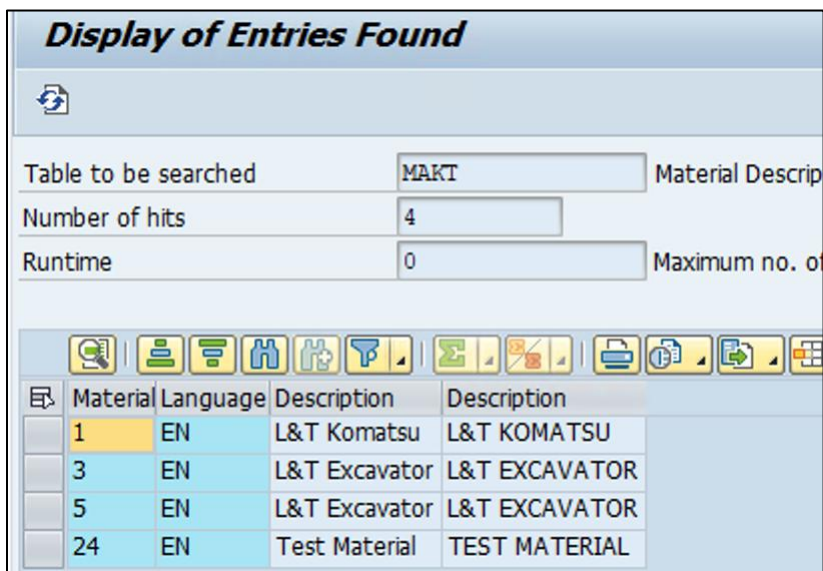
Material 1 *****

General data

Base Unit of Measure TO t Material Group 01

Old material number Ext. Matl Group

- **Masking in SE16 ALV Grid, ALV List, and SE16 Standard List:** The field 'Material Description' will not be masked in the data browser (SE16N).



Display of Entries Found

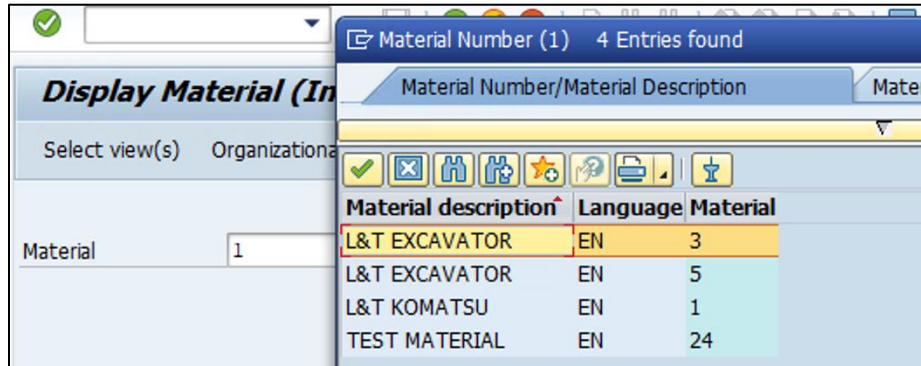
Table to be searched MAKT Material Description

Number of hits 4

Runtime 0 Maximum no. of

Material	Language	Description	Description
1	EN	L&T Komatsu	L&T KOMATSU
3	EN	L&T Excavator	L&T EXCAVATOR
5	EN	L&T Excavator	L&T EXCAVATOR
24	EN	Test Material	TEST MATERIAL

- **Masking in Search Help:** The field 'Material Description' in the search help results will not be masked.

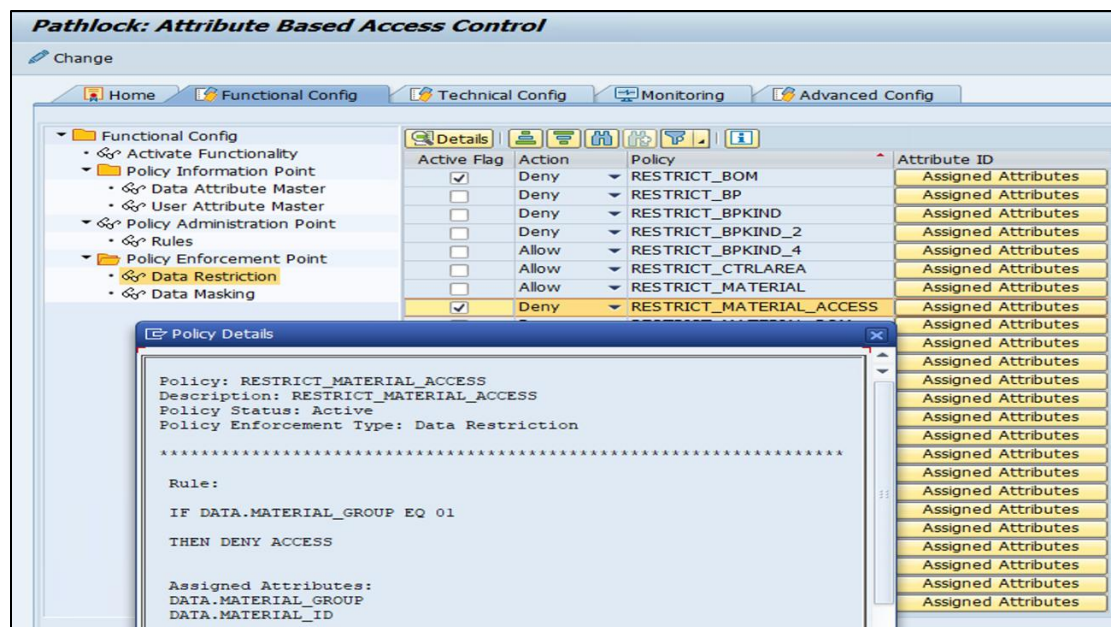


5.4 Use Case 4

Restricting material access based on sensitive material Group with Data Restriction (Input blocking, Filtering, and Output blocking)

To restrict Material creation/change/display based on the sensitive Material Group in transactions, tables, and search helps, follow the same policy applied in [Use Case 2](#). Additionally, configure Data Restriction under Policy Enforcement Point.

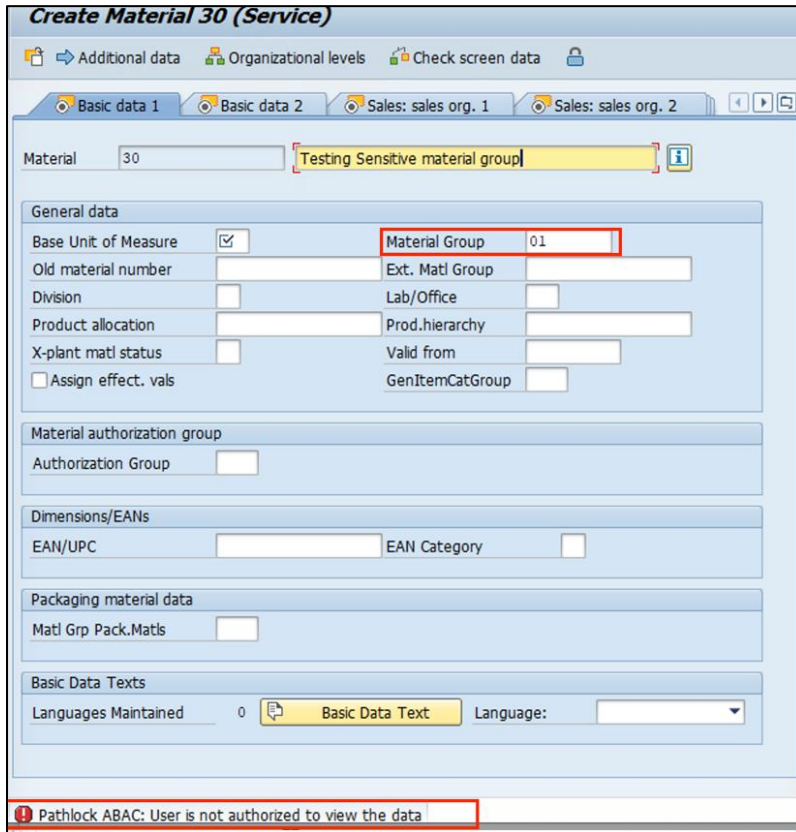
Note: If the same policy is active for both Data masking and Data Restriction, priority will be given to Data Restriction.



Expected Results:

Upon implementing the Data Restriction, the following screenshots demonstrate the expected results of the enforcement. The restrictions will be applied as follows:

- **Input Blocking:** The creation and modification of materials within the sensitive Material group '01' will be restricted.



Create Material 30 (Service)

Additional data | Organizational levels | Check screen data

Basic data 1 | Basic data 2 | Sales: sales org. 1 | Sales: sales org. 2

Material: 30 [Testing Sensitive material group]

General data

Base Unit of Measure	☒	Material Group	01
Old material number		Ext. Matl Group	
Division		Lab/Office	
Product allocation		Prod.hierarchy	
X-plant matl status	☐	Valid from	
Assign effect. vals	☐	GenItemCatGroup	

Material authorization group

Authorization Group

Dimensions/EANs

EAN/UPC EAN Category

Packaging material data

Matl Grp Pack.Matls

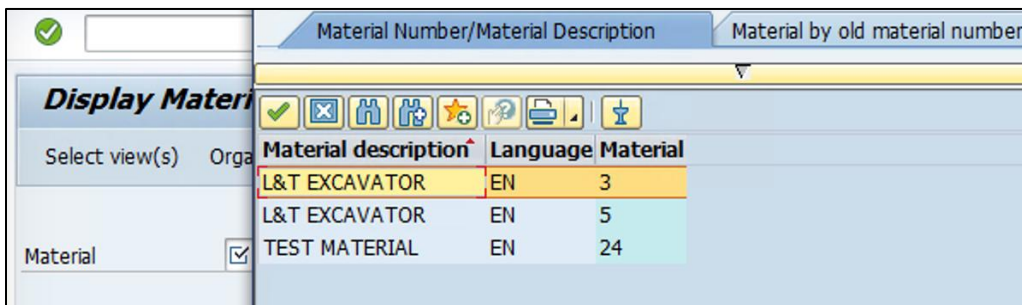
Basic Data Texts

Languages Maintained: 0 [Basic Data Text] Language: [dropdown]

Pathlock ABAC: User is not authorized to view the data

Users will be prohibited from creating or changing materials assigned to the sensitive Material group '01'.

- **Data Filtering:**
 - Materials identified as '1' within the sensitive Material group '01' will be filtered out in Search help.



Display Material

Select view(s) Orga

Material ☒

Material description	Language	Material
L&T EXCAVATOR	EN	3
L&T EXCAVATOR	EN	5
TEST MATERIAL	EN	24

- Materials identified as '1' from the sensitive Material group '01' will also be filtered out in table results.

Display of Entries Found

Table to be searched: Material Descriptions

Number of hits:

Runtime: Maximum no. of hits

Material	Language	Description	Description
3	EN	L&T Excavator	L&T EXCAVATOR
5	EN	L&T Excavator	L&T EXCAVATOR
24	EN	Test Material	TEST MATERIAL

- Output Blocking:** The display of Material '1' belonging to the sensitive Material group '01' will be restricted in the transaction MM03.

Display Material (Initial Screen)

Select view(s) Organizational levels Data

Material:

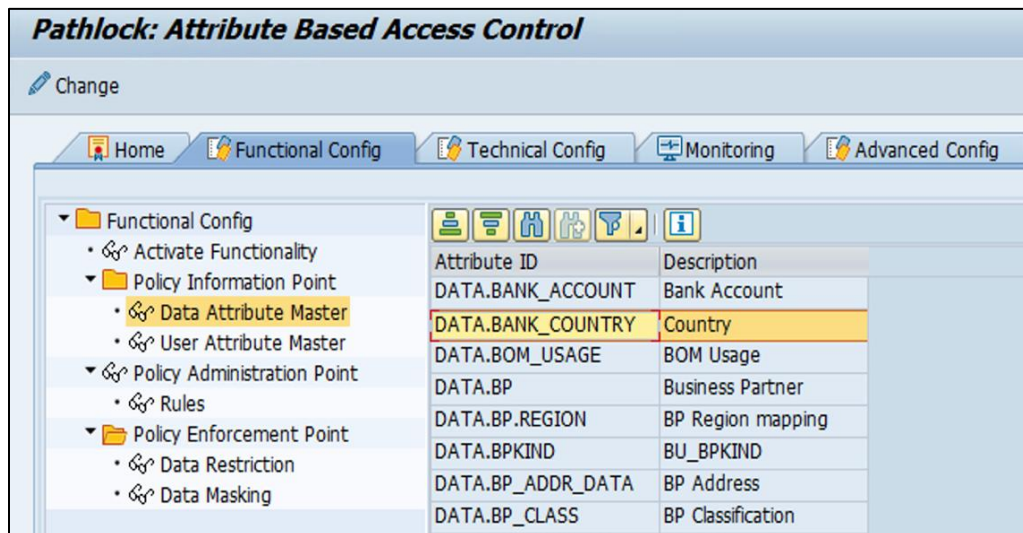
Pathlock ABAC: User is not authorized to view the data

5.5 Use Case 5

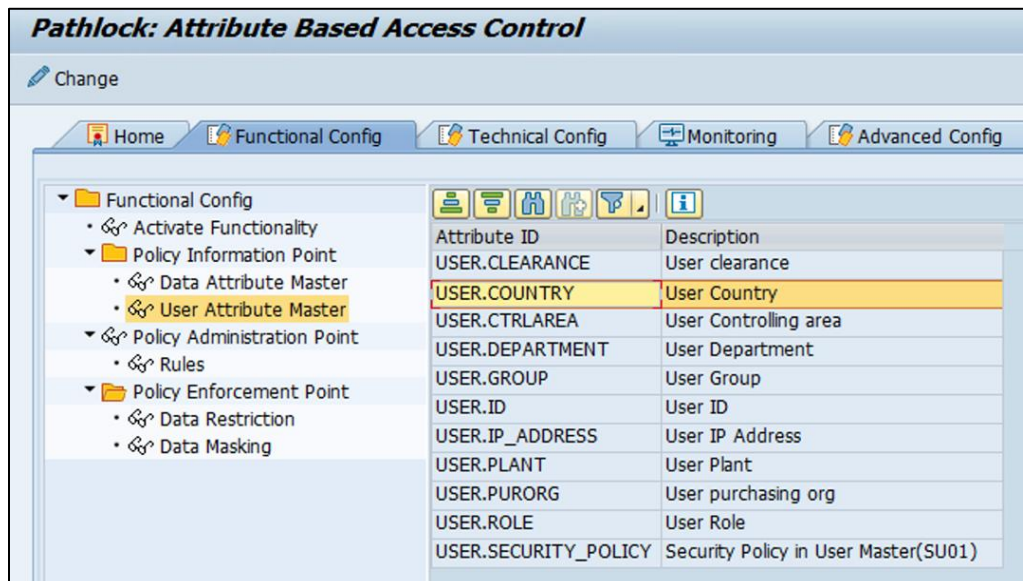
Restrict Bank Account access based on User Country and Bank Country via Data Restriction in native UI5/Fiori Applications

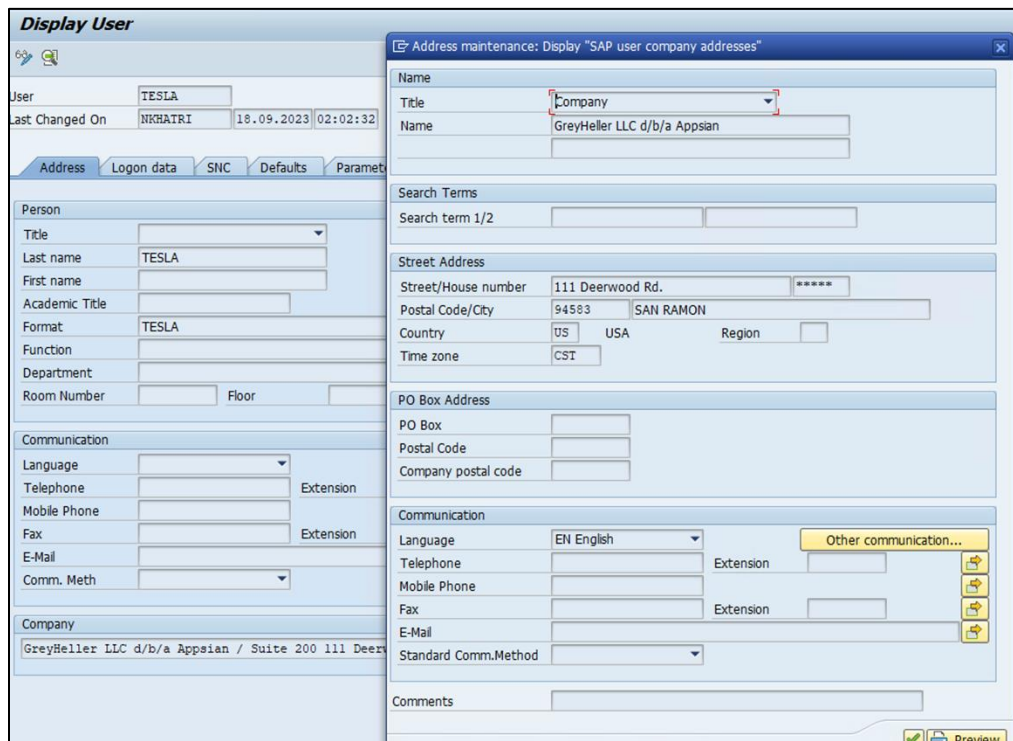
To implement the Data Restriction scenario for the business partner's 'Bank Account,' the following configuration is required:

- Define a Data Attribute with the Attribute ID 'DATA.BANK_COUNTRY' and its description as 'Country'.



- Define a User Attribute named 'USER.COUNTRY' with the description 'User Country' which is available in user master (T-code: SU01).





Display User

User: TESLA
Last Changed On: NRHAIRI 18.09.2023 02:02:32

Address Logon data SNC Defaults Paramet

Person

Title: TESLA
Last name: TESLA
First name:
Academic Title:
Format: TESLA
Function:
Department:
Room Number: Floor:

Communication

Language:
Telephone: Extension:
Mobile Phone:
Fax: Extension:
E-Mail:
Comm. Meth:

Company

GreyHeller LLC d/b/a Appian / Suite 200 111 Deerwood Rd.

Address maintenance: Display "SAP user company addresses"

Name:
Title: Company
Name: GreyHeller LLC d/b/a Appian

Search Terms

Search term 1/2:

Street Address

Street/House number: 111 Deerwood Rd.
Postal Code/City: 94583 SAN RAMON
Country: US USA Region:
Time zone: CST

PO Box Address

PO Box:
Postal Code:
Company postal code:

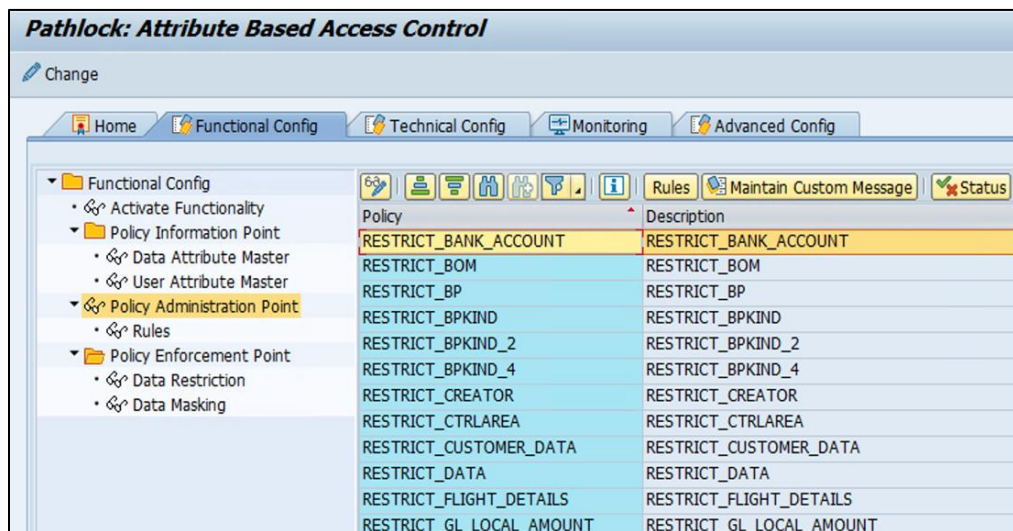
Communication

Language: EN English
Telephone: Extension:
Mobile Phone:
Fax: Extension:
E-Mail:
Standard Comm.Method:

Comments:

Preview

3. Create a Policy named 'RESTRICT_BANK_ACCOUNT' along with its description.



Pathlock: Attribute Based Access Control

Change

Home Functional Config Technical Config Monitoring Advanced Config

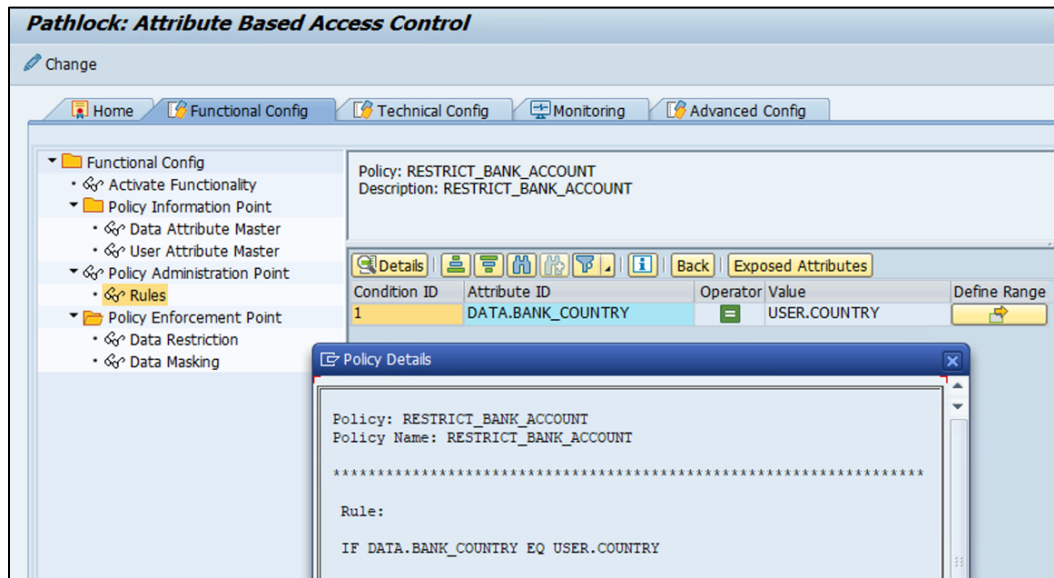
Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
 - Policy Administration Point
 - Rules
 - Policy Enforcement Point
 - Data Restriction
 - Data Masking

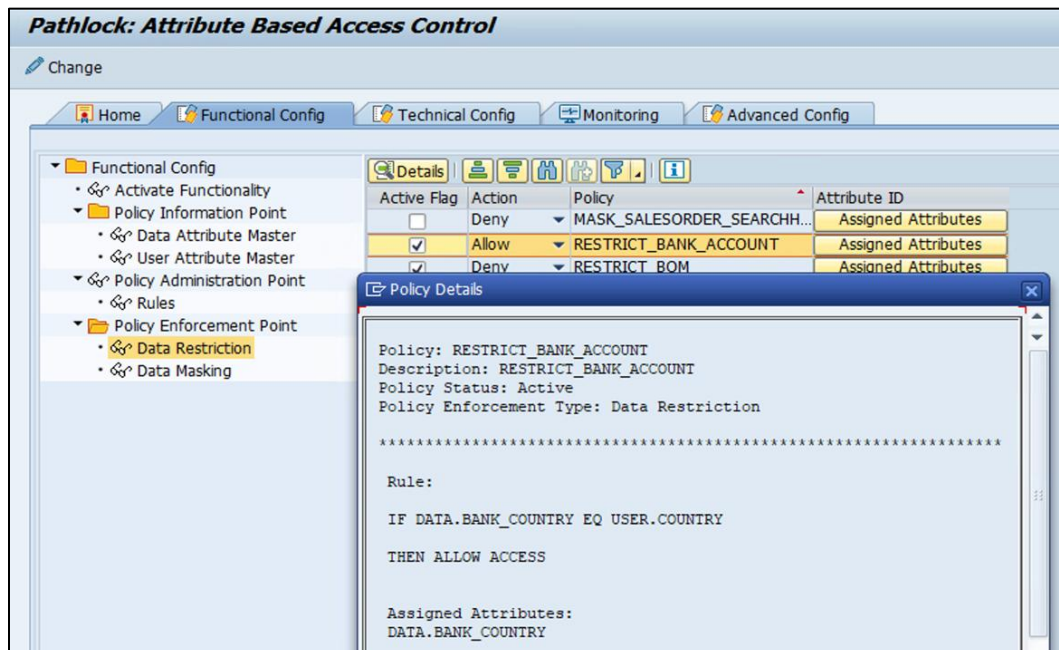
Rules

Policy	Description
RESTRICT_BANK_ACCOUNT	RESTRICT_BANK_ACCOUNT
RESTRICT_BOM	RESTRICT_BOM
RESTRICT_BP	RESTRICT_BP
RESTRICT BPKIND	RESTRICT BPKIND
RESTRICT BPKIND_2	RESTRICT BPKIND_2
RESTRICT BPKIND_4	RESTRICT BPKIND_4
RESTRICT_CREATOR	RESTRICT_CREATOR
RESTRICT_CTRLAREA	RESTRICT_CTRLAREA
RESTRICT_CUSTOMER_DATA	RESTRICT_CUSTOMER_DATA
RESTRICT_DATA	RESTRICT_DATA
RESTRICT_FLIGHT_DETAILS	RESTRICT_FLIGHT_DETAILS
RESTRICT_GL_LOCAL_AMOUNT	RESTRICT_GL_LOCAL_AMOUNT

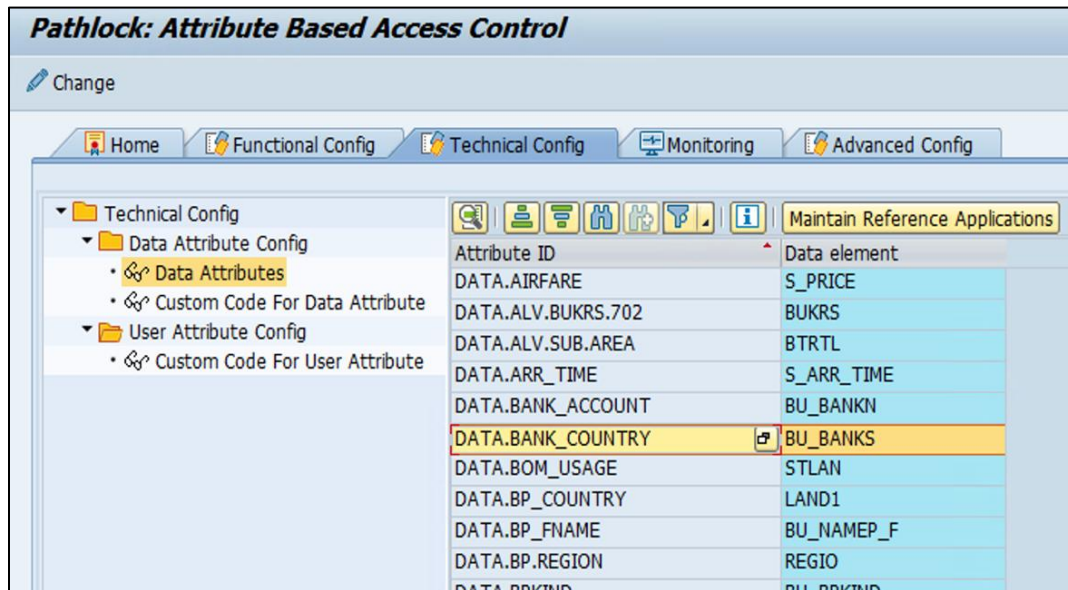
- Define a rule condition using User Attribute 'USER.COUNTRY' to the Policy 'RESTRICT_BANK_ACCOUNT'.



- Configure DAC Control for 'Data Restriction' and link it to the Policy 'RESTRICT_BANK_ACCOUNT'.

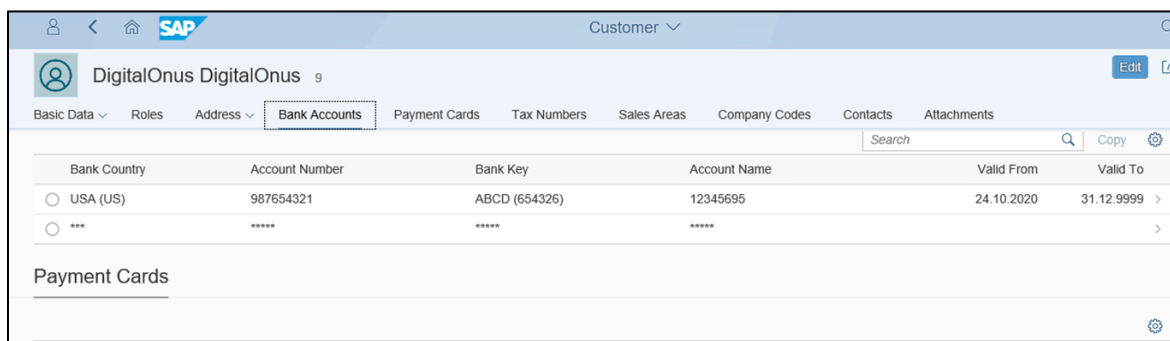


- Under the **Technical Configuration** tab, assign the Attribute 'DATA.BANK_COUNTRY' to the Data Element 'BU_BANKS'.



Expected Results:

Upon implementing the Policy, the following screenshot demonstrate the expected outcomes of the enforcement.

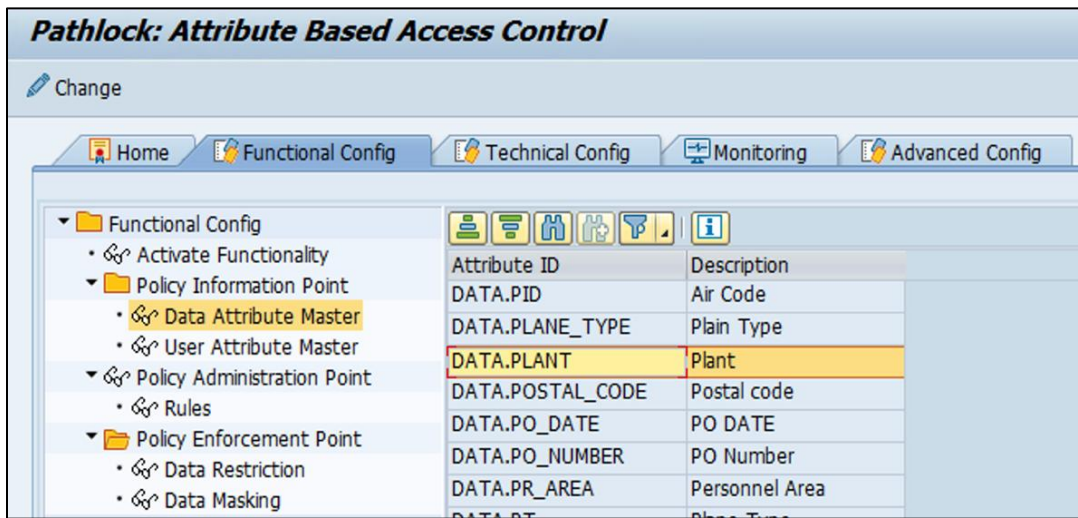


5.6 Use Case 6

Restrict Material based on Data Plant and User Plant, maintained in User Master via Data Restriction

To restrict Material based on Data Plant and User Plant, maintained in User Master Maintenance, the following configuration is required:

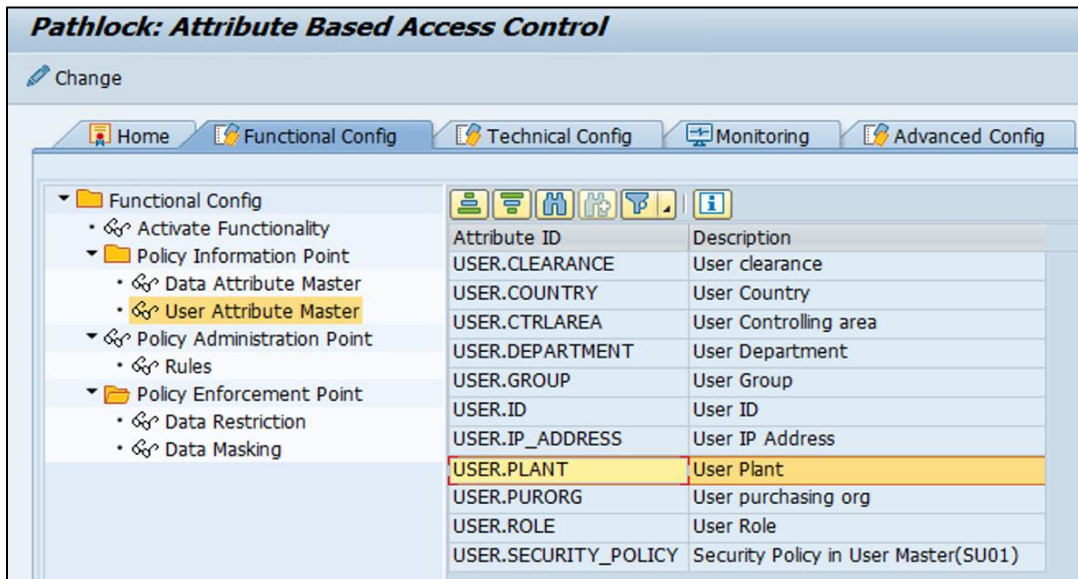
1. Define a Data Attribute with the Attribute ID 'DATA.PLANT' and its description as 'Plant'.



The screenshot shows the 'Pathlock: Attribute Based Access Control' interface. The 'Functional Config' tab is selected. In the left sidebar, 'Data Attribute Master' is highlighted under 'Policy Information Point'. The main table lists attributes with the following data:

Attribute ID	Description
DATA.PID	Air Code
DATA.PLANE_TYPE	Plain Type
DATA.PLANT	Plant
DATA.POSTAL_CODE	Postal code
DATA.PO_DATE	PO DATE
DATA.PO_NUMBER	PO Number
DATA.PR_AREA	Personnel Area

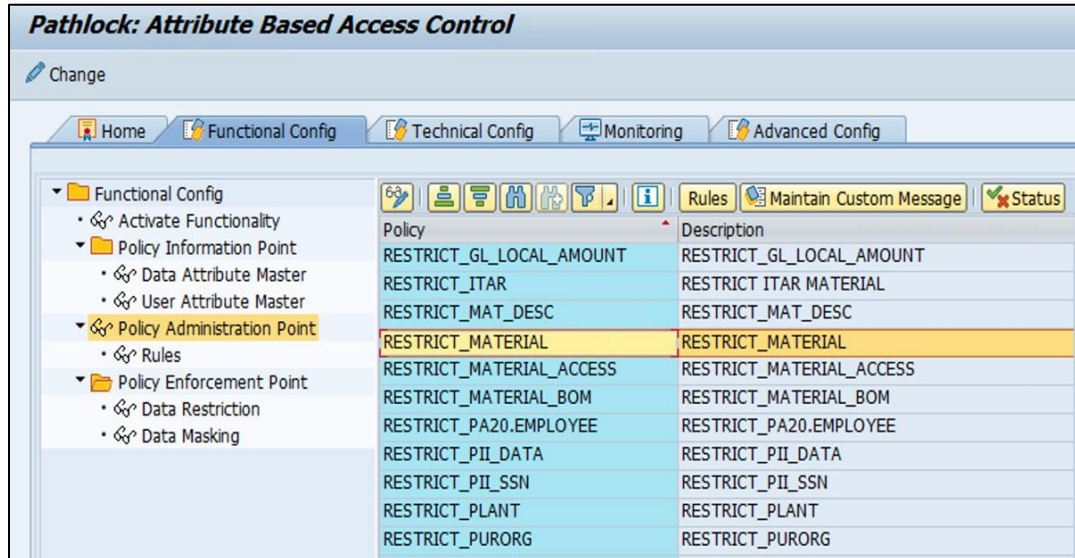
2. Define a User Attribute named 'USER.PLANT' with the description 'User Plant'.



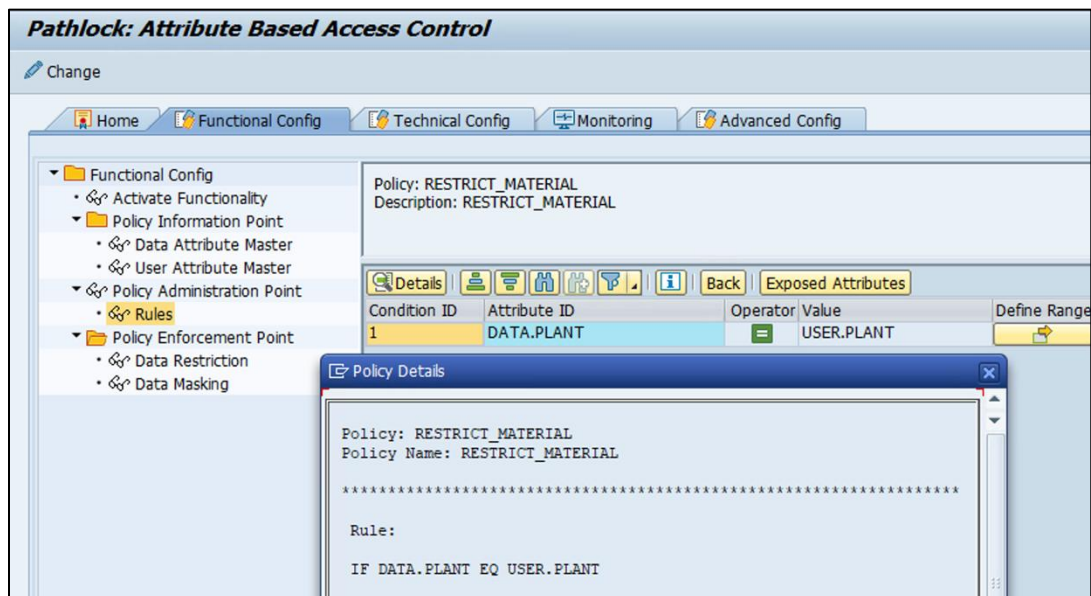
The screenshot shows the 'Pathlock: Attribute Based Access Control' interface. The 'Functional Config' tab is selected. In the left sidebar, 'User Attribute Master' is highlighted under 'Policy Information Point'. The main table lists attributes with the following data:

Attribute ID	Description
USER.CLEARANCE	User clearance
USER.COUNTRY	User Country
USER.CTRLAREA	User Controlling area
USER.DEPARTMENT	User Department
USER.GROUP	User Group
USER.ID	User ID
USER.IP_ADDRESS	User IP Address
USER.PLANT	User Plant
USER.PURORG	User purchasing org
USER.ROLE	User Role
USER.SECURITY_POLICY	Security Policy in User Master(SU01)

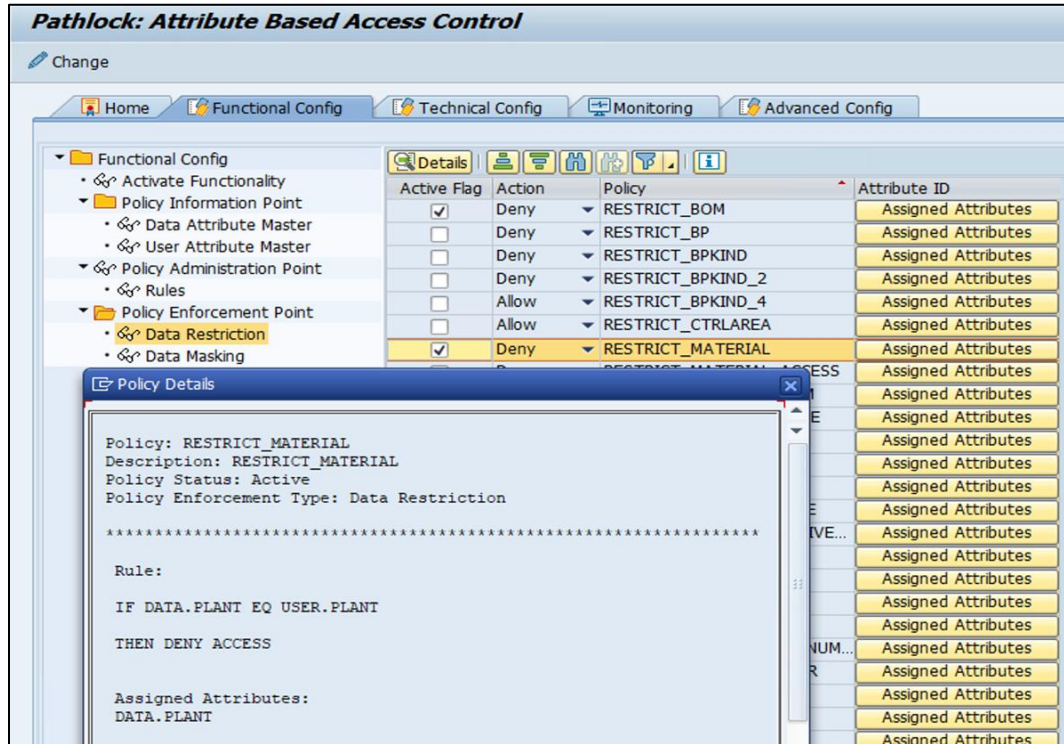
3. Create a Policy named 'RESTRICT_MATERIAL' along with its description.



4. Define rule condition using User Attribute 'USER.PLANT' and Data Attribute 'DATA.PLANT' in the Policy 'RESTRICT_MATERIAL'.



- Configure DAC Control for 'Data Restriction' and link it to the Policy 'RESTRICT_MATERIAL'.



Pathlock: Attribute Based Access Control

Change

Home Functional Config Technical Config Monitoring Advanced Config

Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Restriction**
 - Data Masking

Details

Active Flag	Action	Policy	Attribute ID
☒	Deny	RESTRICT_BOM	Assigned Attributes
☐	Deny	RESTRICT_BP	Assigned Attributes
☐	Deny	RESTRICT BPKIND	Assigned Attributes
☐	Deny	RESTRICT BPKIND_2	Assigned Attributes
☐	Allow	RESTRICT BPKIND_4	Assigned Attributes
☐	Allow	RESTRICT_CTRLAREA	Assigned Attributes
☒	Deny	RESTRICT_MATERIAL	Assigned Attributes

Policy Details

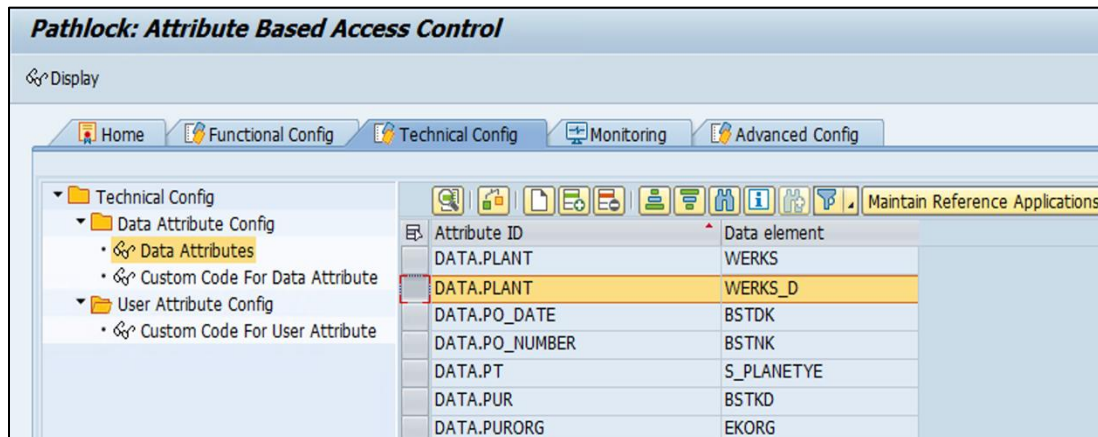
Policy: RESTRICT_MATERIAL
Description: RESTRICT_MATERIAL
Policy Status: Active
Policy Enforcement Type: Data Restriction

Rule:

```
IF DATA.PLANT EQ USER.PLANT
THEN DENY ACCESS
```

Assigned Attributes:
DATA.PLANT

- Under the **Technical Configuration** tab, assign the Attribute 'DATA.PLANT' to the Data Element 'WERKS_D'.



Pathlock: Attribute Based Access Control

Display

Home Functional Config Technical Config Monitoring Advanced Config

Technical Config

- Data Attribute Config
 - Data Attributes**
 - Custom Code For Data Attribute
- User Attribute Config
 - Custom Code For User Attribute

Attribute ID

Attribute ID	Data element
DATA.PLANT	WERKS
DATA.PLANT	WERKS_D
DATA.PO_DATE	BSTDK
DATA.PO_NUMBER	BSTNK
DATA.PT	S_PLANETYE
DATA.PUR	BSTKD
DATA.PURORG	EKORG

7. Maintain USER.PLANT - 'US01' in User Master Maintenance.

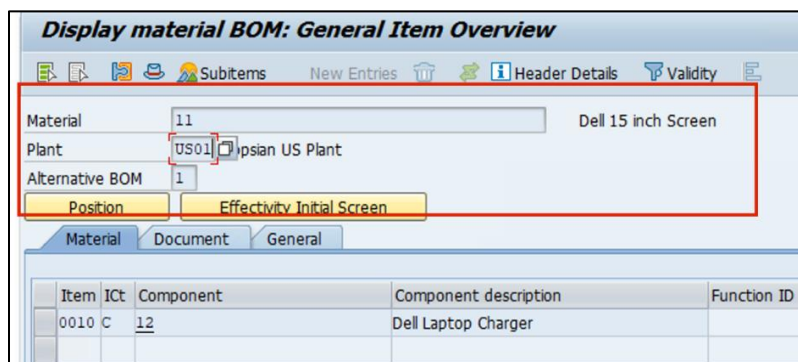


USER ID	Attribute ID	LOW	HIGH
TESLA	USER.PLANT	US01	

Expected Results:

Upon implementing the Policy, the following screenshots demonstrate the expected outcomes of the policy enforcement for User 'Tesla' who has access to Plant – US01:

- For 'US01' Plant:



Item	ICt	Component	Component description	Function ID
0010	C	12	Dell Laptop Charger	

- For 'US02' Plant:

Display material BOM: Initial Screen

Material

Plant

BOM Usage Plant Maintenance

Alternative BOM

Validity

Change Number

Valid From

Valid to

Additional Data

Required quantity

Pathlock ABAC: User is not authorized to view the data

5.7 Use Case 7

Restrict Material BOM Access based on Material Classification determined by Material Number and Plant via Data Restriction

To restrict Material BOM Access based on Material Classification, the following configurations are required:

1. Define a Data Attribute with the Attribute ID 'DATA.MATERIAL_NO' and its description as 'Material Number'.

Pathlock: Attribute Based Access Control

[Change](#)

Home Functional Config Technical Config Monitoring Advanced Config

Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Restriction
 - Data Masking

Attribute ID	Description
DATA.MATERIAL_CLAS...	Material Classification
DATA.MATERIAL_DESC...	Material description
DATA.MATERIAL_GROUP	Material Group
DATA.MATERIAL_ID	Material No
DATA.MATERIAL_NO	Material Number
DATA.MATERIAL_TYPE	Material type
DATA.MAT_BOM_CLASS	Material BOM Class
DATA.MAT_BOM_CLASS	Material BOM Class

2. Define a Data Attribute named 'DATA.PLANT' with the description 'Plant'.

Pathlock: Attribute Based Access Control

Change

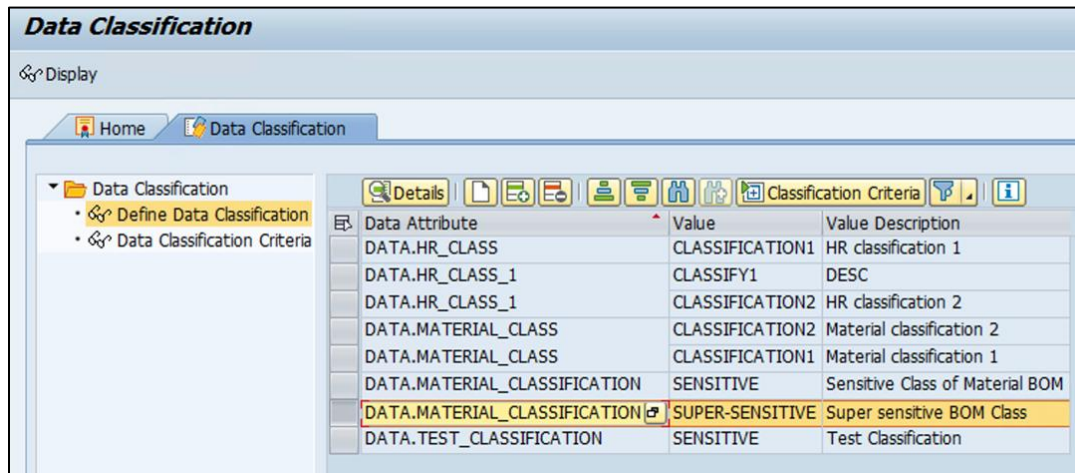
Home Functional Config Technical Config Monitoring Advanced Config

Functional Config

- Activate Functionality
- Policy Information Point
 - Data Attribute Master**
 - User Attribute Master
- Policy Administration Point
 - Rules
- Policy Enforcement Point
 - Data Restriction
 - Data Masking

Attribute ID	Description
DATA.PID	Air Code
DATA.PLANE_TYPE	Plain Type
DATA.PLANT	Plant
DATA.POSTAL_CODE	Postal code
DATA.PO_DATE	PO DATE
DATA.PO_NUMBER	PO Number
DATA.PR_AREA	Personnel Area
DATA.PR...	Plain Type

- Define a Data Attribute with the Attribute ID 'DATA.MATERIAL_CLASSIFICATION' with a value 'SUPER-SENSITIVE' and its description as 'Super sensitive BOM Class'.



Data Classification

Display

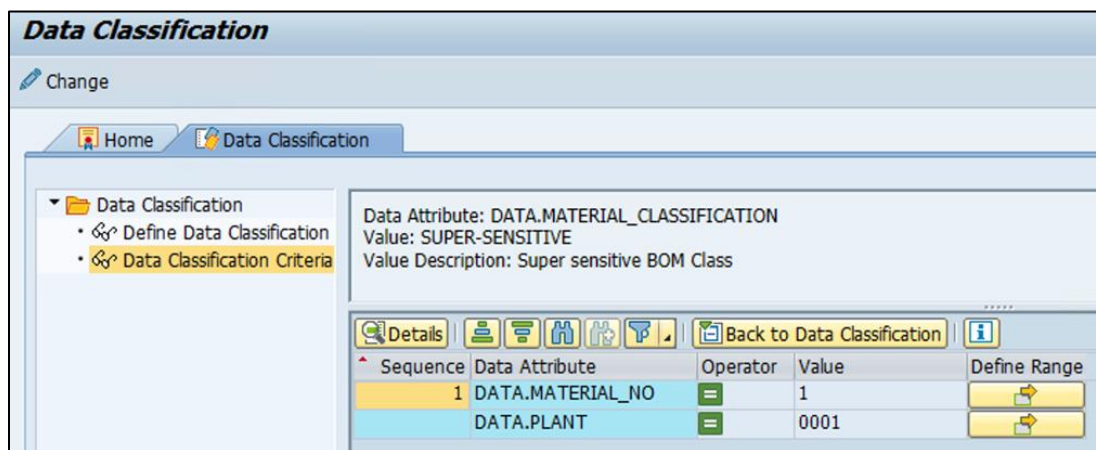
Home Data Classification

Data Classification

- Define Data Classification
- Data Classification Criteria

Data Attribute	Value	Value Description
DATA.HR_CLASS	CLASSIFICATION1	HR classification 1
DATA.HR_CLASS_1	CLASSIFY1	DESC
DATA.HR_CLASS_1	CLASSIFICATION2	HR classification 2
DATA.MATERIAL_CLASS	CLASSIFICATION2	Material classification 2
DATA.MATERIAL_CLASS	CLASSIFICATION1	Material classification 1
DATA.MATERIAL_CLASSIFICATION	SENSITIVE	Sensitive Class of Material BOM
DATA.MATERIAL_CLASSIFICATION	SUPER-SENSITIVE	Super sensitive BOM Class
DATA.TEST_CLASSIFICATION	SENSITIVE	Test Classification

- Define Classification Criteria using the Data Attribute 'DATA.MATERIAL_NO' and 'DATA.PLANT' for 'DATA.MATERIAL_CLASSIFICATION'.



Data Classification

Change

Home Data Classification

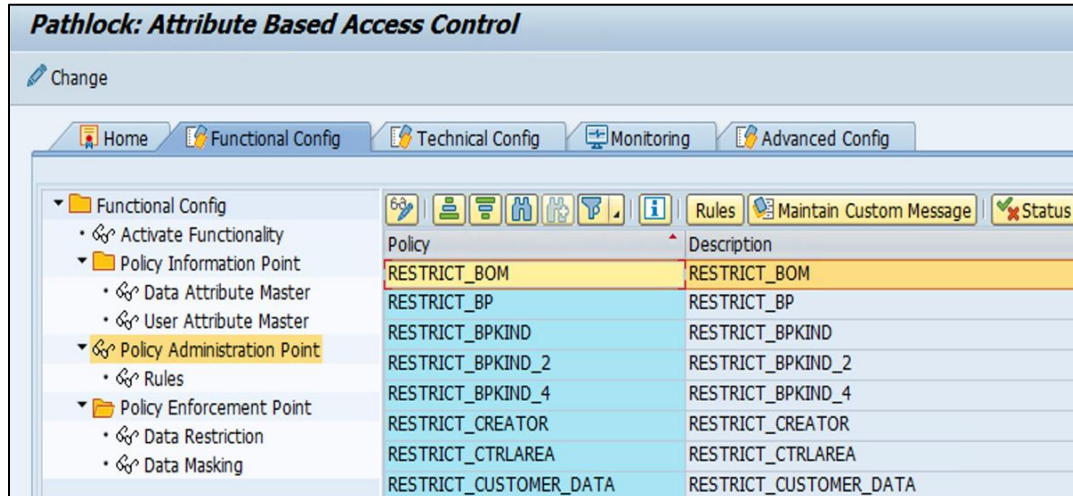
Data Classification

- Define Data Classification
- Data Classification Criteria

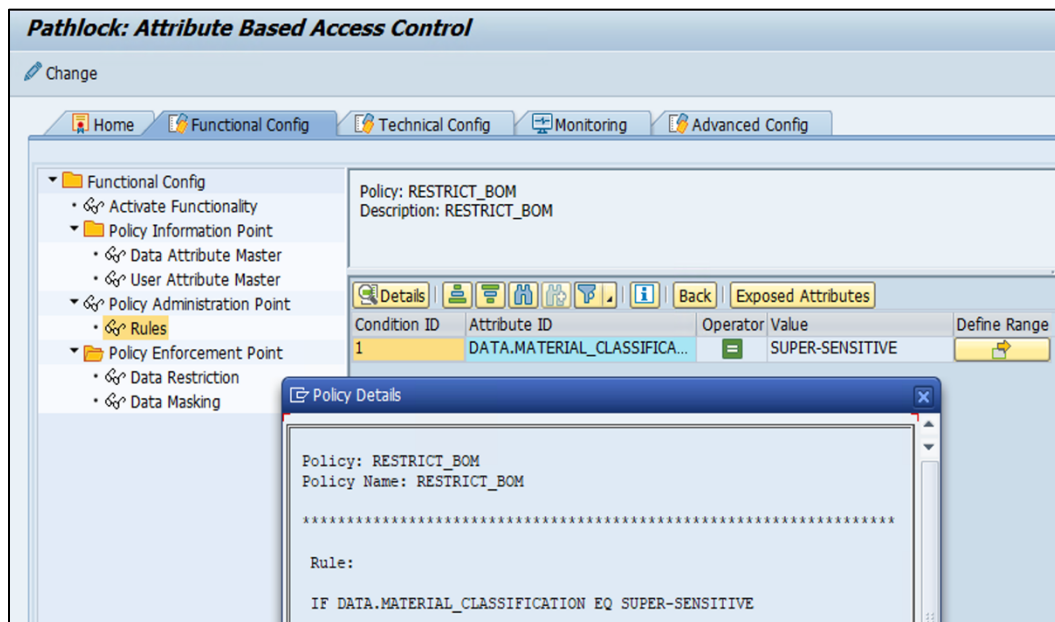
Data Attribute: DATA.MATERIAL_CLASSIFICATION
Value: SUPER-SENSITIVE
Value Description: Super sensitive BOM Class

Sequence	Data Attribute	Operator	Value	Define Range
1	DATA.MATERIAL_NO	=	1	
	DATA.PLANT	=	0001	

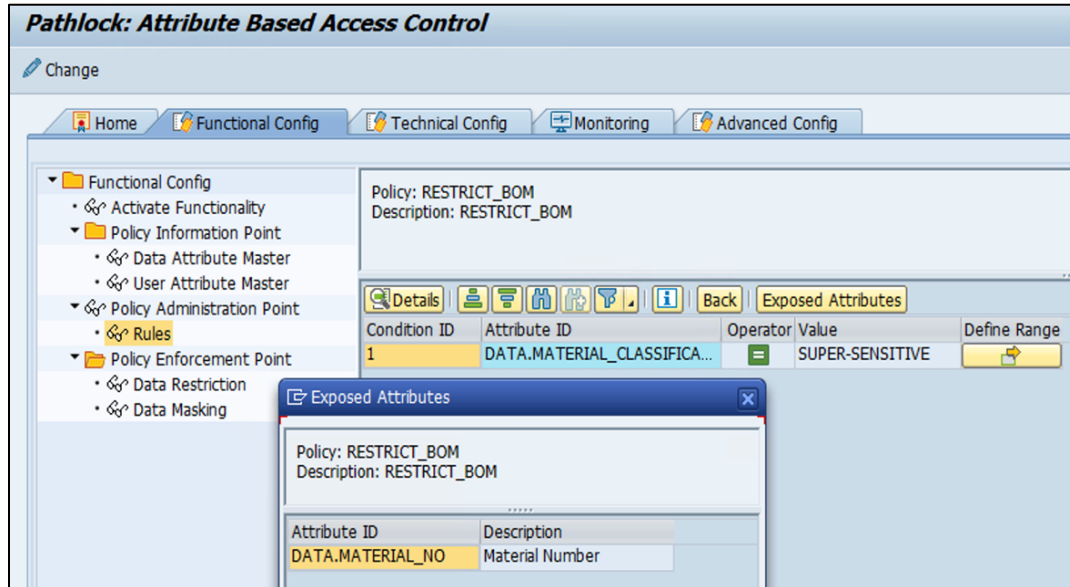
5. Create a Policy named 'RESTRICT_BOM' along with its description.



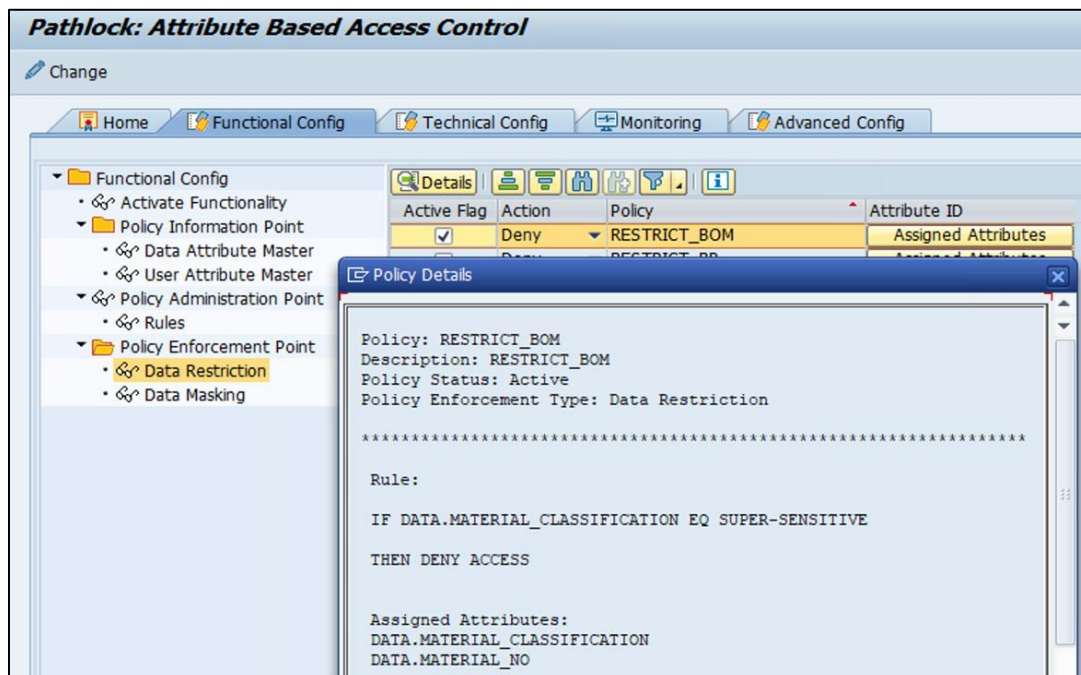
6. Define rule condition using the Data Attribute 'DATA.MATERIAL_CLASSIFICATION' in the Policy 'RESTRICT_BOM'.



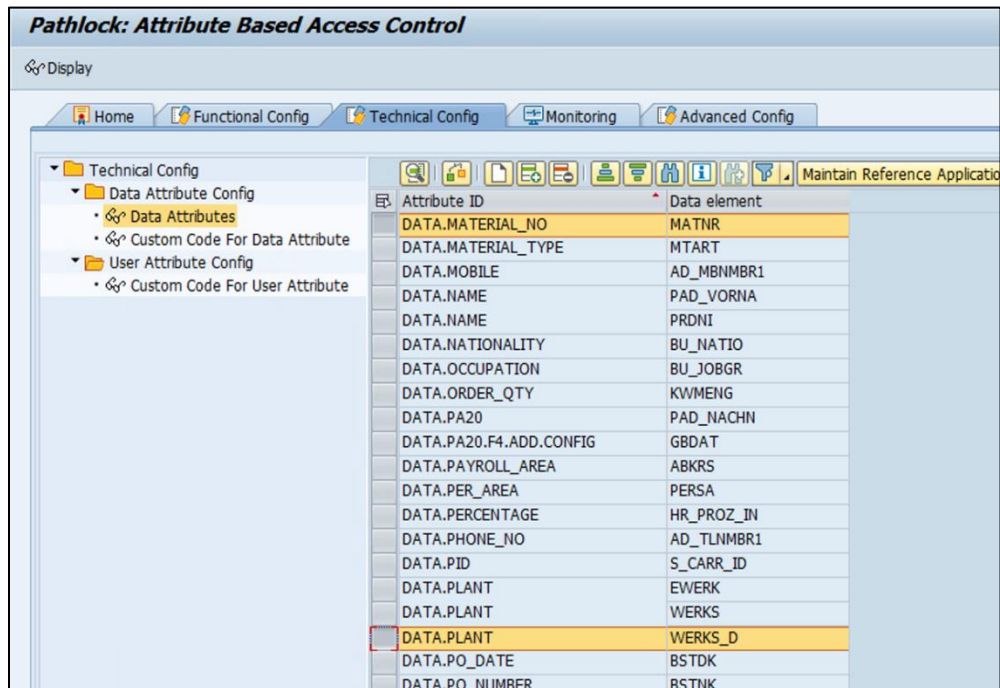
- Define the Data Attribute 'DATA.MATERIAL_NO' in the Exposed Attributes.



- Configure an DAC Control for 'Data Restriction' and link it to the Policy 'RESTRICT_BOM'.

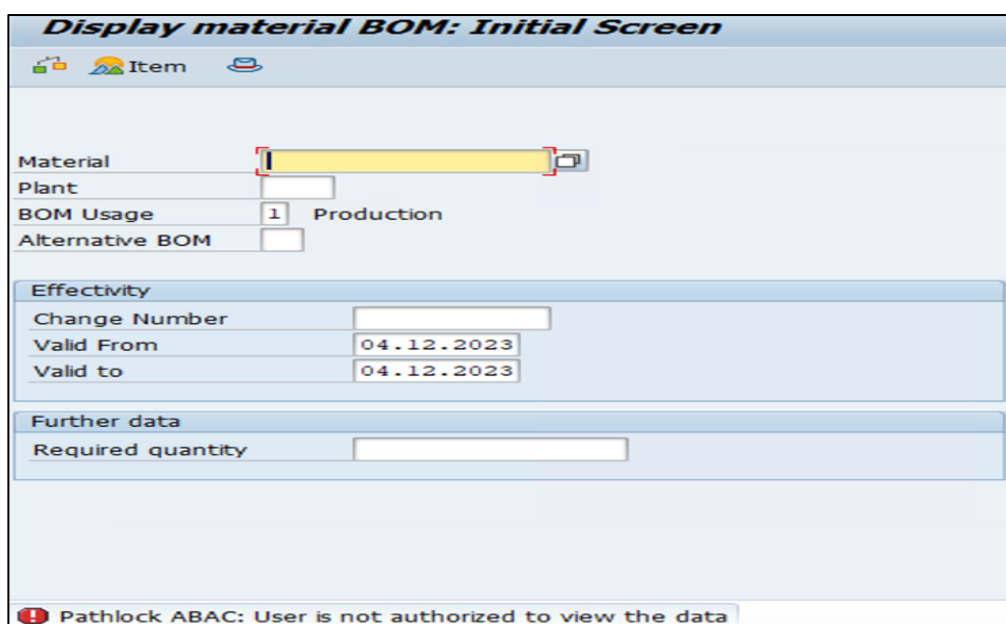


- Under the **Technical Configuration** tab, assign the Data Attribute 'DATA.MATERIAL_NO' to the Data Element 'MATNR' and the Data Attribute 'DATA.PLANT' to the Data Element 'WERKS_D'.



Expected Results:

Upon policy implementation, the provided screenshots illustrate the expected results of policy enforcement for Material Number 1 and Plant 0001.

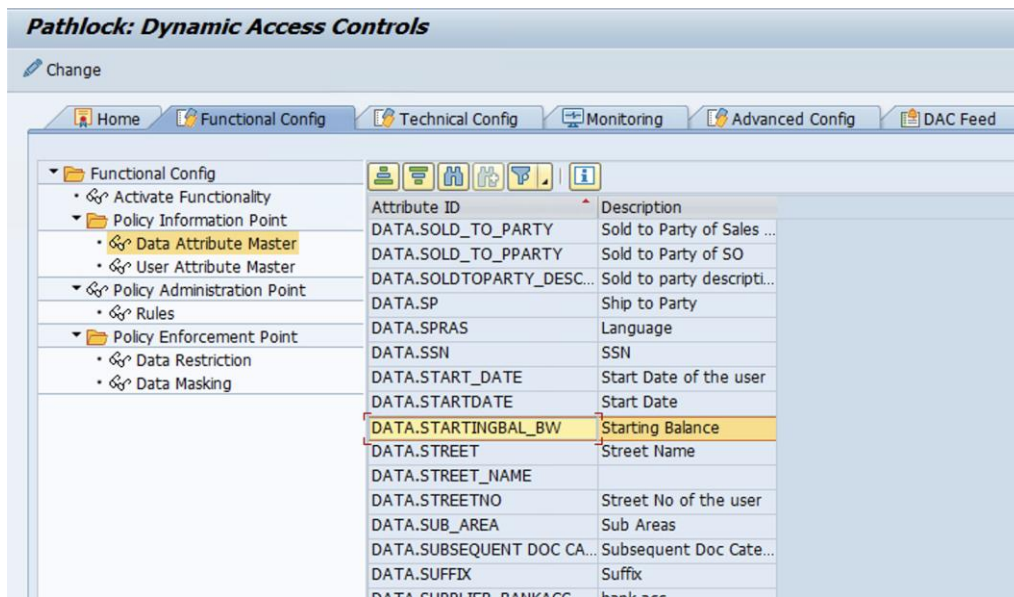


5.8 Use Case 8

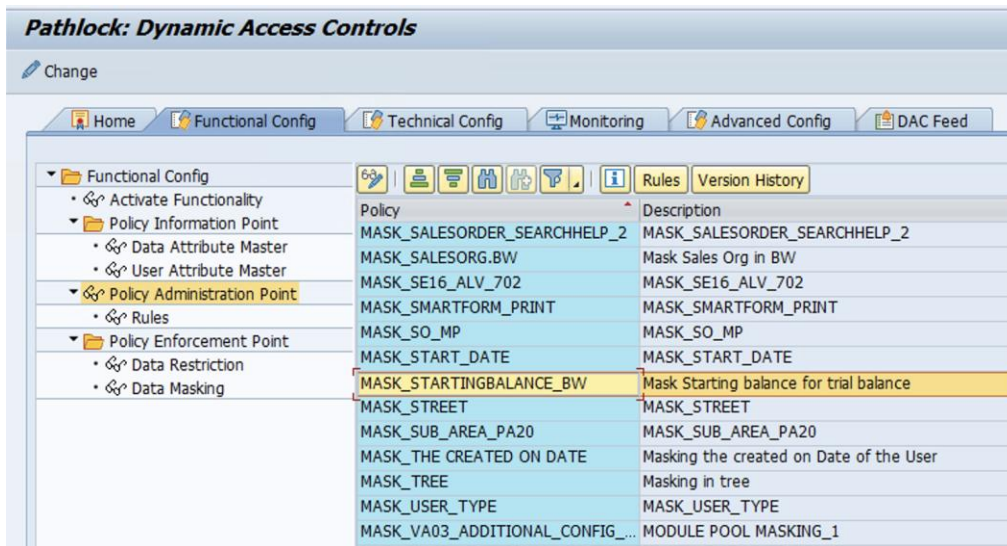
Masking 'Starting Balance' based on 'G/L Account.' In RSRT transaction and Trial Balance Application.

To implement the masking of the 'Starting Balance,' the following configuration is required:

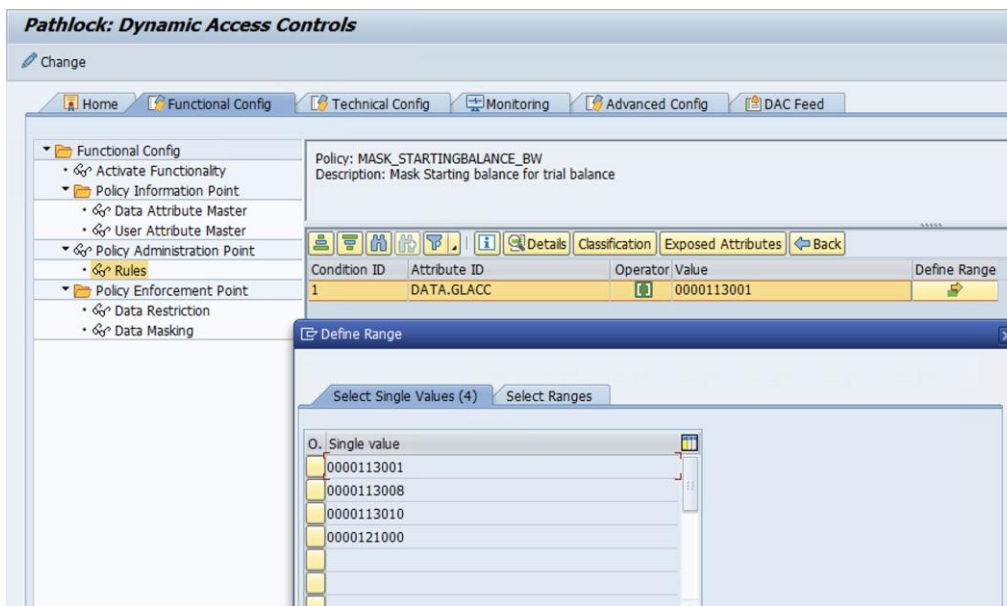
1. Define a Data Attribute with the Attribute ID 'DATA.STARTINGBAL_BW' and its description as 'Starting Balance.'



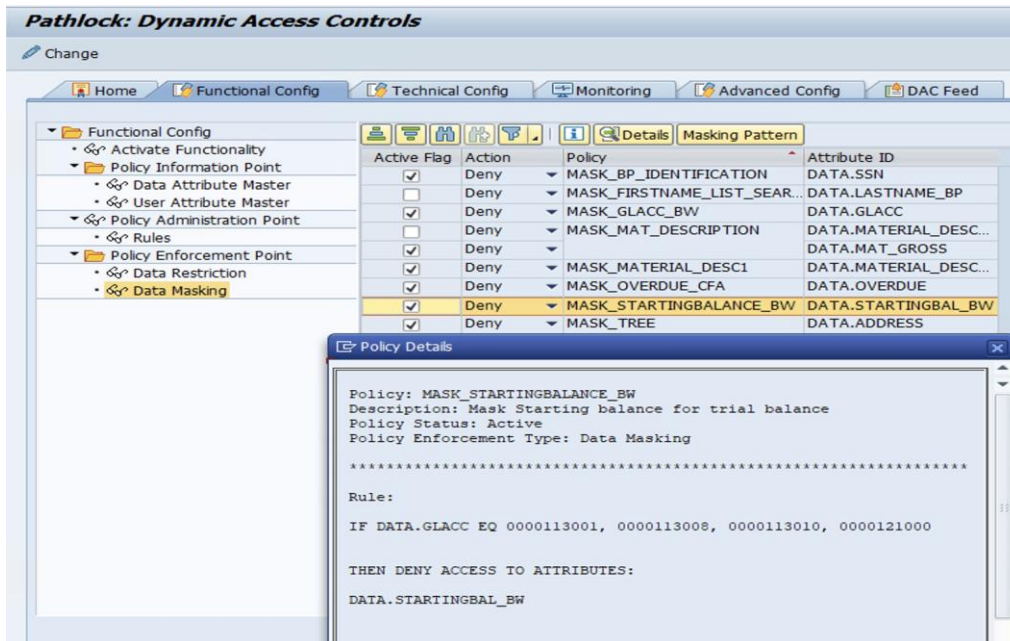
2. Create a Policy named 'MASK_STARTINGBALANCE_BW' along with its description.



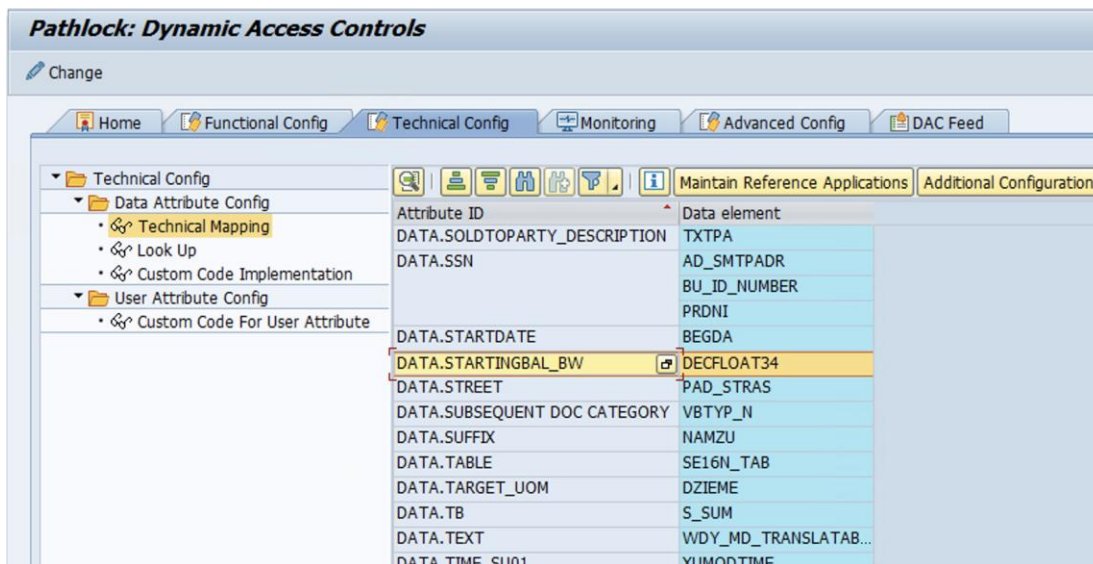
- Define rule condition using Data Attribute 'DATA.GLACC' to the Policy 'MASK_STARTINGBALANCE_BW'.



- Click on the Details button to view the Policy details.



- Under the **Technical Configuration** tab, assign the Attribute 'DATA.STARTINGBAL_BW' to the Data Element 'DECFLOAT34'.



Expected Results:

Upon implementing the Policy, the following screenshots demonstrate the expected outcomes of the enforcement. The masking will be applied across to output options, such as GUI Transactions (RSRT), Fiori/UI5 apps:

Query Monitor

DP_1 2CCFITRIALBALQ0001

Initial State

Information on Query

Static Filter

Dynamic Filter

Variable Values

Navigation Pane

Row Characteristics

Company Code

G/L Account

Column Characteristics

Key Figures

ELTUDCMP1

Free Characteristics

Ledger

G/L Account

Posting Date

Fiscal Year

Ledger Fiscal Year

Fin. Transaction Type

Bus. Transac. Type

Ref. Document Type

Logical System

Ref. Doc. Item Group

Is reversing

Is reversed

Query Trial Balance

Up-to-Dateness of Data: 10.07.2025 07:39:27 UTC

Variables Refresh Swap Axes Documents Conditions Exceptions Dynamic Selections Bookmark

Key Figures

Starting Balance in Company Code CrCY

Company Code	G/L Account	Starting Balance in Company Code CrCY
US01	Appspan USA	
	CANA/111000 Cash Desk	\$ -889.00
	CANA/113001 Bank - outg. checks	\$ 0.00
	CANA/113002 Bank - transfers	\$ -800.00
	CANA/113008 Concentration Acct	\$ 0.00
	CANA/113010 LocLockbox Account	\$ 0.00
	CANA/113030 Paymnt Card Acct	\$ -1,000.00
	CANA/113032 Paymnt Card Acct Out	\$ 3,800.00
	CANA/121000 Customer receivables	\$ 0.00
	CANA/122000 Customer receivables	\$ -6,055.00
	CANA/123020 Customer receivables	\$ -320.00
	CANA/125000 Interest receivable	\$ -300.00
	Result	\$ 0.00
	Overall Result	\$ 0.00

Row 1 - 13 / 13 Column 1 - 1 / 1

SAP Trial Balance

Standard

To show filters here, add them to the filter bar in Filters

Hide Filter Bar Filters (1) Go

Search

Pause

DIMENSIONS

Measures

Account Assgmt ...

Account Assignm...

Account Type

Activity Type

Altern. G/L Account

Asset Value Date

Assignment Refer...

Ast Transaction T...

Balance Trans CrCY

Balance sheet acct

Billing Type

Budget Period

Bus. Transac. Type

Business Area

Business Process

Calendar Month

COLUMNS

Measures

ROWS

Company Code

G/L Account

Company Code	Company Code	G/L Account	G/L Account	Starting Balance in Company Code CrCY
		CANA/111000	Cash Desk	\$ -889.00
		CANA/113001	Bank - outg. checks	\$ 0.00
		CANA/113002	Bank - transfers	\$ -800.00
		CANA/113008	Concentration Acct	\$ 0.00
		CANA/113010	LocLockbox Account	\$ 0.00
		CANA/113030	Paymnt Card Acct	\$ -1,000.00
		CANA/113032	Paymnt Card Acct Out	\$ 3,800.00
		CANA/121000	Customer receivables	\$ 0.00
		CANA/122000	Customer receivables	\$ -6,055.00
		CANA/123020	Customer receivables	\$ -320.00
		CANA/125000	Interest receivable	\$ -300.00
		CANA/125100	Other receivables	\$ -50.00
		CANA/125400	Travel Expenses - Ad	\$ 2,750.00
		CANA/191100	Loans Other - Long T	\$ 100.00
		CANA/211000	Trade Payables - Dom	\$ 7,300.00
		CANA/212000	Trade Payable - Fore	\$ 200.00
		CANA/215160	Employer Benefits	\$ -700.00
		CANA/216020	Corporate Tax Liabil	\$ 2,600.00
		CANA/216200	Tax Account	\$ 0.27
		CANA/217200	Other Liabilities	\$ 320.00
		CANA/223000	Accounts Payable - I	\$ -1,000.00

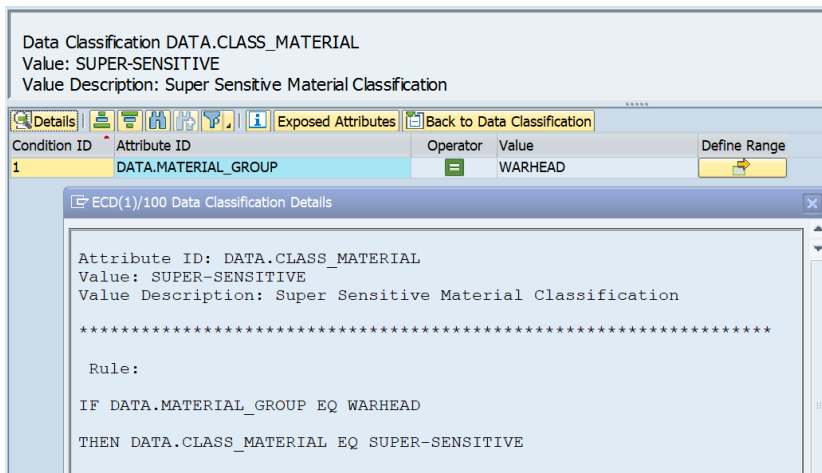
Starting Balance in Company Code Currency is expected to be 0 since it is numeric value

5.9. Use Case 9

Simulate Data Classification in Database tables

Users can now do a simulation of data classification on data base tables. This means that once a data classification has been configured in the system, the users can now check what classification applies to the fields of each record of a database table. Consider a scenario where the below classifications has been made:

1. Classification of material as SUPER-SENSITIVE based on material group.



Data Classification DATA.CLASS_MATERIAL
Value: SUPER-SENSITIVE
Value Description: Super Sensitive Material Classification

Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.MATERIAL_GROUP	=	WARHEAD	

ECD(1)/100 Data Classification Details

```

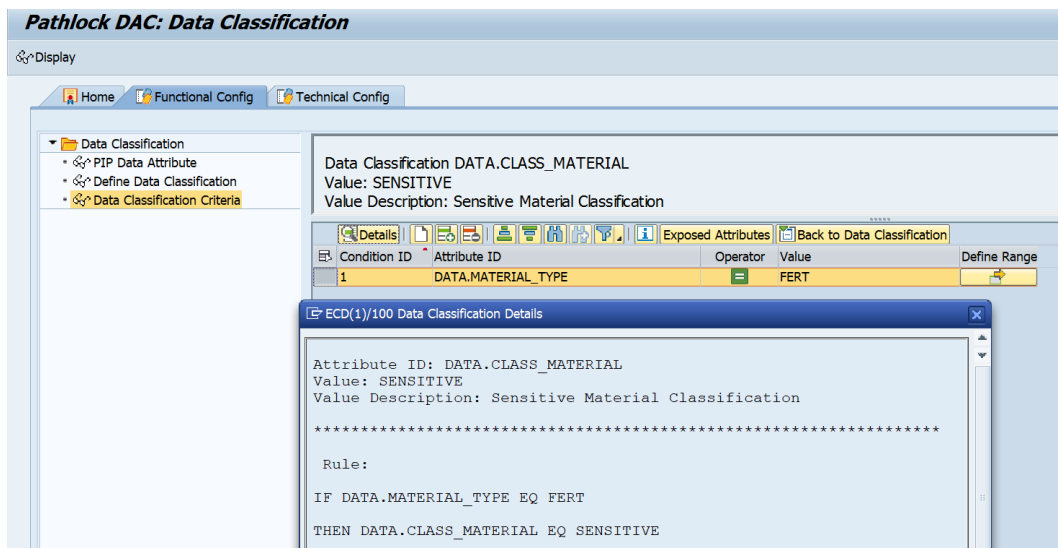
Attribute ID: DATA.CLASS_MATERIAL
Value: SUPER-SENSITIVE
Value Description: Super Sensitive Material Classification

*****

Rule:

IF DATA.MATERIAL_GROUP EQ WARHEAD
THEN DATA.CLASS_MATERIAL EQ SUPER-SENSITIVE
  
```

2. Classification of material as SENSITIVE based on material type



Pathlock DAC: Data Classification

Display

Home Functional Config Technical Config

- Data Classification
 - PIP Data Attribute
 - Define Data Classification
 - Data Classification Criteria

Data Classification DATA.CLASS_MATERIAL
Value: SENSITIVE
Value Description: Sensitive Material Classification

Condition ID	Attribute ID	Operator	Value	Define Range
1	DATA.MATERIAL_TYPE	=	FERT	

ECD(1)/100 Data Classification Details

```

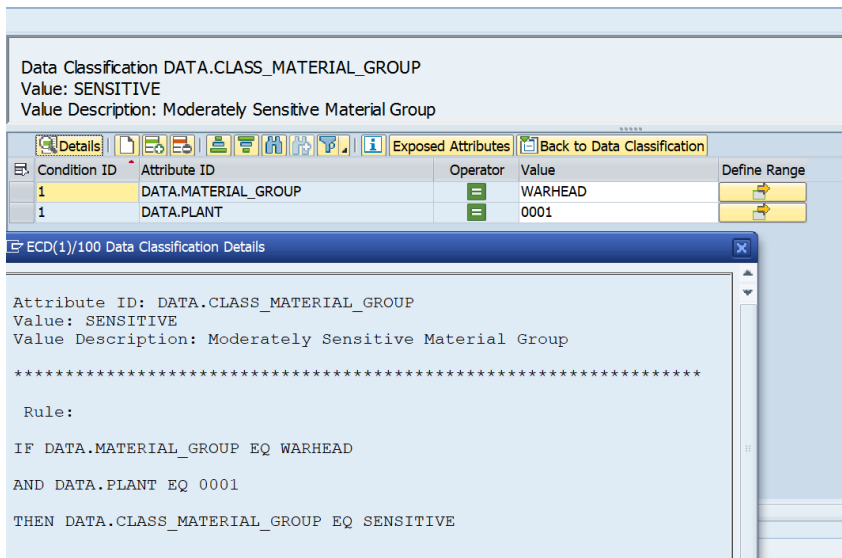
Attribute ID: DATA.CLASS_MATERIAL
Value: SENSITIVE
Value Description: Sensitive Material Classification

*****

Rule:

IF DATA.MATERIAL_TYPE EQ FERT
THEN DATA.CLASS_MATERIAL EQ SENSITIVE
  
```

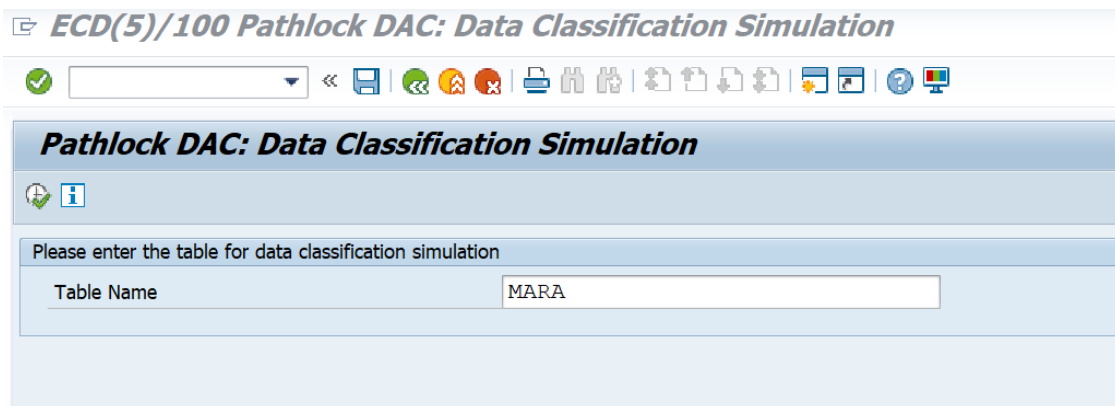
3. Classification of material based on material group and plant



Let us know do a simulation of this classification on the following database tables:

1. MARA
2. MAKT

In order to do a data classification simulation, execute the transaction /APPSDM/DC_SIMU and enter the table name MARA and click on execute button:



A screen similar to SE16N table browser selection screen will be displayed. Enter the filter conditions / limit the number of records to be displayed as shown below and click on the execute button:

ECD(5)/100 General Table Display

General Table Display

Background Number of Entries All Entries

Table: MARA
Text table: MAKT
Layout:
Maximum no. of hits: 10

General Material Data
☐ No texts
☐ Maintain entries

Selection Criteria

Fid name	O...	Fr.Value	To value	More	Output	Technical name
Client						MANDT
Material					☒	MATNR
Created On					☒	ERSDA
Created by					☒	ERNAM
Last Change					☒	LAEDA
Changed by					☒	AENAM
Complete status					☒	VPSTA

As you can see below, a simulation of data classification is displayed along with the key fields and other fields involved in classification

ECD(5)/100 Display of Entries Found

Display of Entries Found

Table to be searched: MARA
Number of hits: 10
Runtime: 0
Maximum no. of hits: 10

General Material Data

Material	MType	Matl Group	Data Classification_1	Value_1	Data Classification_2	Value_2	Data Classification_3	Value_3
1	ERSA	WARHEAD	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE		
3	FERT	WARHEAD	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
31	FERT	WARHEAD	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
32	FERT	01	DATA.CLASS_MATERIAL	SENSITIVE				
33	FERT	01	DATA.CLASS_MATERIAL	SENSITIVE				
34	FERT	02	DATA.CLASS_MATERIAL	SENSITIVE				
35	FERT	WARHEAD	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
36	FERT	01	DATA.CLASS_MATERIAL	SENSITIVE				
37	FERT		DATA.CLASS_MATERIAL	SENSITIVE				
40	FERT		DATA.CLASS_MATERIAL	SENSITIVE				

Repeat the same steps for table MAKT and the result is as given below:

Display of Entries Found

Table to be searched: Material Descriptions

Number of hits:

Runtime: Maximum no. of hits:

Material	Language	Data Classification_1	Value_1	Data Classification_2	Value_2	Data Classification_3	Value_3
1	EN	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE		
3	EN	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
5	EN	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
31	EN	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
33	EN	DATA.CLASS_MATERIAL	SENSITIVE				
34	EN	DATA.CLASS_MATERIAL	SENSITIVE				
35	EN	DATA.CLASS_MATERIAL_GROUP	SENSITIVE	DATA.CLASS_MATERIAL	SUPER-SENSITIVE	DATA.CLASS_MATERIAL	SENSITIVE
36	EN	DATA.CLASS_MATERIAL	SENSITIVE				
37	EN	DATA.CLASS_MATERIAL	SENSITIVE				
40	EN	DATA.CLASS_MATERIAL	SENSITIVE				

As you can see, the values of the fields which are not present on the screen such as plant are also derived by the application.